

IDENTIFICATION

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PRODUCT NAME: RK8E/RK8L DATA RELIABILITY PROGRAM
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TABLE OF CONTENTS

1.	ABSTRACT
2.	RESTRICTIONS
2.1	HARDWARE
2.2	PROGRAM STORAGE
2.3	PRELIMINARY PROGRAMS
2.4	EXECUTION TIME
3.	SWITCH REGISTER SETTINGS
4.	OPERATOR AND/OR PROGRAM ACTION
4.1	STANDARD TEST PROCEDURE
4.2	RK05J DRIVE CARTRIDGE MOUNTING PROCEDURE
4.3	RK05F DRIVE MOUNTING PROCEDURE
4.4	RK0E/RK0L DATA RELIABILITY (ACCEPT MODE)
4.5	RK0E DATA RELIABILITY (MANUAL INTERVENTION MODE)
4.6	CHANGE PROGRAM IOT CODES
5.	ERRORS
5.1	USEFUL INFORMATION
5.2	ERROR HALTS
5.3	ERROR TIMEOUTS
5.4	ERROR RECOVERY AND ERROR DISCONNECT
5.5	STATUS COMPLETE TIMEOUT AND PASS COMPLETE DISCONNECT
5.6	TYPICAL ERROR TIMEOUTS
6.	RESTRICTIONS
7.	TROUBLE SHOOTING INFORMATION
8.	PROGRAM DESCRIPTION (ACCEPT MODE)
9.	CONSOLE PACKAGE ADDENDUM
10.	APT-8 HOOKS
11.	PROGRAM LISTING

1.

ABSTRACT

THE RK0E/RK0L DATA RELIABILITY PROGRAM IS DESIGNED PRIMARILY AS AN ACCEPTANCE TEST TO VERIFY DISK DATA TRANSFERS WITHIN THE DISK SYSTEM.

THE "ACCEPT MODE" OF OPERATION VERIFIES THE CAPABILITY OF TRANSFERRING A TOTAL 3 X 10(9) BITS OF DATA TO AND FROM EACH INDIVIDUAL DISK DRIVE ON THE DISK SYSTEM.

THE "MANUAL INTERVENTION MODE" IS AVAILABLE AS A HARDWARE DEBUGGING AID TO ALLOW THE OPERATOR TO SELECT DATA PATTERNS, TRANSFER LENGTHS, AND ADDRESSING.

(NOTE: LOCATION 0 CONTAINS REVISION LEVEL (IN ASCII) OF PROGRAM ON PROGRAM LOAD).

2.

RESTRICTIONS

THE RK0L CONTROL, WHICH CAN CONTROL UP TO 8 DRIVES, WILL NOT RUN WITH THE DW0E BUS ADAPTER. THE REASON FOR THIS STATEMENT IS THAT THE RK0L CONTROL USES IOTS FOR EXTENDED DRIVES 4-7 WHICH IS NOT AVAILABLE ON THE DW0E.

2.1

HARDWARE

- A. PDP-8/A, 8/E, 8/F, OR 8/M COMPUTER OR OTHER FAMILY OF 8 COMPATIBLE COMPUTER WITH NECESSARY DW0E BUS ADAPTER.
- B. AT LEAST 4K OF READ/WRITE MEMORY. AT LEAST 8K OF MEMORY IS NECESSARY FOR OPERATION OF THE CONSOLE PACKAGE.
- C. ASR-33 TELETYPE OR EQUIVALENT
- D. RK0E OR RK0L DISK CONTROL
- E. RK05J OR RK05F DISK DRIVE(S)
- F. FORMATTED 2200 BPI-16 SECTOR PACK(S).

NOTE: THE RK05F DISK DRIVE IS CONSIDERED AS TWO SEPARATE UNITS. WHEN ANSWERING ALL QUESTIONS THE SEPARATE DRIVES MUST BE SPECIFIED. DSK07, DSK17, DSK27, ETC.

2.2

PROGRAM STORAGE

THE PROGRAM OCCUPIES OR UTILIZES LOCATION 0000 TO LOCATION 7577 OF FIELD 0. ALL EXTENDED MEMORY LOCATIONS, IF AVAILABLE, ARE UTILIZED FOR TESTING.

2.3

PRELIMINARY PROGRAMS

THIS PROGRAM REQUIRES A FORMATTED CARTRIDGE ON ALL DRIVES TO BE TESTED.

ALL BASIC AND EXTENDED MEMORY DIAGNOSTICS SHOULD BE RUN PRIOR TO RUNNING THIS PROGRAM.

RK8E CONTROL: RUN THE RK8E DISKLESS CONTROL TEST AND THE RK8E/RK8L DISK FORMATTER IF THIS DIAGNOSTIC FAILS TO OPERATE PROPERLY.

RK8L CONTROL: RUN THE RK8L INSTRUCTION TEST AND THE RK8E/RK8L FORMATTER IF THIS DIAGNOSTIC FAILS TO OPERATE PROPERLY.

2.4 EXECUTION TIME -----

THE PROGRAM EXECUTION TIME (I.E. PASSING 3×10^9 BITS OF DATA ON A DISK DRIVE), IS APPROX. 4 HOURS PER DISK DRIVE ON A 4K MEMORY SYSTEM OR APPROX. 3.5 HOURS PER DISK DRIVE ON SYSTEMS WITH EXTENDED MEMORY.

3. SWITCH REGISTER SETTINGS -----

SWR0#1 LOOP ON WRITE SEQUENCE.
SWR1#1 LOOP ON READ SEQUENCE.
SWR2#1 INHIBIT ALL ERROR TYPEOUTS
SWR3#1 TYPE "STATUS-COMLETE" REPORT.
SWR4#1 PROGRAM STOP ON HALT.
SWR5#1 DRIVE DISCONNECT AFTER PASS COMPLETION.
SWR6#1 PERFORM ONLY "OVERLAP SEEKS", DO NOT EXECUTE DATA BREAKS.

4. OPERATOR AND/OR PROGRAM ACTION -----

4.1 STANDARD TEST PROCEDURE -----

- A. START AS SPECIFIED THROUGH OUT THIS DOCUMENTATION IS KEY CLEAR AND THEN KEY CONTINUE ON PDP8/E, PDP8/M, AND PDP8/F COMPUTERS.
- B. LOAD THE PROGRAM INTO MEMORY FIELD 0 USING THE STANDARD BINARY LOADER TECHNIQUE.
- C. IF IT IS DESIRED TO CHANGE THE IOT CODES WITHIN THE PROGRAM, FOLLOW THE PROCEDURE IN SECTION 4.6.
- D. RUN THE ACCEPTANCE MODE OF DATA RELIABILITY WITH ALL DRIVES AND MEMORY AVAILABLE BY FOLLOWING THE PROCEDURE

IN SECTION 4.4.

E. THE MANUAL INTERVENTION MODE, SECTION 4.5, MAY BE USED FOR TROUBLE SHOOTING, IF DESIRED.

F. IF POSSIBLE SWR4#1 SHOULD ALWAYS BE USED TO STOP THE PROGRAM.

G. IF THE PROGRAM HAS BEEN STOPPED DUE TO SWR4#1, THE PROGRAM CAN BE RESTARTED, AND THE INITIAL STARTUP QUESTIONS BYPASSED, BY USING 0205 AS THE RESTART ADDRESS.

H. FOR THE ABSOLUTE LOCATIONS OF ALL KNOWN HALTS IN THIS PROGRAM, ACCESS PAGE 1-22 OF THE PROGRAM LISTING.

4.2

RK05J DRIVE CARTRIDGE MOUNTING PROCEDURE

THE FOLLOWING IS THE CORRECT CARTRIDGE MOUNTING PROCEDURE FOR THE RK05J DISK DRIVE. ANY DEVIATION ENCOUNTERED DURING THIS PROCEDURE WILL BE CONSIDERED AN ERROR CONDITION.

A. SET SWITCH LABELED "RUN/LOAD" TO THE "LOAD" POSITION.

B. TURN AC POWER TO DISK DRIVE ON.

C. VERIFY THAT THE LIGHT LABELED "PWR" IS ON.

D. WAIT FOR THE LIGHT LABELED "LOAD" TO COME ON.

E. VERIFY THAT THE LIGHTS LABELED "RDY", "ON CYL", "FAULT", "WT", AND "RD" ARE OFF.

F. OPEN ACCESS DOOR.

G. INSERT CARTRIDGE.

H. CLOSE ACCESS DOOR.

I. SET SWITCH LABELED "RUN/LOAD" TO THE "RUN" POSITION.

J. WAIT FOR THE LIGHTS LABELED "RDY" AND "ON CYL" TO COME ON.

K. TOGGLE SWITCH LABELED "WT PROT" AND VERIFY THAT THE LIGHT LABELED "WT PROT" GOES ON AND OFF.

L. TOGGLE SWITCH LABELED "WT PROT" UNTIL THE LIGHT LABELED "WT PROT" IS OFF.

M. VERIFY THAT LIGHTS LABELED "FAULT", "WT", "RD", AND "LOAD" ARE OFF.

4.3

RK05F DRIVE SETUP PROCEDURE

THE FOLLOWING IS THE CORRECT SETUP PROCEDURE

FOR THE RK05F DISK DRIVE. ANY DEVIATION ENCOUNTERED DURING THIS PROCEDURE WILL BE CONSIDERED AN ERROR CONDITION.

- A. SET SWITCH LABELED "RUN/LOAD" TO THE "LOAD" POSITION.
- B. TURN AC POWER ON.
- C. VERIFY THAT LIGHT LABELED "PWR" IS ON.
- D. WAIT FOR LIGHT LABELED "LOAD" TO COME ON.
- E. VERIFY THAT LIGHTS LABELED "RDY", "ON CYL", "FAULT", "WT", AND "RD" ARE OFF.
- F. SET SWITCH LABELED "RUN/LOAD" TO THE "RUN" POSITION.
- G. WAIT FOR LIGHTS LABELED "RDY" AND "ON CYL" TO COME ON.
- H. TOGGLE SWITCH LABELED "WT PROT" AND VERIFY THAT THE LIGHT LABELED "WT PROT" GOES ON AND OFF.
- I. TOGGLE SWITCH LABELED "WT PROT" UNTIL LIGHT LABELED "WT PROT" GOES OFF.
- J. VERIFY THAT LIGHTS LABELED "FAULT", "WT", "RD", AND "LOAD" ARE OFF.

4.4

RK05E/RK05L DATA RELIABILITY (ACCEPT MODE)

- A. MAKE READY ALL DRIVES TO BE TESTED USING THE RK05J DRIVE CARTRIDGE MOUNTING PROCEDURE SECTION 4.2 OR THE RK05F DRIVE PROCEDURE IN SECTION 4.3.
- B. SET SWITCH LABELED "RUN/LOAD" TO THE "LOAD" POSITION ON ALL DRIVES NOT BEING TESTED.
- C. VERIFY THAT AC POWER IS ON, ON ALL DRIVES NOT BEING TESTED.
- D. SET THE SWITCH REGISTER TO 0200 AND PRESS LOAD ADDRESS.
- E. SET THE SWITCH REGISTER TO 0000 AND PRESS START.
- F. THE OPERATOR MAY SET SWRS=1 IF IT IS DESIRED TO HAVE THE PROGRAM AUTOMATICALLY DISCONNECT EACH DISK DRIVE AS EACH MAKE THEIR PASS COMPLETION. (NOTE: IF SWRS=0, ALL DISK DRIVES WILL CONTINUE TO RUN AFTER THEIR PASS COMPLETION)
- G. THE TTY WILL PRINT THE FOLLOWING PROGRAM NAME AND QUESTION.

RK05E/RK05L DATA RELIABILITY
EXTENDED R/W MEMORY (0-7)?

THE OPERATOR SHOULD THEN TYPE THE AMOUNT OF EXTENDED READ/ WRITE MEMORY BANKS NUMBERED SEQUENTIALLY FROM BANK 0, AS INDICATED BY THE TTY QUESTION.

- H. THE TTY WILL PRINT THE FOLLOWING QUESTION(S), ASKING THE

DESIRED DISK DRIVE(S) TO BE USED IN TESTING.

EXERCISE DISK07
EXERCISE DISK17
EXERCISE DISK27
EXERCISE DISK37
EXERCISE DISK47
EXERCISE DISK57
EXERCISE DISK67
EXERCISE DISK77

FOR THE QUESTION(S) ABOVE, TYPE Y FOR YES, IF IT IS DESIRED TO TEST THE DISK DRIVE IN QUESTION, OTHERWISE, TYPE N FOR NO.

I. THE TTY WILL PRINT THE FOLLOWING QUESTION.

ACCEPT MODE?

THE OPERATOR SHOULD THEN TYPE Y FOR YES TO RUN THE ACCEPTANCE MODE OF OPERATION.

J. THE TTY WILL PRINT THE FOLLOWING QUESTION.

ARE YOU SURE?

IF THE OPERATOR IS CERTAIN OF THE AMOUNT OF MEMORY, THE DISK DRIVE(S) SELECTED, AND THE MODE OF OPERATION, TYPE Y FOR YES. TYPING N FOR NO WILL RESULT IN A REPEAT OF ALL MESSAGES AND QUESTIONS ENCOUNTERED THUS FAR.

K. THE PROGRAM SHOULD START TESTING THE DISK DRIVE(S) AND MEMORY SELECTED.

L. THE "STATUS=COMPLETE" TYPEOUT SHOULD OCCUR UPON PASS COMPLETION OF EACH DISK DRIVE. ALL OTHER TYPEOUTS OR HALTS WILL BE CONSIDERED AS AN ERROR CONDITION. SEE SECTION 5.5 FOR "STATUS=COMPLETE" TYPEOUT.

M. A SUCCESSFUL PASS COMPLETE ON A DISK DRIVE WILL BE CONSIDERED AS NO "HARD" ERRORS AND NO MORE THAN ONE (1) "SOFT" ERROR PER PASS COMPLETE.

N. IF ANY ERRORS DO OCCUR, THE OPERATOR SHOULD ACCESS SECTION 5 IN THIS DOCUMENTATION.

4.5

RK8E/RK8L DATA RELIABILITY (MANUAL INTERVENTION MODE)

THE MANUAL INTERVENTION MODE IS AVAILABLE AS A TROUBLE SHOOTING AID AND SHOULD ONLY BE USED FOR SUCH PURPOSES, IF DESIRED.

A. MAKE READY ALL DISK DRIVES TO BE TESTED USING THE RK05J DRIVE CARTRIDGE MOUNTING PROCEDURE SECTION 4.2. OR THE RK05F DRIVE PROCEDURE SECTION 4.3.

B. SET SWITCH LABELED "RUN/LOAD" TO THE "LOAD" POSITION ON

ALL DRIVES NOT BEING TESTED.

- C. VERIFY THAT AC POWER IS ON, ON ALL DRIVES NOT BEING TESTED.
- D. SET THE SWITCH REGISTER TO 0200 AND PRESS LOAD ADDRESS.
- E. SET THE SWITCH REGISTER TO 0000 AND PRESS START.
- F. THE TTY WILL PRINT THE FOLLOWING PROGRAM NAME AND QUESTION.

RK0E/RK0L DATA RELIABILITY
EXTENDED R/W MEMORY (0-7)?

THE OPERATOR SHOULD THEN TYPE THE AMOUNT OF EXTENDED READ/
WRITE MEMORY BANKS NUMBERED SEQUENTIALLY FROM BANK 0, AS
INDICATED BY THE TTY QUESTION.

- G. THE TTY WILL PRINT THE FOLLOWING QUESTION(S), ASKING THE
DESIRED DISK DRIVE(S) TO BE USED IN TESTING.

EXERCISE DISK0?
EXERCISE DISK1?
EXERCISE DISK2?
EXERCISE DISK3?
EXERCISE DISK4?
EXERCISE DISK5?
EXERCISE DISK6?
EXERCISE DISK7?

FOR THE QUESTION(S) ABOVE, TYPE Y FOR YES, IF IT IS DESIRED
TO TEST THE DISK DRIVE IN QUESTION, OTHERWISE, TYPE N FOR
NO.

- H. THE TTY WILL PRINT THE FOLLOWING QUESTION.

ACCEPT MODE?

THE OPERATOR SHOULD THEN TYPE N FOR NO TO RUN THE MANUAL
INTERVENTION MODE OF OPERATION.

- I. THE TTY WILL THEN PRINT THE FOLLOWING QUESTION, ASKING
IF THE OPERATOR DESIRES TO SELECT A CONSTANT MEMORY FIELD,
RATHER THAN THE NORMAL RANDOM FIELD SELECTION.

FIELD?

IF THE OPERATOR DESIRES TO SELECT A CONSTANT FIELD,
TYPE Y FOR YES, OTHERWISE, TYPE N FOR NO. IF Y WAS TYPED THE
TTY WILL SPACE OUT UNCE AND WAIT FOR THE OPERATOR TO TYPE
THE DESIRED FIELD IN OCTAL (0-7).

- J. THE TTY WILL PRINT THE FOLLOWING QUESTION, ASKING IF THE
OPERATOR DESIRES TO SELECT A CONSTANT TRACK, RATHER THAN
THE NORMAL RANDOM TRACK SELECTION.

TRACK?

IF THE OPERATOR DESIRES TO SELECT A CONSTANT TRACK, TYPE Y FOR YES, OTHERWISE, N FOR NO. IF Y WAS TYPED, THE TTY WILL SPACE OUT ONCE AND WAIT FOR THE OPERATOR TO INPUT THE DESIRED TRACK ADDRESS (00000-14537).

- K. THE TTY WILL PRINT THE FOLLOWING QUESTION, ASKING IF THE OPERATOR DESIRES TO SELECT HALF BLOCK OR FULL BLOCK TRANSFERS, RATHER THAN THE NORMAL RANDOM SELECTION.

BLOCK LENGTH?

IF THE OPERATOR DESIRES TO CHANGE THE BLOCK LENGTH, TYPE Y FOR YES, OTHERWISE, N FOR NO. IF Y WAS TYPED THE TTY WILL SPACE OUT ONCE AND WAIT FOR THE OPERATOR TO TYPE THE BLOCK LENGTH DESIRED (0-256 WORD BLOCK OR 1-128 WORD BLOCK).

- L. THE TTY WILL PRINT THE FOLLOWING QUESTION, ASKING IF THE OPERATOR DESIRES TO SELECT A CONSTANT NUMBER OF SECTORS TO BE TRANSFERED, RATHER THAN THE NORMAL RANDOM SECTOR SELECTION.

EXTRA SECTORS?

IF THE OPERATOR DESIRES TO SELECT A CONSTANT AMOUNT OF SECTORS, TYPE Y FOR YES, OTHERWISE, N FOR NO. IF Y WAS TYPED THE TTY WILL SPACE OUT ONCE, AND WAIT FOR THE OPERATOR TO TYPE IN THE EXTRA SECTORS DESIRED (00-17). (NOTE: IF THE FIELD AND THE BLOCK LENGTH PREVIOUSLY SELECTED WAS 0, THE AMOUNT OF EXTRA SECTORS WILL BE LIMITED TO 07. OTHERWISE THE MAXIMUM AMOUNT IS LIMITED TO 17.)

- M. THE TTY WILL PRINT THE FOLLOWING QUESTION, ASKING IF THE OPERATOR DESIRES TO SELECT A DATA PATTERN, RATHER THAN NORMAL RANDOM DATA SELECTION.

DATA?

IF THE OPERATOR DESIRES TO SELECT A DATA PATTERN, TYPE Y FOR YES, OTHERWISE, N FOR NO. IF Y WAS TYPED, THE TTY WILL DO A "CRLF" AND WAIT FOR THE OPERATOR TO TYPE IN 12 OCTAL DATA WORDS TO BE USED IN TESTING.

- N. THE TTY WILL PRINT THE FOLLOWING QUESTION.

ARE YOU SURE?

IF THE OPERATOR IS CERTAIN OF THE INFORMATION SELECTED, TYPE Y FOR YES. TYPING N FOR NO WILL RESULT IN A REPEAT OF ALL MESSAGES AND QUESTIONS ENCOUNTERED THUS FAR.

- P. THE PROGRAM SHOULD START EXECUTING THE OPERATIONS SELECTED.

- R. IF ERRORS ARE ENCOUNTERED, ACCESS SECTION 5 IN THIS DOCUMENTATION.

THE PROGRAM NORMALLY RECOGNIZES DEVICE IOT CODE X74X. TO CHANGE THE DEVICE IOT CODES WITHIN THE PROGRAM:

- A. SET THE SWITCH REGISTER TO 0204 AND PRESS LOAD ADDRESS.
- B. SET THE SWITCH REGISTER TO 0000, SET SWITCH REGISTER BITS 3-8 TO THE DESIRED DEVICE IOT CODE, AND PRESS START.
- C. THE PROGRAM WILL CHANGE THE DEVICE IOT CODES WITHIN THE PROGRAM AND THEN HALT.
- D. PRESSING KEY CONTINUE WILL START THE PROGRAM AT LOCATION 0200 (SEE SECTIONS 4.4 OR 4.5 FOR OPERATION INSTRUCTIONS).

5. ERRORS -----

5.1 USEFUL INFORMATION -----

ALL STATUS ERRORS WILL BE REPORTED AS STATUS ERRORS. ALL DATA ERRORS WILL BE REPORTED AS DISK DATA ERRORS.

WHEN DATA IS BEING READ OFF THE DISK AND A CRC ERROR OCCURS THE PROGRAM WILL REPORT THE ERROR AS A READ STATUS ERROR. THE PROGRAM WILL THEN CHECK THE DATA READ FOR DATA ERRORS. IF DATA ERRORS EXIST THEY WILL BE REPORTED AS DISK DATA ERRORS.

5.2 ERROR HALTS -----

ERROR HALTS FOR WHICH THERE ARE NO ERROR TYPEOUTS ARE LISTED AND DEFINED AS FOLLOWS.

BIGSTP	MASTER ERROR HALT FOR ALL OF THE FOLLOWING ERROR STOPS. WHEN THE COMPUTER HALTS THE AC REGISTER WILL INDICATE THE PC OF THE FAILING ERROR STOP.
INTER1	NO DISK INTERRUPT
ERHLT0	SKIP TRAP FOR IOT "DLSC"
ERHLT2	SKIP TRAP FOR IOT "DCLR"
ERHLT3	SKIP TRAP FOR IOT "DLAG"
ERHLT5	SKIP TRAP FOR IOT "DRST"
ERHLT6	SKIP TRAP FOR IOT "DLDC"
BADHLT	CHECKSUM FAILED BUT WORD-BY-WORD COMPARE WORKED

NODSKS

NO DISKS AVAILABLE TO RUN

FLOHLT

PROGRAM WILL ONLY RUN IN FIELD 0

FOR THE ABSOLUTE LOCATIONS OF THE HALTS LISTED ABOVE,
ACCESS PAGE 1-22 OF THE PROGRAM LISTING.

5.3

ERROR TYPEOUTS

WHEN AN ERROR OCCURRES THE PROGRAM WILL PRINT AN
"ERROR HEADER" WHICH WILL SPECIFY THE PARTICULAR TYPE
OF ERROR FOUND AT THE TIME OF THE FAILURE.

POSSIBLE "ERROR HEADERS" ARE AS FOLLOWS.

WRITE STATUS ERROR
READ STATUS ERROR
DISK DATA ERROR
RECALIBRATE STATUS ERROR

AFTER THE "ERROR HEADER" MENTIONED ABOVE IS TYPED, THE
PROGRAM WILL PRINT THE FOLLOWING ERROR INFORMATION
FOUND AT THE TIME OF THE FAILURE, PERTAINING TO THE
FAILURE. POSSIBLE TYPEOUTS ARE AS FOLLOWS.

PC: PROGRAM LOCATION OF THE ACTUAL FAILURE.

ST: CONTENTS OF THE STATUS REGISTER.

EX: EXTENDED DRIVE BIT

CM: SOFTWARE COMMAND REGISTER.

IA: INITIAL SOFTWARE DISK ADDRESS REGISTER OR THE
CYLINDER, SURFACE, AND SECTOR BITS.

DA: FINAL SOFTWARE DISK ADDRESS REGISTER OR THE
CYLINDER, SURFACE, AND SECTOR BITS.

CA: SOFTWARE INITIAL CURRENT ADDRESS

WC: SOFTWARE INITIAL WORD COUNT

FW: SOFTWARE FINAL WORD COUNT

AS: SECTOR IN ERROR ON THE PARTICULAR CYLINDER
AND SURFACE IN QUESTION.

WA: WORD ADDRESS WITHIN THE SECTOR IN ERROR

AD: BREAK ADDRESS OF DATA BREAK IN COMPUTER.

DG: EXPECTED DATA

DB: DATA FOUND DURING DATA BREAK.

5,4

ERROR RECOVERY AND ERROR DISCONNECT

WHEN A READ, WRITE, OR DISK DATA ERROR OCCURS (SEE SECTION 5.3), THE PROGRAM WILL TRY TO REPEAT THE FAILING SEQUENCE FOUR (4) TIMES. IF THE ERROR HAS OCCURRED FOUR (4) TIMES SIMULTANEOUSLY, THE ERROR WILL BE CONSIDERED AS A NON-RECOVERABLE ERROR, THE "ERROR HEADER" WILL BE CHANGED TO INDICATE "NON-RECOVERABLE" ERROR, ANOTHER DISK ADDRESS WILL BE SELECTED FOR TESTING. IF A "SOFT" ERROR SHOULD OCCUR ON A TRACK, THE PROGRAM WILL RETRY THE READ SEQUENCE (64) TIMES BEFORE SELECTING ANOTHER TRACK FOR TESTING. (NOTE: THIS 64 RETRY'S ON "SOFT" ERRORS WILL BE TERMINATED ON A "HARD" ERROR).

POSSIBLE NON-RECOVERABLE ERROR HEADERS ARE AS FOLLOWS.

NON-RECOVERABLE READ STATUS ERROR
NON-RECOVERABLE WRITE STATUS ERROR
NON-RECOVERABLE DISK DATA ERROR

IF A NON-RECOVERABLE READ OR WRITE ERROR SHOULD OCCUR, THE DISK IN QUESTION WILL THEN BE RECALIBRATED (RESTORED TO CYLINDER 0). IF THE RECALIBRATE SEQUENCE FAILS, THE DISK DRIVE IN ERROR WILL BE DISCONNECTED BY THE PROGRAM AND NO LONGER BE TESTED.

THE FOLLOWING "DISCONNECT" AND "STATUS-COMPLETE" TYPEOUTS SHOULD OCCUR.

RECALIBRATE ERROR DISCONNECT!
DISK X DISCONNECTED!
DSK HARD SOFT COMP
X 0030 0010 0001
X 0240 5670 0001

IF ALL DISKS ON THE SYSTEM HAVE BEEN DISCONNECTED DO TO RECALIBRATE ERRORS THE FOLLOWING TYPEOUT WILL OCCUR AND THE PROGRAM WILL HALT.

DISK SYSTEM SHUT DOWN, NO DISKS TO RUN!

5,5

STATUS-COMPLETE TYPEOUT AND PASS COMPLETE DISCONNECT

ALL ERRORS AND PASS COMPLETES ARE TALLIED BY THE PROGRAM PER DISK DRIVE.

THE FOLLOWING IS AN EXAMPLE OF THE "STATUS-COMPLETE" TYPEOUT THAT WILL OCCUR WHEN SWR3=1 INDICATING TYPE THIS REPORT, A PASS COMPLETE OCCURS ON A DRIVE UNDER TEST, OR A DRIVE IS DISCONNECTED DO TO A RECALIBRATE ERROR.

DSK HARD SOFT COMP
X XXXX XXXX XXXX
X XXXX XXXX XXXX
X XXXX XXXX XXXX

X XXXX XXXX XXXX

THE TYPEOUT AS MENTIONED ABOVE IS DESCRIBED AS FOLLOWS.

DSK DISK DRIVE IN QUESTION.

HARD ALL ERRORS OTHER THAN THAT DEFINED AS
A SOFT ERROR.

SOFT A READ CRC STATUS ERROR WITH BAD DATA PER
TRANSFER WITH RECOVERY POSSIBLE WITHIN FOUR (4)
RETRYs. (NOTE: FOUR (4) CONSECUTIVE RETRYs WILL
BE CONSIDERED AS A NON-RECOVERABLE ERROR OR A
"HARD" ERROR).

COMP PASS COMPLETES. <3 X 10(9) BITS>

IF SHRS=1 INDICATING "DISCONNECT ON PASS COMPLETION", AND
A DISK DRIVE UNDER TEST MAKES A PASS COMPLETION, THE
FOLLOWING TYPEOUT WILL OCCUR AND THE DRIVE WILL BE
DISCONNECTED.

DISK X PASS COMPLETE!
DISK X DISCONNECTED!
DSK HARD SOFT COMP
X XXXX XXXX XXXX
X XXXX XXXX XXXX

IF SHRS=0 INDICATING DON'T "DISCONNECT ON PASS COMPLETION",
AND A DISK DRIVE UNDER TEST MAKES A PASS COMPLETION, THE
FOLLOWING TYPEOUT WILL OCCUR AND THE DRIVE WILL CONTINUE
TO RUN.

DISK X PASS COMPLETE!
DSK HARD SOFT COMP
X XXXX XXXX XXXX
X XXXX XXXX XXXX

IF SHRS=1 AND ALL DRIVES HAVE MADE THEIR PASS COMPLETION
AND HAVE BEEN DISCONNECTED, THE FOLLOWING TYPEOUT WILL
OCCUR AND THE COMPUTER WILL HALT.

DISK SYSTEM SHUT DOWN, NO DISKS TO RUN!

5.6

TYPICAL ERROR TYPEOUTS

THE FOLLOWING IS AN EXAMPLE OF AN "ERROR HEADER" AND
ERROR TYPEOUT THAT COULD HAVE OCCURRED ON A WRITE STATUS
ERROR. (NOTE CRC IN THE STATUS INDICATOR "ST!")

WRITE STATUS ERROR
PC:2371 ST:4010 EX:0001 CM:4000 IA:0001 DA:0002
CA:3600 WC:7000 FW:0000

THE FOLLOWING IS AN EXAMPLE OF AN ERROR TYPEOUT THAT COULD
HAVE OCCURRED IF THE STATUS REGISTER FAILED ON A SEEK
ONLY FUNCTION.

SEEK STATUS ERROR
PC:12076 ST:14002 EX:10001 CM:13000 DA:14007

THE FOLLOWING IS A TYPICAL EXAMPLE OF AN "ERROR HEADER"
AND ERROR TYPEOUT THAT COULD HAVE OCCURRED ON A DISK
DATA ERROR. (NOTE: ADDITION DATA ERRORS IN BUFFER)

DISK DATA ERROR
PC:11674 ST:14010 EX:10001 CM:11432 IA:11035 DA:11021
CA:10001 WC:15000 FW:17400
AS:10015 WA:10007 AD:10010 DG:10537 DB:10536
AS:10015 WA:10077 AD:10100 DG:17777 DB:17776
AS:10016 WA:10002 AD:10405 DG:16167 DB:16166

6. RESTRICTIONS

ALL DISK DRIVES SHOULD BE SET TO THE LOAD POSITION
THAT ARE NOT BEING TESTED.

7. TROUBLE SHOOTING INFORMATION

IOT	FUNCTION
---	-----
6740 DLSC	LOAD SECTOR COUNTER AND EXTENDED DRIVE BIT FOR RK8L.
AC	
--	
0-3	LOAD THE DESIRED AMOUNT OF SECTORS TO BE TRANSFERRED WITH THE BINARY VALUE IN AC BITS 0-3.
4	EXTENDED DRIVE BIT FOR DRIVES 4-7
6741 DSKP	"SKIP" SKIP IF TRANSFER DONE FLAG OR ERROR FLAG IS SET.
6742 DCLR	"CLEAR" FUNCTION IS REGULATED BY AC BITS 10 AND 11. THE AC IS THEN CLEARED.
AC10 AC11	
----- -----	
0 0	CLEAR THE AC AND STATUS REGISTER.
0 1	CLEAR THE AC, CONTROL, AND MAJOR REGISTERS. THIS INSTRUCTION WILL STOP THE CONTROL EVEN IF IT IS WRITING A HEADER. THIS IS THE ONLY INSTRUCTION THAT CLEARS MAINTENANCE MODE.

1 0

CLEAR AC, RECALIBRATE DISK DRIVE,
AND CLEAR STATUS REGISTER.

6743 DLAG

"LOAD DISK ADDRESS AND GO" LOAD THE
DISK CYLINDER, SURFACE, AND SECTOR
FROM THE AC, CLEAR THE AC, AND DO
THE COMMAND IN THE COMMAND REGISTER.

AC
--

0-6

CYLINDER

7

SURFACE (1=UPPER) (0=LOWER)

8-11

SECTOR

6744 DLCA

"LOAD CURRENT ADDRESS" LOAD THE
CURRENT ADDRESS FROM AC. THE AC
IS THEN CLEARED.

AC
--

0-11

CURRENT ADDRESS

6745 DRST

"READ STATUS" CLEAR THE AC AND
READ THE CONTENTS OF THE STATUS
REGISTER INTO THE AC.

AC
--

0

TRANSFER DONE

1

READY TO SEEK, READ, OR WRITE.

2

NOT USED

3

SEEK FAIL

4

DISK FILE READY

5

CONTROL BUSY ERROR

6

TIME OUT ERROR

7

WRITE LOCK ERROR

8

CRC ERROR

9

DATA RATE ERROR

10

DRIVE STATUS ERROR

11

CYLINDER ADDRESS ERROR

6746 DLDC

"LOAD COMMAND" LOAD THE COMMAND
REGISTER FROM AC, CLEAR THE AC,
AND CLEAR THE STATUS REGISTER.

AC
--

0-2#0

READ DATA

0-2#1

READ ALL

0-2#2

WRITE LOCK

0-2#3

SEEK ONLY

0-2#4

WRITE DATA

0-205	WRITE ALL
0-206	NOT USED
0-207	NOT USED
3	ENABLE INTERRUPT
4	ENABLE SET TRANSFER DONE ON SEEK DONE
5	HALF BLOCK 120 WORDS
6	EXTENDED MEMORY ADDRESS
7	EXTENDED MEMORY ADDRESS
8	EXTENDED MEMORY ADDRESS
9	UNIT SELECT
10	UNIT SELECT
11	EXTENDED CYLINDER ADDRESS

6747 DMAN

"MAINTENANCE IOT" LOAD THE MAINTENANCE REGISTER FROM THE AC. THE FUNCTION IS REGULATED BY THE AC BITS. MAINTENANCE MODE CAN ONLY BE CLEARED BY DCLK "CLEAR CONTROL".

AC
--

0	ENTER MAINTENANCE MODE
1	ENABLE SHIFT TO LOWER BUFFER
2	AC BIT 10, CRC REGISTER, AND THE LOWER DATA BUFFER ARE CONNECTED AS A SHIFT REGISTER. AC BIT 10 DATA SHIFTS TO THE CRC, THE CRC SHIFTS TO THE LOWER DATA BUFFER.
3	SHIFT COMMAND REGISTER TO THE LOWER DATA BUFFER.
4	SHIFT THE SURFACE AND SECTOR REGISTER TO THE LOWER DATA BUFFER.
5	SHIFT AC 10 DATA TO THE UPPER DATA BUFFER. THE UPPER BUFFER SHOULD SINK IN THE SILO WHEN FULL.
6	ONE SINGLE CYCLE BREAK REQUEST. DIRECTION IS REGULATED BY FUNCTION IN THE COMMAND REGISTER.
7	CLEAR AC THEN READ THE LOWER DATA BUFFER TO THE AC.
8	NOT USED.
9	NOT USED.
10	USED AS DATA WITH OTHER BITS IN THE MAINTENANCE MODE.
11	NOT USED

8.

PROGRAM DESCRIPTION (ACCEPT MODE)

THE FOLLOWING IS BRIEF DESCRIPTION OF THE STEPS TAKEN BY THE PROGRAM WHEN RUNNING THE ACCEPT MODE.

- A. A RANDOM FIELD IS GENERATED. IF FIELD GENERATED IS A NON-EXISTING FIELD, THE MAXIMUM FIELD AVAILABLE WILL BE USED.

- B. A RANDOM BLOCK LENGTH IS GENERATED (128 OR 256 WORD SECTORS).
- C. A RANDOM AMOUNT OF SEQUENTIAL SECTORS TO TRANSFER IS GENERATED. IF THE FIELD PREVIOUSLY SELECTED WAS AN EXTENDED FIELD OR IF HALF BLOCK TRANSFERS WERE SELECTED (128 WORD SECTORS), THE AMOUNT OF SECTORS WILL BE LIMITED TO 17(8). IF THE FIELD SELECTED WAS FIELD 0 AND IF FULL BLOCK TRANSFERS WERE SELECTED (256 WORD SECTORS), THE AMOUNT OF SECTORS WILL BE LIMITED TO 7(8).
- D. A RANDOM STARTING SECTOR WILL BE GENERATED. THE RANDOM AMOUNT OF EXTRA SECTORS PREVIOUSLY GENERATED WILL BE ADDED TO THIS STARTING SECTOR, DETERMINING THE ACTUAL LENGTH OF THE DATA TRANSFER. IF THE STARTING SECTOR WAS 14 AND THE AMOUNT OF EXTRA SECTORS WAS 6, SECTORS 14, 15, 16, 17, 00, 01, AND 02 WILL BE USED FOR TRANSFERING DATA.
- E. AN INITIAL SOFTWARE WORD COUNT WILL BE CALCULATED.
- F. AN INITIAL RANDOM CURRENT ADDRESS WILL BE GENERATED. IF THE FIELD PREVIOUSLY GENERATED WAS FIELD 0, THE CURRENT ADDRESS WILL BE LIMITED WITHIN THE END OF THE PROGRAM +4000 LOCATIONS.
- G. THE BUFFER SELECTED WILL BE FILLED WITH RANDOM DATA, CHECKSUMMED, AND THE CHECKSUM SAVED. (NOTE: BUFFER IS DEPENDENT ON FIELD, WORD COUNT, BLOCK LENGTH, AND CURRENT ADDRESS PREVIOUSLY SELECTED.)
- H. THE PROGRAM WILL THEN POLE THE DISK DRIVES. DRIVE SELECTION IS SEQUENTIAL, THAT IS DISK0, DISK1, DISK2, ETC.
- I. DATA WILL BE WRITTEN ON THE SELECTED DISK DRIVE TO COMPLETE THE SEEK OPERATION USING THE RANDOM PARAMETERS GENERATED ABOVE. AS DATA IS WRITTEN, A BACK GROUND PROGRAM WILL CLEAR THE BUFFER AREA ALREADY WRITTEN ON THE DISK.
- J. WHEN THE WRITE AND CLEAR IS COMPLETE, DATA WILL BE READ OFF THE CURRENT DRIVE INTO THE BUFFER AREA. AS DATA IS READ, A BACK GROUND PROGRAM WILL CHECKSUM THE BUFFER INFORMATION ALREADY READ OFF THE DISK.
- K. WHEN THE READ AND CHECKSUM IS COMPLETE, THE CHECKSUM FOUND WILL BE COMPARED TO THE CHECKSUM SAVED PREVIOUS TO THE WRITE OPERATION. IF CHECKSUMS DO NOT COMPARE OR IF A CRC ERROR HAS OCCURRED, A WORD BY WORD COMPARE WILL BE MADE TO DETERMINE AND TYPE OUT THE BAD DATA FOUND.
- L. STEPS A-H WILL BE REPEATED AND THE DRIVE POLE WILL BE STARTED AT THE CURRENT DRIVE +1.
- M. FOR ALL POSSIBLE ERRORS, SEE SECTION 5 IN THIS DOCUMENT.

9.

CONSOLE PACKAGE ADDENDUM

9.1. DESCRIPTION -----

THE CONSOLE PACKAGE HAS BEEN ADDED TO THIS DIAGNOSTIC TO ALLOW THE PROGRAM TO RUN WITH NO HARDWARE SWITCH REGISTER AND TO HAVE COMMUNICATIONS WITH THE DIAGNOSTIC VIA A TERMINAL. THE DIAGNOSTIC CAN BE RUN IN TWO MODES WITH THE CONSOLE PACKAGE . 1) RUNNING WITH THE CONSOLE PACKAGE ACTIVE . THIS ALLOWS THE OPERATOR CONTROL OF THE DIAGNOSTIC THROUGH THE TERMINAL. THE DIAGNOSTIC WILL ASK FOR THE VALUE OF THE PSEUDO SWITCH REGISTER, BEFORE CONTINUING WITH EXECUTION OF THE DIAGNOSTIC. ALL ERRORS AND PASS COMPLETES WILL BE PRINTED AT THE TERMINAL. NO HALTS WILL BE EXECUTED. 2) CONSOLE PACKAGE NOT ACTIVE-THIS WILL RESULT IN THE NORMAL STANDALONE OPERATION OF THE PROGRAM AS DISCRIBED IN SECTIONS 1 THROUGH 8 OF THIS DUCUMENT.

9.2 RESTRICTIONS -----

- 1) RUNNING THE CONSOLE PACKAGE REQUIRES THAT THE PSEUDO SWITCH REGISTER BE USED.
- 2) ONCE RUNNING THE CONSOLE PACKAGE NONACTIVE AND NOW DESIRE TO RUN IT ACTIVE, ONE MUST RELOAD THE DIAGNOSTIC AND INITILIZE FOR A ACTIVE CONSOLE PACKAGE.

9.3 INITIALIZATION -----

FOR A ACTIVE CONSOLE PACKAGE -----

- 1.) SET LOCATION 21 BIT0=0 TO INDICATE USE PSEUDO SWITCH REGISTER.
- 2.) SET LOCATION 22 BIT3=1 TO INDICATE CONSOLE PACKAGE ACTIVE.

FOR A NON ACTIVE CONSOLE PACKAGE -----

- 1.) SET LOCATION 21 BIT0=1 TO INDICATE NOT TO USE PSEUDO SWITCH REGISTER, BUT TU USE HARDWARE SWITCHES.
- 2.) SET LOCATION 22 BIT3=0 TO INDICATE CONSOLE PACKAGE NOT ACTIVE.

9.4 CONTROL CHARACTERS -----

CONTROL CHARACTERS ARE USED TO GIVE THE OPERATOR THE

ABILITY TO PERFORM THE FOLLOWING FUNCTIONS.
NOTE: THE PROGRAM WILL RESPOND TO THE CONTROL
CHARACTER IN FIVE (5) SECONDS OR LESS.

CONTROL C

THIS WILL START THE LOADER THAT IS
IN LOCATION 7600.

CONTROL R

THIS WILL RESTART THE PROGRAM AND
REASK THE SWITCH REGISTER QUESTION AS
DESCRIBED IN SETION 9.6.

CONTROL E

THIS WILL CONTINUE THE PROGRAM FROM
AN ERROR IF ALLOWED BY THE DIAGNOSTIC
OR FROM A WAITING STATEMENT.

CONTROL L

THIS WILL SWITCH THE TERMINAL MESSAGES
FROM THE DISPLAY TO A LINE PRINTER.
TO RESTORE THE MESSAGES ON THE TERMINAL
CONTROL L MUST BE TYPED AGAIN. IF
NO PRINTER IS AVAILBLE AND CONTROL L
IS TYPED THE RESULT WILL BE THAT THE
CONSOLE PACKAGE WILL WAIT FOR CONTROL C OR R.
THE CONTROL L WILL OUTPUT TO THE LINE
PRINTER AND THE PROGRAM WILL
ATTEMPT TO CONTINUE AS IF A CONTROL E
WAS TYPED IN.

CONTROL D

THIS WILL ALLOW THE ABILITY TO CHANGE
THE SWITCH REGISTER DURING PROGRAM
OPERATION. TYPING THIS CHARACTER WILL RESULT
IN AN INTERIGATION OF THE SWITCH REGISTER
QUESTION AS DESCRIBED IN SECTION 9.6.

CONTROL S

THIS WILL STOP PROGRAM EXECTUION AND WAIT IN A
LOOP FOR A CONTINUE. THE ONLY WAY TO CONTINUE
WILL BE TO TYPE A CONTROL Q, R OR C .
THIS IS A NONPRINTING CHARACTER.

CONTROL Q

THIS IS TO CONTINUE A PROGRAM AFTER A CONTROL
S IS TYPED. THIS IS A NONPRINTING CHARACTER.

9.5

WAITING MESSAGE

THE WAITING MESSAGE IS USED TO ALLOW THE OPERATOR TIME
TO MAKE A DECISION AS TO WHAT CONTROL CHARACTER
TO TYPE. THIS MESSAGE MAY APPEAR AT THE END
OF PASS MESSAGE IF THE HALT ON PASS BIT IS SET. THE CONTROL
CHARACTERS MAY NOW BE USED TO PERFORM THE NEEDED FUNCTION.

THE WAITING MESSAGE MAY BE PRINTED AFTER A ERROR MESSAGE
IF THE HALT ON ERROR BIT IS SET, HERE AGAIN THE CONTROL
CHARACTERS MAY BE USED.
THE WAITING MESSAGE MAY BE PRINTED IF OPERATOR INTERVENTION
IS REQUIRED.

9.6 SWITCH REGISTER MESSAGE

THIS MESSAGE IS USED TO SETUP THE PSEUDO SWITCH REGISTER
BEFORE PROGRAM EXECUTION TAKES PLACE, THE SWITCH REGISTER
IS SETUP WHEN THE FOURTH CHARACTER IS ENTERED OR A CARRIAGE
RETURN IS TYPED

SR=0000 4000

UNDER SCORING INDICATES OPERATOR RESPONSE

9.7 END OF PASS

THE NORMAL PASS COMPLETE TYPEOUT AS DESCRIBED IN SECTION
5.5 IS USED.

9.8 ERRORS

THE STANDARD ERROR REPORTS AS DESCRIBED IN SECTION 5
OF THIS DOCUMENT WILL BE USED.

9.9 SWITCH REGISTER SETTINGS

THE STANDARD SWITCH SETTINGS AS DESCRIBED IN SECTION 3
OF THIS DOCUMENT WILL BE USED.

9.10 PARAMETER CONTROL WORDS

THE CONSOLE PACKAGE USES THE LOCATIONS 20 21 22 FOR THE
FOLLOWING PURPOSES.

LOCATION 20
PSEUDO SWITCH REGISTER

LOCATION 21
HARDWARE IDENTIFIER 1

LOCATION 22

LOCATION 0021

BIT ---	OCTAL VALUE -----	FUNCTION WHEN 0 -----	FUNCTION WHEN 1 -----
0	4000	USE PSEUDO SWITCHES	USE HARDWARE SWITCHES
1	2000	NO OPTION 1	HAS OPTION 1
2	1000	NO OPTION 2	HAS OPTION 2
3	400	NO 8A SIMULATOR	HAS 8A SIMULATOR
4	200	NO OPTION SIMULATOR	HAS OPTION SIMULATOR
5	100	NOT ON 8A XOR	ON 8A XOR
6	40	NOT PDP8-E TYPE CPU	PDP8-E TYPE CPU

7-11

8A MEMORY SIZE EX. 1K=00
2K=01
7K=06

32K-31

LOCATION 0022

BIT ---	OCTAL VALUE -----	FUNCTION WHEN 0 -----	FUNCTION WHEN 1 -----
0	4000	NOT ON ACT 8A LINE	ON ACT 8A LINE
1	2000	NOT ON ACT 8E LINE	ON ACT 8E LINE
2	1000	NOT YET DEFINED	
3	400	DEACTIVE CONSOLE PACKAGE	ACTIVE CONSOLE PACKAGE

9.11

LOCATION CHANGES

THE FOLLOWING FIELD 1 LOCATIONS CAN BE CHANGED TO MEET THE SPECIFIC NEED FOR MODIFICATION OF THE DIAGNOSTIC.

0246

IS THE LOCATION FOR THE VALUE OF THE
NUMBER OF PROGRAM PASSES NEED TO
PRINT THE END OF PASS MESSAGE.

1037

IS THE LOCATION SET FOR THE NUMBER OF
FILLER CHARACTERS AFTER A CRLF SET TO FOUR (4)

10.

APT-8 HOOKS

10.1 DESCRIPTION

TWO INTERFACES HAVE BEEN PROVIDED WHICH ALLOW THIS DIAGNOSTIC TO RUN UNDER THE STANDARD APT-8 SYSTEM. THESE INTERFACES ARE:

1. TIMING INTERFACE

2. ERROR INTERFACE

EACH WILL BE EXPLAINED IN DETAIL.

10.2 SETUP

THE FOLLOWING INFORMATION MUST BE INDICATED DURING THE INITIAL PROGRAM START UP.

1. SINGLE OR MULTIPLE DRIVE TESTING.
2. DRIVE OR DRIVES TO BE TESTED.
3. DIAGNOSTIC RUNNING UNDER APT-8.
4. THE AMOUNT OF MEMORY IN 1K INCREMENTS.

IF SINGLE DRIVE TESTING BIT 5 OF ADDRESS 22 MUST BE SET TO A ONE (1) WITH BITS 6-11 CONTAINING THE DRIVE TO BE TESTED. IF MULTIPLE DRIVES ARE TO BE DONE BIT MUST BE SET TO A ZERO (0) AND BIT 6-11 CONTAINING THE HIGHEST NUMBER DRIVE TO BE TESTED. WHEN MULTIPLE DRIVE TESTING ONLY A SPECIFIC NUMBER OF DRIVES CAN BE INDICATED. THE PROGRAM ASSUMES THE DRIVES ARE TO BE DONE BEGINNING WITH DRIVE ZERO (0) AND FINISHING WITH THE HIGHEST DRIVE INDICATED. IF MULTIPLE DRIVES OTHER THAN CONSECUTIVELY NUMBERED DRIVES BEGINNING WITH DRIVE ZERO (0) ARE TO BE DONE, THEY MUST BE DONE AS SINGLE DRIVES AND TESTED INDEPENDANTLY.

THE PROGRAM ALLOWS DRIVES 0-7 TO BE SELECTED. USER SHOULD NOT EXCEED 0-3 DRIVES FOR THE MK8E CONTROL.

BIT ZERO OF ADDRESS 22 MUST BE SET TO A ONE TO INDICATE THAT THE PROGRAM WILL RUN UNDER APT-8.

NOTE: IT SHOULD BE NOTED AT THIS TIME THAT WHILE RUNNING UNDER APT-8 THE HARDWARE SWITCH REGISTER IS INOPERATIVE. ONLY THE HALT AND SINGLE STEP SWITCH WILL EFFECT THE PROGRAM RUN.

AMOUNT OF MEMORY IN 1K INCREMENTS SHOULD BE STORED IN BITS 7-11 OF LOCATION 21. AN ADDITION OF 1 TO THE NUMBER OF BITS IN 7-11 INCREASES MEMORY SIZE BY 1K. EX. 4K=3; 8K=7. REMEMBER TO RETAIN STATUS OF BITS WHEN MODIFYING LOCATION 21.

APT-8 INTERFACES:

10.3.1. TIMING

APT-8 IS NOTIFIED OF PROGRAM RUN BETWEEN .2 SEC AND 2.0 SEC ON A 1.2 MICROSECOND MEMORY CYCLE. THIS WILL ALLOW THE DIAGNOSTIC TO RUN WITHOUT CAUSING AN APT-8 TIMEOUT ERROR IF THE DIAGNOSTIC IS TO BE RUN ON THE SLOWER MOS MEMORY.

10.3.2. ERRORS

ONLY THE DRIVE IN ERROR IS REPORTED TO APT-8 SYSTEM.
SYSTEM. ERRORS WHICH CAUSE A PROGRAMMED HALT CAUSE A TIMEOUT
ERROR. IF A PROGRAMMED HALT SHOULD OCCUR, THE ERROR PC WILL
APPEAR IN THE AC ON THE DEVICE UNDER TEST. PROGRAMMED HALTS
ARE EXPLAINED EARLIER IN THIS DOCUMENT.

11. PROGRAM LISTING

```
/RK8E/RK8L DATA RELIABILITY PROGRAM: MU=88-DHRKC=M
/
/MAINDEC=88-DHRKC=M-L
/
/COPYRIGHT 1972,1975,1976,1977 DIGITAL EQUIP. CORP.
/
/MAYNARD, MASS. 01754
/
0001      FIELD 1
/
/CONSOL SRC=V2=RB= CONSOLE PACKAGE
/
/THE PROGRAM SHOULD CHECK FOR A CONTROL CHARACTER FROM THE TERMINAL
/EVERY FIVE(5) SECONDS OR SOONER.

/LOCATIONS THAT NEED TO BE SET UP FOR USING THE CONSOLE PACKAGE.

/CNTVAL IN XC8PASS THIS LOCATION DETERMINES THE NUMBER OF
/PROGRAM COMPLETIONS THAT ARE NEEDED BEFORE THE PASS MESSAGE IS TYPED
/THE VALUE SHOULD PUT THE PASS MESSAGE OUT IN THE RANGE OF 1 TO 5 MINUTES.
/THIS SHOULD BE A POSITIVE NUMBER.

/C8STRT THIS IS FOUND IN CNTRL ROUTINE CONTROL R PART
/IT IS THE RETURN WHEN CONTROL R IS ENTERED (RESTART PROGRAM)
/THE RETURN JUMPS TO XD8SW WHICH CONTAINS C8STRT SO PUT THE LABEL C8STRT
/WHERE YOU WANT TO RESTART THE PROGRAM.

/SETUP1 IN XC8ERR THIS IS THE MASK BIT FOR HALT ON ERROR
/PLACE THE CORRECT BIT IN THIS LOCATION FOR HALTING ON ERRORS.

/SETUP2 IN XC8PASS THIS IS THE MASK FOR HALT A END OF PASS.

/THE CALL TABLE IS A CONDITIONAL ASSEMBLY.
/TO ASSEMBLE THE CALL REMOVE THE / BEFORE CONSOL=0.
/IN COMBINING THE CONSOL PACKAGE TO A DIAGNOSTIC.
/THE CALL TABLE IS TO BE AT THE BEGINNING OF A PROGRAM.

0000      CONSOL=0
0001      PSKP= 0001
0002      PCLP= 0002
0003      PSKE= 0003
0004      PSTB= 0004
0005      PSIE= 0005
0006      GTP= 0006
0007      ACL= 7701
0008      CAP= 0007
0009      MQL= 7421
0010      MGA= 7501
/
0020      *20
/
0020 0000      F10P2, 0
0021 4000      F10P1, 4000
```

```
0022 0000      F10P2, 0
/
/IFDEF CONSOL <

0024      *24

0024 0000      C8PASS= JMS I .
0025 0000      XC8PAS . /C8 PASS COMPLETION ROUTINE
0026 0000      C8CKSW= JMS I . /CHECK SW REG SETTING
0027 0000      C8TTYI= JMS I . /FETCH CONSOL CHAR
0028 0000      XC8TTY .
0029 0000      C8CNTR= JMS I . /CHECK FOR CONTROL CHAR
0030 0000      XC8CNT .
0031 0000      C8PRNT= JMS I . /C8 PRINT A BUFFER
0032 0000      XC8PNT .
0033 0000      C8SWIT= JMS I . /SET UP PSEUDO SW, REG
0034 0000      XC8PSW .
0035 0000      C8OCTA= JMS I . /CONVERT TO ASCII AND PRINT
0036 0000      XC8OCT .
0037 0000      C8CRLF= JMS I . /DO A CARRIAGE RETURN+ LINE FEED
0038 0000      XC8CRL .
0039 0000      C8ECHO= JMS I . /CHECK INPUT CHAR
0040 0000      XC8ECH .
0041 0000      C8TYPE= JMS I . /C8 PRINT ONE CHAR
0042 0000      XC8TYP .
0043 0000      C8ERR= JMS I . /C8 ERROR HANDLER
0044 0000      XC8ERR .
0045 0000      C8INQU= JMS I . /LOOK FOR OPERATOR INTERVENTION
0046 0000      XC8INQ .
0047 0000      C8CKPA= JMS I . /CHECK IF CONTROL CHAR
0048 0000      XC8CKP .
0049 0000      C8PAUS= JMS I . /IF CONSOL PACKAGE RETURN CALL PLUS ONE
0050 0000      XC8PAU . /IF NOT USING CONSOL REPLACE CALL WITH
/ A HLT AND THEN GO TO THE HALT

/*****
/ *20 /PSEUDO SWITCH REGISTER
/
/ *21 /HARDWARE INDICATORS
/ 4000=USE FRONT PANEL SWITCH REGISTER
/ 8000=USE THE PSEUDO SWITCH REGISTER LOC.20
/
/ *22 /SYSTEM CONFIGURATION
/ 4000=CONSOL PACKAGE SET ACTIVE
/ 8000=CONSOL PACKAGE SET DEACTIVE
/
/ *23 /RESERVED FOR FUTURE USE
/
0000      *200
/
/*****
/C8PASS
```

```

/THIS IS CALLED AT THE END OF EACH PROGRAM COMPLETION
/THE VALUE OF** CNTVAL** WILL BE DETERMINED BY THE TIME IT TAKES
/THE PROGRAM TO COMPLETE THIS MANY C0PASS TO BE IN THE 1 TO 4 MINUTE
/RANGE
/      C0PASS=JMS      XC0PAS
/EX. OF CALL      C0PASS
/                  HLT
/                  JMP      START1      /HALT IF NON CONSOL PACKAGE
/                                      /CONTINUE RUNNING THIS PROGRAM

```

```

/RETURN TO LOCATION CALL PLUS ONE WITH THE AC=0 IF NON CONSOL PACKAGE AND HLT
/IF CONTINUE TO RUN THEN RETURN TO CALL PLUS2 AC=0
/THE LOCATION SETUP2 IS THE MASK BIT FOR THE HALT AT END OF PASS
/CHECK THAT IT IS CORRECT FOR THE CURRENT PROGRAM

```

/CALLS USED BY XC0PAS ARE CHKCLA-XC0CNLF-XC0OCTA-XC0SW-XC0PNT-XC0INO-

```

0200 0000      XC0PAS, 0
0201 7200      CLA
0202 4777      JMS      CHKCLA      /IS WORD 22 BIT 3 ACTIVE CONSOLE?
0203 5212      JMP      DOPACK      /IS CLASSIC
0204 4776      JMS      C0GET      /GET REGISTERS.
0205 4262      JMS      XC0SW      /DEACTIVE CONSOL CHECK SR SETTING
0206 0375      AND      (400      /FOR HALT ON END OF C0PASS
0207 7640      SZA CLA      /1= HALT & CONTINUE
0210 5600      JMP I      XC0PAS      /GO TO HALT
0211 5230      JMP      C0BY1      /CONTINUE ON RUNNING PROGRAM
0212 4232      DOPACK, JMS      CKCOUT      /CLASS CHECK C0PASS COUNT
0213 5230      JMP      C0BY1      /C0PASS COUNT NOT DONE REDO PROGRAM
0214 2250      ISZ      PASCNT      /C0PASS COUNT DONE SET C0PASS COUNT
0215 4774      JMS      XC0CNLF
0216 4303      JMS      XC0PNT      /C0PNT BUFFER
0217 0253      MESPAS
0220 1250      TAD      PASCNT      /
0221 4773      JMS      XC0OCTA      /GET NUMBER
0222 4774      JMS      XC0CNLF      /CONVERT IT TO ASCII
0223 4776      JMS      C0GET      /DO A CARRIAGE RETURN
0224 4262      JMS      XC0SW      /GET REGISTERS.
0225 0375      SETUP2, AND      (400      /CHECK A HALT AT END OF C0PASS
0226 7640      SZA CLA      /MASK BIT
0227 4772      JMS      XC0INO      /HALT =1 NO SKIP CONTINUE =0
0230 2200      C0BY1, ISZ      XC0PAS      /STOP PROGRAM EXECUTION-LOOK FOR INPUT
0231 5600      JMP I      XC0PAS      /BUMP RETURN
0232 0000      CKCOUT, 0
0233 1251      TAD      DOSET      /CHECK IF SET UP NEEDED
0234 7640      SZA CLA      /0=SET UP C0PASS COUNT VALUE
0235 5242      JMP      NOSET      /1=C0PASS COUNT VALUE OK
0236 1252      TAD      CNTVAL      /C0PASS COUNT VALUE ON
0237 7040      CMA      /SET COUNT VALUE FOR THIS PROG
0240 3247      DCA      DOCNT      /SET TO NEGATIVE
0241 2251      ISZ      DOSET      /STORE IN HERE
0242 2247      NOSET, ISZ      DOCNT      /INDICATE VALUE SET UP
0243 5230      JMP      C0BY1      /COUNT THE NUMBER OF PASSES
/EXIT FOR ANOTHER PASS

```

```

0244 3251      DCA      DOSET      /SET TO C0PNT C0PASS
0245 2232      ISZ      CKCOUT      /BUMP RETURN FOR
0246 5632      JMP I      CKCOUT      /C0PASS C0TYPE OUT
0247 0000      DOCNT, 0
0250 0000      PASCNT, 0
0251 0000      DOSET, 0
0252 0000      CNTVAL, 0
0253 0410      MESPAS, TEXT      "DHRKC PASS "
0254 2213
0255 0310
0256 4040
0257 2001
0260 2323
0261 4000

```

/*****

/C0CKSW

```

/THIS ROUTINE CAN BE USED INPLACE OF A READ THE SWITCHES LAS.
/ROUTINE THAT WILL CHECK WHERE TO READ THE
/C0 SWITCHES FROM IE. FROM PANEL OR PSEUDO SWITCH REGISTER
/THE SELECTION IS DETERMINED BY THE STATE OF BIT 0 IN LOCATION 21.

```

```

/C0CKSW=      JMS XC0SW
/EX.      JMS      XC0SW      /READ THE C0SWIT REGISTER
/RETURN WITH THE CONTENTS OF SWITCH REGISTER

```

/RETURN TO NEXT LOCATION FOLLOWING CALL WITH THE AC= TO VALUE OF C0SWIT SETTING

/CALLS USED ARE-XC0CKPA-

```

0262 0000      XC0SW, 0
0263 4771      JMS      XC0CKPA      /GO CHECK THE IF ANY CONTRL
0264 7000      NOP
0265 1021      TAD      21      /GET MU FUN INDICATOR
0266 7710      SPA CLA      /CHECK IF FROM PANEL 4000
0267 7614      TAD      7614      /DO LAS AND SKIP GET FROM PANEL WITH LAS
0270 1020      TAD      20      /PSEUDO SWITCH
0271 5662      JMP I      XC0SW      /EXIT WITH STATUS BIT IN AC.

```

/*****

/C0TTYI

```

/THIS ROUTINE WILL LOOK FOR A INPUT FROM THE TERMINAL
/AND REMOVE ANY PARITY BITS, THEN MAKE IT 8 BIT ASCII.
/      C0TTYI= JMS XC0TTYI
/EX.      JMS      XC0TTYI
/

```

```

/READ CHAR FROM THE CONSOL DEVICE
/RETURN TO CALL PLUS ONE AC CONTAINS THE CHAR

```

```

/CALLS USED=NONE- BUT C6CHAR IS OFF PAGE AND IN ROUTINE CALLED XC6ECHO
/
0272 0000      XC6TTY, 0
0273 0031      KSF                                /LOOK FOR KEYBOARD FLAG
0274 5273      JMP      .-1
0275 0036      KRB                                /GET CHAR
0276 0370      AND      (177                      /MASK FOR 7 BITS
0277 1367      TAD      (200                      /ADD THE EIGHTH BIT
0300 3766      DCA      C6CHAR                    /STORE IT
0301 1766      TAD      C6CHAR
0302 5672      JMP I   XC6TTY                      /EXIT

```

[illegible]

/CSPRINT

```

/THIS ROUTINE WILL TYPE THE CONTENTS OF THE CB PRINT BUFFER, THE LOCATION
/OF THE BUFFER WILL BE IN THE ADDR8 FOLLOWING THE CALL,  PRINTING OF THE BUFFER
/Will stop when a 00 char is detected, characters are packed 2 per word,

```

/ C0PNT0 JMS XC0PNT

```

/EX.      JMS      XCBPNT      /CBPNT THE CONTENTS OF THE FOLLOWING BUFFER
/          MESS77      /LOCATION OF CBPNT BUFFER

```

```

/COBPRNT WILL USE THE LOCATION FOLLOWING THE CALL AS THE POINTER FOR THE
/COBPRNT ROUTINE.RETURN TO CALL PLUS TWO WITH AC= 0

```

/CALLS USED ARE-XCSTYPE-XCSPNT

```

0303 0000      XCOPNT, @
0304 7300      CLA CLL
0305 1703      TAD I      XCOPNT      /GET C0PNT BUFFERS STARTING LOCATION
0306 3336      DCA      PTSTOR      /STORE IN PTSTOR
0307 2303      ISZ      XCOPNT      /BUMP RETURN
0310 1736      C8D01, TAD I      PTSTOR      /GET DATA WORD
0311 0365      AND      (7700)      /MASK FOR LEFT BYTE
0312 7450      SNA      /CHECK IF 00 TERMINATE
0313 5703      JMP I      XCOPNT      /EXIT
0314 7500      SMA      /IS AC MINUS
0315 7020      CML      /MAKE CHAR A 300 AFTER ROTATE
0316 7001      IAC      /MAKE CHAR A 200 AFTER ROTATE
0317 7012      RTR
0320 7012      RTR
0321 7012      RTR
0322 4764      JMS      XC0TYPE      /PUT CHAR IN BITS 4-11 MAKE IT 0 BIT ASCII
0323 1736      TAD I      PTSTOR      /C0PNT IT ON CONSOLE
                                /GET DATA WORD

```

```

0324 0363 AND      (B077      /MASK FOR RIGHT BYTE
0325 7450 BNA      /CHECK IF 00 TERMINATOR
0326 5703 JMP I    XC0PNT  //EXIT
0327 1362 TAD      (3740      /ADD FUDGE FACTOR TO DETERMINE IF 200
0330 7500 BNA      /OR 300 IS TO BE ADD TO CHAR
0331 1361 TAD      (100      /ADD 100
0332 1360 TAD      (240      /ADD 200
0333 4764 JMS      XC0TYPE  /C0TYPE ONLY BITS 4-11
0334 2336 I82      PTSTOR   /BUMP POINTER FOR NEXT WORD
0335 5310 JMP      C00D1     /DO AGAIN
0336 0000 PTSTOR, 0      /STON FOR C0PNT BUFFER

```

—————

/CSPAUS

```

/THIS ROUTINE WILL CHECK IF THE CONSOL PACKAGE IS ACTIVE,IF ACTIVE
/IT WILL RETURN TO CALL PLUS ONE ADDRESS, AND DO THAT INSTRUCTION.
/IF THE CONSOL PACKAGE IS NOT ACTIVE THE CALL WILL BE REPLACED
/WITH A 7482 HALT AND THEN RETURN TO THE HALT.

```

/ CBPAUS# JMS XCOPAUS

```

/EX.      JMS      XC8PAUS      /CHECK IF ON ACTIVE CONSOLE IF NOT HALT HERE
/          ANYTHING      /RETURN HERE IF ON ACTIVE CONSOLE

```

/CALLS USED ARE=CHKCLA=

```

0337 0000      XC0PAU, 0
0340 7300      CLA CLL
0341 4777      JMS      CHKCLA      /CHECK LOC 22 BIT 3 CONSOLE BIT
0342 5350      JMP      C0D03        /GO TO CONSOLE PART RETURN CALL+1
0343 7040      CMA
0344 1337      TAD      XC0PAU      /DEACTIVE CONSOLE PACKAGE PUT MLT IN CALL
0345 3337      DCA      XC0PAU      /GET CORRECT RETURN ADDR
0346 1357      TAD      (7402
0347 3737      DCA I   XC0PAU      /SET UP RETURN
0348 5737      JMP I   XC0PAU      /GET CODE FOR MLT
                                /PUT MLT IN CALL LOCATION
0350 5737      C0D03, JMC I   XC0PAU /GO TO HALT OR RETURN TO NEXT LOCATION

```

0357	7402
0360	0240
0361	0100
0362	3740
0363	0077
0364	1077
0365	7700
0366	1075
0367	0200
0370	0177
0371	1041

0372	0635
0373	1000
0374	1023
0375	0400
0376	0624
0377	1200
	0400

PAGE

/*****

46CENTK

/THIS ROUTINE WILL CHECK FOR THE PRESENCE OF CONTROL CHARACTERS

IT WILL CHECK FOR THE FOLLOWING CHAN C-R-W-L-S

/ CBCNTR= JMS XCBCNT

```

/EX.      JMS      XC8CNTR      /CHECK FOR CONTROL CHARACTER
/          JMP      ANYTHING     /LOC FOLLOWING CALL IS FOR CONTINUING THE PROGRAM
/          JMP      ANYTHING     /LOC. IS FOR RETURN IF INMODE SET AND NOT CNTRL CHAR

```

```

/RETURN IS TO CALL PLUS ONE IF CONTINUE
/RETURN IS TO CALL PLUS TWO IF INMODE SET AND NOT CONTROL CHAR
/RETURN IS TO CALL PLUS TWO IF INMODE IS NOT SET AND NO
/CONTROL CHAR .,THIS WILL PRINT THE CHARACTER AND A ?
/CLEAR THE AC AND RETURN CALL+2.

```

/CALLS USED ARE=CHKCLA=XCBTYPE=XCBCLRF=CBGET=UPAROW=XCBTYT=XCBBSW

1

1

```

0400 0000      XCSNT, 0
0401 3777'    DCA      ACSAVE      /SAVE THE AC
0402 4776'    JMS      CHKCLA      /CHECK LOC.22 BIT3 FOR CONSOLE BIT
0403 5206'    JMP      +3          /ON ACTIVE CONSOLE
0404 1777'    TAD      ACSAVE      /DEACTIVE CONSOLEGET AC FOR RETURN
0405 5600'    JMP I    XCSNT      /EXIT NOT UN ACTIVE CONSOLE
0406 6804'    GTF
0407 3775'    DCA      FLSAVE
0410 7501'    MQA
0411 3774'    DCA      MQSAVE      /SAVE THE MQ
0412 3255'    OCA      INDEXA     /SET DISPLACEMENT INTO TABLE 0
0413 1257'    TAD      XTABLA     /GET ADURS OF TABLE A
0414 3256'    DCA      GETDAT     /CONTAINS POINTER TO CONTROL CHAR
0415 1656'    REDDA, TAD I    GETDAT /GET CONTROL CHAR FROM TABLE
0416 7450'    SNA
0417 5226'    JMP      DONEA      /CHECK FOR A 0 END OF TABLE
0420 1773'    TAO      COCHAR     /END OF TABLE NO CONTROL CHAR
0421 7650'    SNA      CLA        /COMPARE CHAN TO CONTROL CHAR
0422 5243'    JMP      /0 IF MATCH
0423 2255'    ISZ      GOITA      /MATCH
0424 2256'    ISZ      INDEXA     /NO MATCH NOT END OF TABLE REDD
0425 5215'    JMP      GETDAT     /BUMP INDEX FOR EXIT WHEN CONTROL FOUND
0426 1772'    DONEA, TAD      REDDA /BUMP GETDAT FOR COMPARE OF NEXT CNTRL CHAR.
0427 7648'    SNA      CLA        /CHECK IF PROGRAM EXPECTS CHAR
0430 5240'    JMP      EXITA      /1=CHAN EXPECTED 0= NO CHAR EXPECTED
                                /CHAN EXPECTED

```

```

0431 1773*      TAO      C8CHAR      /GET CHAR= NOT CONTROL+ NOT EXPECTED
0432 4771*      JMS      XC8TYPE    /C8PNNT CHAR
0433 1370       TAO      C277       /GET CUDE FOR "?"
0434 4771*      JMS      XC8TYPE
0435 4767*      JMS      XC8CRLF
0436 2200       ISZ      XC8CNT      /BUMP RETURN
0437 5600       JMP I      XC8CNT      /EXIT CALL+2
0440 2200       EXITA, ISZ      XC8CNT      /BUMP RETURN FOR MAIN PROGRAM CHECK OF CHAR
0441 1773*      TAO      C8CHAR      /PUT CHAR IN AC,
0442 5600       JMP I      XC8CNT      /EXIT
0443 1773*      GOITA, TAO      C8CHAR      /GET THE CONTENTS OF CHAR
0444 1366       TAO      C100       /ADD 100 TO FORM A GOOD ASCII CHARACTER
0445 3773*      DCA      C8CHAR      /RESTORE CORRECT CHAR
0446 1260       TAO      XTABLB      /GET START OF TABLE B
0447 1255       TAO      INDEXA      /GET NUM F \ INTO TABLE
0450 3254       DCA      G0TOA      /STORE IT
0451 1654       TAO I      G0TOA      /GET THE ROUTINE STARTTING ADDRESS
0452 3254       DCA      G0TOA      /STORE IT IN HERE
0453 5654       JMP I      G0TOA      /G0TU CONTROL CHAR ROUTINE
0454 0000       G0TOA, 0000      /ADD OF CNTRL ROUTINE TO EXECUTE
0455 0000       INDEXA, 0000      /DISPLACEMENT INTO CNTRL TABLE
0456 0000       GETDAT, 0000      /LOCATION OF ADDRS OF CONTROL CHAR,
0457 0461       XTABLA, TABLA      /ADDMS OF TABLEA
0460 0471       XTABLB, TABLB      /ADDMS OF TABLEB
0461 7575       TABLA, 7575      /CNTRL C BACK TO MONITOR 203
0462 7564       7564            /CNTRL L SWITCH ERROR PRINTING DEVICE 214
0463 7557       7557            /CNTRL Q START DISPLAYING CHAR, AGAIN 221
0464 7556       7556            /CNTRL R BACK TO BEGINNING OF PROGRAM 222
0465 7555       7555            /CNTRL S STOP SENDING CHAR TO DISPLAY WAIT FOR CNTRL Q 223
0466 7573       7573            /CNTRL E CONTINUE WITH PROGRAM 205
0467 7574       7574            /CONTRL D CHANGE SWITCH REGISTER ON FLY
0470 0000       0000
0471 0551       TABLB, CNTRLC
0472 0537       CNTRL L
0473 0500       CNTRL Q
0474 0511       CNTRL R
0475 0521       CNTRL S
0476 0545       CNTRL E
0477 0600       CNTRL D
/
/CNTRL Q
/START SENDING CHAR, TO THE DISPLAY
/THIS WILL RETURN CONTROL TO CALL THAT WAS SET BY
/THE CALL FOR CONTROL S,
/
0500 3772*      CNTRLQ, DCA      INMODE      /SET SUFT FLAG FOR UNEXPECTED CHAR
0501 1335       TAD      C8SETS      /CHECK IF CONTROL S TYPED IN
0502 7640       SZA CLA
0503 5306       JMP      BYNETR      /CONTROL S TYPED IN
0504 4765*      JMS      C8GET      /NO CONTROL S TYPED PREVIOUSLY
0505 5600       JMP I      XC8CNT      /LEAVE VIA CNTR ENTRY ADDRESS
0506 3335       BYNETR, DCA      C8SETS      /CLEAN THE SUFT FLAG
0507 4765*      JMS      C8GET      /RESTORE REGISTERS
0510 5736       JMP I      C8METH      /EXIT TO ADDRESS SET BY CONTROL S

```

```

/
/CONTROL R
/GO TO THE QUESTION C8SWIT
0511 3764' CNTRLR, DCA TTYLPT /CLEAR THE TYPE FLAG SET TO TTY
0512 3335 DCA C8SETS /CLEAR SOFT FLAG FOR CNTRL S
0513 3772' DCA INMODE
0514 4765' JMS UPAROW /PRINT THE " AND C8CHAN
0515 3762' C8BY4, DCA C8SWST /CLEAR FLAG FOR CNTRL D OR R
0516 6203 CIP CIP 0
0517 5720 JMP I XDUSW /GO TO ADJMS OF C8SWIT
0520 0200 XDUSW, BGN /DOSW IS LABEL FOR C8SWIT QUESTION
/
/CONTROL S
/STOP SENDING CHAR. TO DISPLAY UNTIL A "Q IS RECEIVED
/
0521 1335 CNTRL8, TAD C8SETS /IF1 DO NOT STORE IN C8RETR
0522 7640 SZA CLA
0523 5327 JMP C8D07 /DON'T SET UP C8RETR
0524 7001 IAC /MAKE RETURN CALL PLUS 2
0525 1200 TAD XC8CNT /GET RETURN FOR THIS CALL
0526 3336 DCA C8RETR /STORE IT HERE FOR USE BE CNTRL Q
0527 2335 C8D07, ISZ C8SETS /SET FLAG TO SAVE CALL
0530 4761' JMS XC8TTYI /LOOK FOR THE INPUT
0531 4765' JMS C8GET /GET REGISTERS
0532 4200 JMS XC8CNTR /CHECK FOR THE CONTROL CHAR
0533 7200 CLA
0534 5321 JMP CNTRL8 /IF NOT A CNTRL Q R C REASK
0535 0000 C8SETS, 0
0536 0000 C8RETR, 0
/
/SWITCH OUTPUT FROM ONE OUTPUT DEVICE TO ANOTHER- THE TWO OUTPUTS ARE THE
/CONSOLE AND THE PRINTER WITH DEVICE CODE 06.
/
0537 1764' CNTRL8, TAD TTYLPT /GET PRESENT C8SWIT INDICATOR
0540 7040 CMA /COMPLEMENT IT
0541 3764' DCA TTYLPT /STORE NEW C8SWIT
0542 4763' JMS UPAROW /C8PNT " AND CHAR ON NEW DEVICE
0543 4765' JMS C8GET /RESTORE THE REGISTERS
0544 5600 JMP I XC8CNT /EXIT
/
/CONTROL E
/CONTINUE RUNNING FROM A INQUIRE OR ERROR
/
0545 4763' CNTRL8, JMS UPAROW /PRINT THE CONTROL CHAR
0546 3762' DCA C8SWST /CLEAR ENTRY FLAG.
0547 4765' JMS C8GET /GET THE REGISTERS
0550 5600 JMP I XC8CNT /RETURN TO CALL PLUS ONE
/
/CONTROL C

```

```

/RETURN TO MONITOR CONTROL C
0551 3764' CNTRL8, DCA TTYLPT /CLEAR THE LPT FLAG TO PRINT ON DISPLAY
0552 3762' DCA C8SWST /CLEAR ENTRY FLAG.
0553 4763' JMS UPAROW /C8PNT " AND LETTER IN CHAR
0554 6203 CIP CIP 0 /GO TO 0 FLD
0555 6007 CAF /CLEAR THE WORLD
0556 3760 JMP I (7000) /GO TO DIAGNOSTIC MONITOR
/*****
/
/
0560 7600
0561 0272
0562 0745
0563 0615
0564 1121
0565 0624
0566 0100
0567 1023
0570 0277
0571 1077
0572 1076
0573 1075
0574 1346
0575 1347
0576 1200
0577 1345
0600
PAGE
/
/CONTROL D
/CHANGE THE SWITCH REGISTER ANYTIME CNTRL D AND RETURN TO
/THE PROGRAM RUNNING.
0600 4215 CNTRL8, JMS UPAROW
0601 1213 TAD C8SETD /CHECK IF THE RETURN ADDR IS SAFE
0602 7640 SZA CLA
0603 5207 JMP C8D011 /DO NOT CHANGE THE RETURN ADDR
0604 1777' TAD XC8CNT /GET THE RETURN ADDR AND SAVE IT
0605 3214 DCA C8RETR /SAVE THE RETURN HERE
0606 2213 ISZ C8SETD /INDICATE RETURN SAVED DONT DESTROY
0607 4256 C8D011, JMS XC8PSW /GO CHANGE THE SWITCH REGISTER
0610 3213 DCA C8SETD /CLEAR THE FLAG
0611 4224 JMS C8GET /RESTORE THE AC MU LINK ETC
0612 5614 JMP I C8RETR /RETURN TO THE PROGRAM
/
0613 0000 C8SETD, 0
0614 0000 C8RETR, 0
/
/THIS WILL TYPE A UP ARROW AND THE CHAN IN C8CHAN.
0615 0000 UPAROW, 0 /C8PNT THE " AND THE CHAR C8TYPED IN

```

```

0616 1376      TAD      (336      /CODE FOR "
0617 4775"     JMS      XC8TYPE
0620 1774"     TAD      C8CHAR      /C8TYPE THE CHAR
0621 4775"     JMS      XC8TYPE
0622 4773"     JMS      XC8CRLF
0623 5615      JMP I    UPAROW      /EXIT

```

```

/*****

```

```

0624 0000      C8GET, 0
0625 7200      CLA
0626 1772"     TAD      MQSAVE
0627 7421"     MQL      /RESTORE MQ
0630 1771"     TAD      FLSAVE
0631 7804      RAL      /RESTORE THE LINK
0632 7200      CLA
0633 1770"     TAD      AC8AVE      /RESTORE THE AC
0634 5624      JMP I    C8GET      /GET THE REGISTERS

```

```

/*****

```

```

/C8INGU
/C8INGU ROUTINE WILL PRINT A WAITING
/AND THE PROGRAM IS EXPECTING A CONTROL CHAR INPUT
/IF CONTINUE FROM CONTROL CHAR RETURN IS CALL PLUS ONE
/IF NO CONTROL CHAR ENTERED THEN WAITING IS REPRINTED
/AND PROGRAM WAITS FOR A CONTROL CHAR AGAIN.

```

```

/      C8INGU =      JMS XC8ING

```

```

/EX.      JMS      XC8ING      /C8 WILL PRINT A WAITING AND WAIT FOR INPUT
/      DO ANYTHING      /RETURN IS CALL PLUS ONE AC = 0 CONTINUE

```

```

/CALLS USED ARE=CHKCLA=XC8PNT=XC8TYI=C8GET=XC8CNTR=

```

```

0635 0000      XC8ING, 0
0636 7300      CLA CLL
0637 4767"     JMS      CHKCLA      /CHECK LOC 22 BIT 3 CONSOLE BIT
0640 7410      SKP
0641 5635      JMP I    XC8ING      /ACTIVE CONSOLE PACKAGE
0642 4766"     JMS      XC8PNT      /NOT CONSOLE LEAVE
0643 0651      WATME8
0644 4765"     JMS      XC8TYI      /INQUIRY WAITING
0645 4224      JMS      C8GET      /GET CHARACTER
0646 4777"     JMS      XC8CNTR      /CHECK IF CONTROL CHARACTER
0647 5635      JMP I    XC8ING      /EXIT AND CONTINUE
0650 5236      JMP      XC8ING+1    /REASK
0651 2701      WATME8, TEXT "WAITING "
0652 1124
0653 1116

```

```

0654 0740
0655 0000

```

```

/*****

```

```

/C8SWIT

```

```

/ROUTINE WILL CHECK IF CONSOLE IS ACTIVE IF IT IS ACTIVE DISPLAY
/8W QUESTION. IN NOT ACTIVE IT WILL NOT PRINT THE SW QUESTION BUT
/RETURN TO CALL PLUS ONE AC=0.
/C8SWIT WILL SET UP THE PSEUDO SWITCH
/REGISTER WITH THE NEW DATA ENTERED

```

```

/      C8SWIT =      JMS XC8PSW

```

```

/EX.      JMS      XC8PSW      /SET UP PSEUDO C8SWIT REGISTER IF
/ON THE CONSOLE PACKAGE. RETURN IS CALL PLUS ONE AC = 0

```

```

/CALLS USED ARE=CHKCLA=XC8PSW=XC8PNT=XC8OCTA=XC8TYPE=

```

```

0656 0000      XC8PSW, 0
0657 4767"     JMS      CHKCLA      /CHECK LOC 22 BIT 3 CONSOLE BIT
0660 7410      SKP
0661 5656      JMP I    XC8PSW      /ACTIVE CONSOLE
0662 1345      TAD      C8SWST      /DEACTIVE CONSOLE PACKAGE
0663 7640      SZA CLA      /RETURN WITHOUT ASKING PSEUDO SWITCH
0664 5764"     JMP      C8BY4      /IS THE SW FLAG SET FOR SWITCH?
0665 2345      ISZ      C8SWST      /SKIP IF ONE ENTRY AT A TIME OK
0666 4766"     C8WDPS, JMS      XC8PNT      /SECOND ENTRY WITH OUT A EXIT GO TO SW QUESTION
0667 0747      MESA
0670 1020      TAD      20
0671 4763"     JMS      XC8OCTA      /FIRST ENTRY SET FLAG
0672 1362      TAD      (40      /C8PNTI SW
0673 4775"     JMS      XC8TYPE      /GET CONTENTS OF SW
0674 2761"     ISZ      INMODE      /CONVERT IT TO ASCII
0675 4760"     JMS      XC8ECHO      /GET SPACE
0676 4315      JMS      TSTCHA      /SET FLAG FOR CHAR EXECTED
0677 1774"     TAD      C8CHAR      /LOOK FOR INPUT
0700 3020      DCA      20          /NOT CONTROL TEST IT IS LEGAL
0701 1357      TAD      (=3        /STORE NEW CHAR IN SW REG
0702 3346      DCA      THPCNT      /GET A MINUS 3
0703 4760"     JMS      XC8ECHO      /STORE IN TEMP COUNT
0704 4315      JMS      TSTCHA      /GET NEXT CHAR
0705 1020      TAD      20          /CHECK IF CR+ GOOD CHAR
0706 7106      RTL CLL      /GET C8SWIT REGISTER
0707 7004      RAL      /ROTATE IT LEFT 3 PLACES
0710 1774"     TAD      C8CHAR      /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0711 3020      DCA      20          /SAVE NEW CONTENTS
0712 2346      ISZ      THPCNT      /BUMP COUNT
0713 5303      JMP      GETCH1      /JMP BACK+ GET NEXT CHAR
0714 5342      JMP      ENDIT      /END 4 CHAR C8TYPED IN

```

```

0715 0000 TSTCHA, 0
0716 7041 CIA
0717 1356 TAD (215
0720 7050 SNA CLA
0721 5342 JMP ENDIT
0722 1774 TAD C0CHAR
0723 1355 TAD (-260
0724 7710 SPA CLA
0725 5336 JMP ERR1
0726 1774 TAD C0CHAR
0727 1354 TAD (-270
0730 7700 SNA CLA
0731 5336 JMP ERR1
0732 1774 TAD C0CHAR
0733 0353 AND (7
0734 3774 DCA C0CHAR

0735 5715 JMP I TSTCHA
0736 1352 ERR1, TAD (277
0737 4775 JMS XC0TYPE
0740 4773 JMS XC0CRLF
0741 5266 JMP C0NOP3
0742 4773 ENDIT, JMS XC0CRLF
0743 3345 DCA C0HST
0744 5656 JMP I XC0PSW
0745 0000 C0HST, 0

0746 0000 TMPCNT, 0
0747 2322 MESA, TEXT "SR= "
0750 7540
0751 0000

0752 0277
0753 0007
0754 7510
0755 7520
0756 0215
0757 7775
0760 1063
0761 1076
0762 0040
0763 1000
0764 0515
0765 0272
0766 0303
0767 1200
0770 1345
0771 1347
0772 1346
0773 1023
0774 1075
0775 1077
0776 0336
0777 0400

```

```

1000 PAGE
      /C0OCTA
      /OCTAL TO ASCII CONVERSION
      /THIS ROUTINE WILL TAKE THE OCTAL NUMBER IN THE AC AND CONVERT IT TO ASCII
      /THE RESULT WILL BE PRINTED ON THE CONSOLE TERMINAL
      / C0OCTA= JMS XC0OCT
      /
      /EX. JMS XC0OCTA /AC CONTAINS NUMBER TO BE CHANGE
      / RETURN IS TO CALL PLUS ONE AC=0
      /
      /CALLS USED ARE=XC0TYPE=

1000 0000 XC0OCT, 0
1001 7106 CLL RTL
1002 7006 RTL
1003 3221 DCA C0TMP1
1004 1377 TAD (-4
1005 3222 DCA C0CKP
1006 1221 C0O04, TAD C0TMP1
1007 0376 AND (0007
1010 1375 TAD (260
1011 4277 JMS XC0TYPE
1012 1221 TAD C0TMP1
1013 7006 RTL
1014 7004 RAL
1015 3221 DCA C0TMP1
1016 2222 ISZ C0CKP
1017 5206 JMP C0O04
1020 5600 JMP I XC0OCT
1021 0000 C0TMP1, 0
1022 0000 C0CKP, 0

      /*****
      /C0CRLF
      /C0TYPE CR AND LF WITH FILLERS FOLLOWING EACH LF AND CR
      /
      / C0CRLF= JMS XC0CRL
      /
      /EX. JMS XC0CRLF /C0PHNT A CR AND LF WITH FILL
      / /RETURN TO CALL PLUS ONE AC =0
      /CALLS USED ARE=XC0TYPE=

1023 0000 XC0CRLF, 0
1024 7300 CLA CLL
1025 1374 TAD (215
1026 4277 JMS XC0TYPE
1027 1237 TAD FILLER
1030 7040 CMA
1031 3240 DCA FILCNT

```

```

1032 1373      TAD      (212      /GET CODE FOR LF
1033 4277      C000R, JMS      XC8TYPE
1034 2240      ISZ      FILCNT      /CHECK ON FILLER CHAR
1035 3233      JMP      C0002      /TYPE A NON PRINTING CHAR
1036 3623      JMP I      XC8CRL      /EXIT
1037 0004      FILLER, 0004      /FILLER SET FOR 4 CHAR
1040 0000      FILCNT, 0      /COUNTER FOR FILL

```

```

//*****
/C00KPA
/THIS ROUTINE WILL CHECK IF A CHARACTER WAS ENTERED FROM THE
/TERMINAL. IF THE FLAG IS SET AND THE CONSOLE PACKAGE IS
/ACTIVE A CHECK IS MADE TO DETERMINE IF IT IS A CONTROL CHAR.
/IF IT WAS A CONTROL CHAR THEN ITS CONTROL FUNCTION IS PERFORMED.
/IF NOT A CONTROL CHARACTER OR A CONTROL E-D-L-0- IT WILL DO
/THE CONTROL FUNCTION AND RETURN TO CALL PLUS 2.
/IF A NON CONTROL CHARACTER WILL BE PRINTED AND A "3" IT WILL RETURN TO
/CALL PLUS 2.
/IF NO FLAG IS SET OR THE CONSOLE IS NOT ACTIVE THE RETURN IS TO
/CALL PLUS 1.

```

```

/      C00KPA= JMS      XC0CKP

```

```

/EX.      JMS      XC0CKPA      /CALL TO CHECK IF CONTROL CHAR SET
/      ANYTHING(SKIP)      /RETURN IF NOT FLAG OR NOT CONSOLE ACTIVE
/      ANYTHING(JMP EXIT SKIP CHAIN)      /RETURN IF NOT CONTROL OR CONTINUE CONTROL

```

```

/CALLS USED ARE=XC0TTYI=XC0CNR=C00EY=

```

```

1041 0000      XC0CKP, 0
1042 3772      DCA      AC0AVE      /SAVE THE AC
1043 6004      GTF      /SAVE THE FLAG
1044 3771      DCA      FL0AVE      /SAVE THE FLAG
1045 7501      MGA      /PUT MQ IN AC
1046 3770      DCA      MQ0AVE      /SAVE THE MQ
1047 0031      KSF      /CHECK THE KEYBOARD FLAG
1050 5201      JMP      C00Y3      /EXIT TO CALL PLUS 1
1051 4767      JMS      CHKCLA      /CHECK LOC 28 BIT 3 CONSOLE BIT
1052 7410      SKP      /ACTIVE CONSOLE PACKAGE
1053 5201      JMP      C00Y3      /EXIT TO CALL PLUS 1
1054 4766      JMS      XC0TTYI      /GET THE CHAR
1055 4765      JMS      C00ET      /GET THE FLAG
1056 4764      JMS      XC0CNR      /CHECK IF CONTROL CHAR.
1057 7000      NOP      /RETURN IF A CONTINUE CHAR.
1060 2241      ISZ      XC0CKP      /BUMP RETURN FOR CALL PLUS 2
1061 4765      C00Y3, JMS      C00ET      /GET REGISTERS
1062 5641      JMP I      XC0CKP      /BAY GOOD BY

```

```

//*****

```

```

/C00ECH
/THIS ROUTINE WILL LOOK FOR A CHAR FROM THE KEYBOARD. STORE IT IN LOCATION CHAR
/CHECK IF IT WAS A CONTROL CHARACTER= SET INMODE= PRINT CHARACTER

```

```

/      C00ECHO =      JMS XC0ECH
/EX.      JMS      XC0ECHO      /LOOK FOR CONSOLE CHAR C0PRNT IT
/      /RETURN CALL PLUS ONE AC = CHAR C0TYPED IN

```

```

/CALLS USED ARE=XC0TTYI=XC0CNR=C00ET=XC0ECH=XC0TTYE

```

```

/
XC0ECH, 0
1063 0000      JMS      XC0TTYI      /WAIT FOR CHAR FROM KEYBOARD
1064 4766      JMS      C00ET      /RESTORE THE REGISTERS
1065 4765      ISZ      INMODE      /SET INMODE IDENTIFYING THIS AS A EXPECTED CHAR
1066 2276      JMS      XC0CNR      /GO CHECK IF IT IS A CONTROL CHAR
1067 4764      JMP I      XC0ECH      /WAS A CONTROL CHAR= CONTINUE RUNNING
1070 5663      JMS      XC0TYPE      /NOT A CONTROL CHAR C0PRNT IT
1071 4277      DCA      INMODE      /CLEAR FLAG THAT CHAR EXPECTED
1072 3276      TAD      C0CHAR      /GET CHAR IN AC
1073 1275      TAD      C0CHAR      /EXIT
1074 5663      JMP I      XC0ECH
1075 0000      C0CHAR, 0
1076 0000      INMODE, 0

```

```

//*****

```

```

/C0TYPE
/THIS ROUTINE WILL C0PRNT ON THE CONSOLE OR THE LPT WITH DEVICE CODE 66.
/

```

```

/      C0TYPE= JMS XC0TYP

```

```

/EX.      JMS      XC0TYPE      /C0PRNT THE CHAR IN THE AC.
/      /RETURN CALL PLUS ONE AC =0000
/      /DO NOT CLEAR THE LINK IN THIS ROUTINE NEEDED BYC0OCT

```

```

/CALLS USED ARE=C0HANG=XC0CNR=XC0PNT=XC0CHLP=XC0INQU=

```

```

1077 0000      XC0TYP, 0
1080 3320      DCA      PNTBUF      /STORE CHAN
1081 1321      TAD      TTYLPT      /CHECK 0=TTY 7777=LPT
1082 7640      SZA CLA      /DO OUT PUT ON LPT
1083 5312      JMP      XDULPT
1084 1320      TAD      PNTBUF
1085 0046      TLS
1086 0041      TSF
1087 5306      JMP      .-1
1088 0042      TCF
1089 5316      JMP      C00Y5
1090 1320      XDULPT, TAD      PNTBUF      /GET CHAR
1091 0046      PCLF      /C0PRNT IT
1092 4322      JMS      C0HANG      /CHECK KEYBOARD IF HUNG
1093 0042      PCLF      /CLEAR THE FLAG
1094 7600      C00Y5, 7600      /CLEAR THE AC

```

```

1117 5677      JMP I      XC8TYP      /EXIT
1120 0000      PNTBUF, 0
1121 0000      TTYLPT, 0

1122 0000      C8HANG, 0
1123 7200      CLA
1124 1316      TAD          C8BY5      /
1125 3320      OCA          PNTBUF      /GET CONSTANT 7600
1126 6661      PSKF          /PNTBUF IS NOW A COUNTER
1127 7410      SKP          /SKIP ON PRINTER DONE
1130 5722      JMP I      C8HANG      /NOT DONE YET
1131 2345      I8Z          C8CONT      /SAW FLAG DONE
1132 5326      JMP          C8CONT      /FIRST COUNTER FAST ONE
1133 2320      I8Z          PNTBUF      /CHECK IF FLAG SET YET
1134 5331      JMP          C8CONT      /MADE 4096 COUNTS ON FAST COUNTER
1135 1764      TAD          XC8CNTR      /KEEP IT UP FOR 5 SEC
1136 3322      OCA          C8HANG      /GET THE RETURN ADDRESS IN CONTROL
1137 3321      OCA          TTYLPT      /SAVE IT IN HANG
1140 4763      JMS          XC8PNT      /ALLOW PRINTING ON TTY
1141 1146      MESHANG
1142 4223      JMS          XC8CRLF      /LPT ERROR
1143 4762      JMS          XC8INQU      /PRINT WAITING
1144 5722      JMP I      C8HANG      /CONTINUE TO SAVE ADDRESS
1145 0000      C8CONT, 0      /COUNTER FOR TIMER
1146 1420      MESHANG,TEXT      "LPT ERROR"
1147 2440
1150 0522
1151 2217
1152 2200

1162 0635
1163 0303
1164 0400
1165 0624
1166 0272
1167 1200
1170 1346
1171 1347
1172 1345
1173 0212
1174 0215
1175 0260
1176 0007
1177 7774
1200

```

PAGE

```

/*****
/*****
/*****

/THIS ROUTINE WILL CHECK LOCATION 22 THE HARD WARE CONFIG WORD.
/TO SEE IF THE CONSOLE BIT 3 (400) IS SET IF SET THEN RETURN
/TO CALL PLUS TWO FOR A ACTIVE CONSOLE PACKAGE AC=0
/IF NOT SET THEN TO CALL PLUS ONE FOR A DEACTIVE CONSOLE PACKAGE.

```

```

1200 0000      CHKCLA, 0
1201 7200      CLA
1202 1022      TAD          22          /GET THE CONTENTA OF LOCATION 22
1203 0377      AND          (400)      /MASK FOR BIT 3 (400)
1204 7650      SNA CLA
1205 2200      ISZ          CHKCLA      /
1206 5600      JMP I      CHKCLA      /ACTIVE CONSOLE PACKAGE RETURN
                                      /CALL PLUS ONE (1) FOR ACTIVE
                                      /DEACTIVE CONSOLE PACKAGE RETURN
                                      /CALL PLUS TWO (2)

/C8ERR
/THIS ROUTINE WILL DETERMINE WHAT TO DO WHEN A C8ERR IS ENCOUNTERED
/Will check if classic system, will check C8SWIT registers.
/      C8ERR= JMS XC8ERR
/EX.      JMS      XC8ERR      /GO TO C8ERR CALL IF NOT CONSOLE
                                      /RETURN IS CALL PLUS ONE AC =0000

/CALLS USED ARE-CHKCLA-XC8CRLF-XC8SW-XC8INQU-XC8PNT-XC8OCTA-

1207 0000      XC8ERR, 0
1210 6002      IOF
1211 3345      DCA          ACSAVE      /SAVE AC
1212 6004      GTF
1213 3347      DCA          FLSAVE      /SAVE THE FLAGS
1214 7501      MGA
1215 3346      DCA          MQSAVE      /SAVE THE MQ
1216 7340      CLA CLL CMA      /SUBTRACT A 1 FOR TRUE LOCATION
1217 1207      TAD          XC8ERR      /GET RETURN LOCATION
1220 3344      OCA          PCSAVE      /SAVE ADD OF C8ERR CALL
1221 6201      CDF
1222 7340      CLA CLL CMA
1223 1776      TAD I      (CLASIK)      /GET REAL PC.
1224 3316      DCA          REALPC      /SAVE IT.
1225 6211      CDF          10
1226 4200      JMS          CHKCLA      /CHECK LOC.22 BIT 3 CONSOLE BIT
1227 7410      SKP          /ACTIVE CONSOLE PACKAGE
1230 5270      JMP          NTCLAS      /NOT CLASSIC SYSTEM
1231 4775      JMS          C8GET      /GET REGISTERS.
1232 4774      JMS          XC8SW      /CHECK SWITCH REG FOR BIT THAT INDICATES
                                      /NO ERROR MESSAGE
1233 0373      SETUP1, AND      0000      /MASK FOR BIT FOR NO ERROR PRINTING
                                      /IF THIS ERROR MESSAGE IS TO ALWAYS
                                      /BE PRINTED LEAVE AND VALUE AT 0000
                                      /SKIP IF BIT IS 0 PRINT ERROR MESSAGE
                                      /DO NOT PRINT

1234 7640      SZA CLA
1235 5262      JMP          C8U010      /
1236 4772      JMS          XC8CRLF      /
1237 4771      JMS          XC8PNT      /
1240 1320      ERHMS
1241 4771      JMS          XC8PNT      /PRINT THE ERROR MESSAGE
1242 1330      MESPC
1243 1316      TAD          REALPC      /PRINT THE PC STATEMENT
1244 4770      JMS          XC8OCTA      /GET PL
1245 4771      JMS          XC8PNT      /CONVERT 4 DIGIT PC TO ASCII
1246 1333      MESAC

```

/PRINT THE AC MESS

```

1247 1345      TAD      AC8AVE
1250 4770"     JMS      XC8OCTA
1251 4771"     JMS      XC8PNT
1252 1336      MESMQ
1253 1346      TAD      MQ8AVE
1254 4770"     JMS      XC8OCTA
1255 4771"     JMS      XC8PNT
1256 1341      MESFL
1257 1347      TAD      FL8AVE
1260 4770"     JMS      XC8OCTA
1261 4772"     JMS      XC8CRLF
1262 4775"     C80010, JMS      C8GET
1263 4774"     JMS      XC8SW
1264 7610      SKP CLA
1265 5300      JMP      C8BY2
1266 4767"     JMS      XC8INQ
1267 5300      JMP      C8BY2
1270 4775"     NTCLAS, JMS      C8GET
1271 4774"     JMS      XC8SW

1272 7610      SKP CLA
1273 5607      JMP I      XC8ERR
1274 1366      TAD      (7402
1275 3744      DCA I      PC8AVE
1276 4775"     JMS      C8GET
1277 5744      JMP I      PC8AVE
1280 4775"     C8BY2, JMS      C8GET
1301 5607      JMP I      XC8ERR

/
1302 7402      /ROUTNS, MLT
1303 7000      NOP
1304 3317      DCA      MYAC
1305 6201      CDF      0
1306 1020      TAD      SWH
1307 3765      DCA I      (8WR)
1310 1776      TAD I      (CLASIK)
1311 3315      DCA      CLRTRN
1312 1317      TAD      MYAC
1313 6202      CIF      0
1314 5715      JMP I      CLRTRN

/
1315 0000      CLNTRN, 0
1316 0000      REALPC, 0
1317 0000      MYAC, 0

/
1320 0410      ERNMES, TEXT "DMRKC=H FAILED "
1321 2213
1322 0310
1323 4040
1324 0601
1325 1114
1326 0504
1327 4000
1330 4040      MESPC, TEXT " PC:"

```

/PUT INSTRUCTION TO EXECUTE HERE.

/SAVE AC

/RETURN TO FIELD 0,

```

1331 2003
1332 7200
1333 4040      MESAC, TEXT " AC:"
1334 0103
1335 7200
1336 4040      MESMQ, TEXT " MQ:"
1337 1521
1340 7200
1341 4040      MESFL, TEXT " FL:"
1342 0614
1343 7200
1344 7777      PC8AVE, 7777
1345 7777      AC8AVE, 7777
1346 7777      MQ8AVE, 7777
1347 7777      FL8AVE, 7777

/
1365 0020
1366 7402
1367 0635
1370 1000
1371 0303
1372 1023
1373 0000
1374 0262
1375 0624
1376 1514
1377 0400
0000      FIELD 0

```

0000	00000000	00000000	11101111	11111111	11000000	00000000	00000000	00000000
0100	00000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
0200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0300	11111111	11111111	11111111	11111111	11111111	10000001	11111111	11111111
0400	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0500	11111111	11111111	11111111	11111111	11111111	11111110	11111111	11111111
0600	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0700	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1000	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1100	11111111	11111111	11111111	11111111	11111111	11100000	00111111	11111111
1200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1300	11111111	11111111	11111111	11111111	11111111	00000000	00000111	11111111
1400								
1500								
1600								
1700								
2000								
2100								
2200								
2300								
2400								
2500								
2600								
2700								
3000								
3100								
3200								
3300								
3400								
3500								
3600								
3700								

4000
4100
4200
4300
4400
4500
4600
4700
5000
5100
5200
5300
5400
5500
5600
5700
6000
6100
6200
6300
6400
6500
6600
6700
7000
7100
7200
7300
7400
7500
7600
7700

```

/
/NOTE: LOCATION 0 CONTAINS THE REVISION
/LEVEL (IN ASCII) ON PROGRAM LOAD.
/
/ALL KNOWN HALTS
/
1400 0556 ERMLT0 /SKIP TRAP DLSC
1401 0563 ERMLT2 /SKIP TRAP DCLR
1402 2561 ERMLT3 /SKIP TRAP DLG
1403 2544 ERMLT5 /SKIP TRAP DRST
1404 0547 ERMLT6 /SKIP TRAP DLDC
1405 3130 INTER1 /NO DISK INTERRUPT
1406 2362 INTER2 /UNDEFINED INTERRUPT
1407 0206 FLDHLT /PROGRAM WILL ONLY RUN IN FIELD 0
1410 2702 NODSKS /NO DISKS AVAILABLE TO RUN
1411 0603 STPHLT /PROGRAM STOP FROM SHR4=1
1412 2755 CHNHLT /IOT CHANGE HALT
1413 1707 BADHLT /COMPUTER MUST BE DOWN, CHECKSUM FAILED
/BUT WORD-BY-WORD COMPARE WORKED.
/STOP FOR ALL ERROR HALTS.

1414 3136 BIGSTP
/
6740 DLSC=6740 /LOAD SECTOR COUNTER
6741 DSKP=6741 /SKIP ON TRANSFER DONE OR ERROR
6742 DCLR=6742 /CLEAR DISK CONTROL LOGIC
6743 DLG=6743 /LOAD ADDRESS AND GO
6744 DLCA=6744 /LOAD CURRENT ADDRESS
6745 DRST=6745 /READ STATUS REGISTER
6746 DLDC=6746 /LOAD COMMAND REGISTER
/
4406 LAB=JMS I XLAS
4407 CLASIC=JMS I XCLAS
4408 RANDAT=JMS I XRNWRD
4409 DISCON=JMS I XDUMP
4410 SPACE=JMS I XSPAC
4411 ONEIN=JMS I XOCT1
4412 FORIN=JMS I XOCT4
4413 SETGEN=JMS I XSTGEN

4435 SETFLD=JMS I XSTFLD
4437 YESNO=JMS I XCHKYN

4436 SELCHK=JMS I XCKPOT
4440 RANGEN=JMS I XRNDOO
4442 RESRAN=JMS I XRSRAN
4441 DISKGO=JMS I XDSKGO
4443 RECAL=JMS I XRESTR
4444 RECEIV=JMS I XWAIT
4446 ERROR=JMS I XERRO
4447 ROSTAT=JMS I XROST
4453 LOADD=JMS I XLDD
4450 DSKSKP=JMS I XDSKSP

```

```

4451 LDCMD=JMS I XLDCM
4452 LDCUR=JMS I XLDCA
4454 CLRALL=JMS I XCLDR
4455 PRNTER=JMS I XPRN
4456 OCTEL=JMS I XFROCT
4445 TYPE=JMS I XPRINT
4457 CRLF=JMS I XCRLF
4426 GENDAT=JMS I XGNDAT
4424 CHK22=JMS I XCHK22
4425 XTICK=JMS I XXTCK
/
0000 *0
/
0000 0310 310 /REVISION "H" INTERRUPT SERVICE RETURN
0001 5001 5001 /DCA SAVAC SAVE AC AT INT.
0002 0002 0002 /RAL SHIFT LINK AT TIME OF INT.
0003 0003 0003 /DCA SYLNK SAVE LINK AT TIME OF INT.
0004 0004 0004 /JMP I 5 RETURN TO INT. SERVICE
0005 0005 0005 /RETURN POINTER

0006 1546 XLAS, MYLAS
0007 1514 XCLAS, CLASIK
/
0010 *10
/
0010 0000 AUTO10, 0
/
0011 0000 AUTO11, 0
/
0012 0000 AUTO12, 0
/
0013 0004 K0004, 0004
0014 0070 K0070, 0070
0015 0100 K0100, 0100
0016 0200 K0200, 0200
0017 0400 K0400, 0400
/
0020 *20
/
0020 0000 SWK, 0
0021 4000 OP1, 4000
0022 0000 OP2, 0
/
0023 2156 KAERRO, AERROR
0024 0523 XCHK22, CHK22
0025 1154 XTICK, XXTICK
0026 1737 XGNDAT, GNDAT
0027 2000 XRNWRD, RNWRD
0030 2637 XDUMP, DUMP
0031 1506 XSPAC, SPAC
0032 2400 XOCT1, OCT1
0033 2430 XOCT4, OCT4
0034 1753 XSTGEN, STGEN
0035 2703 XSTFLD, STFLD

```

```

0036 2060 XCKPOT, CHKPOT
0037 2035 XCHKYN, CHKYN
0040 1715 XRNDOM, RANDOM
0041 2200 XD0KGO, D0KGO
0042 1761 XRSRAN, RSRAN
0043 3052 XRESTR, RESTOR
0044 2000 XWAIT, WAIT
0045 2620 XPINT, PRINT
0046 1200 XENRO, ENRO
0047 2541 XROST, ROST
0050 0751 X0KXP, 00KXP
0051 0542 XLOCM, LOCM
0052 2550 XL0CA, LOCA
0053 2554 XL0AD, LOAD
0054 0560 XCLDR, CLDR
0055 1450 XPRN, PRN
0056 1426 XFROCT, FROCT
0057 1414 XCMLF, UPONE
0060 0000 AMOUNT, 0
0061 0001 K0001, 0001
0062 0003 K0003, 0003
0063 0006 K0006, 0006
0064 0007 K0007, 0007
0065 0010 K0010, 0010
0066 0017 K0017, 0017
0067 0260 K0260, 0260
0070 0277 K0277, 0277
0071 0770 A0770, 0770
0072 7007 A7007, 7007
0073 4000 K4000, 4000
0074 4100 K4100, 4100
0075 1000 K1000, 1000
0076 1777 K1777, 1777
0077 7700 K7700, 7700
0100 7760 K7760, 7760
0101 7777 K7777, 7777
0102 0077 K0077, 0077
0103 6201 KCDF, CDF
0104 7400 K7400, 7400
/
/
DECIMAL
/
0105 7764 M12, -12
/
OCTAL
/
0106 7774 M4, -4
0107 7770 M10, -10
0110 7775 K7775, 7775
/
/
0111 0000 TRASH1, 0
0112 0000 TRASH2, 0
0113 0000 TRASH3, 0
0114 0000 UPDATE, 0
0115 0000 POLDSK, 0

```

```

0116 0000 OPNTAL, 0
0117 0000 BUFTAL, 0
0120 0000 PCNEG, 0
0121 0000 STNEG, 0
0122 0000 EXNEG, 0
0123 0000 CMNEG, 0
0124 0000 INTDA, 0
0125 0000 DAREG, 0
0126 0000 CAMEG, 0
0127 0000 WCNEG, 0
0130 0000 FWNEG, 0
0131 0000 ASNEG, 0
0132 0000 WAREG, 0
0133 0000 ADREG, 0
0134 0000 DGREG, 0
0135 0000 DBREG, 0
0136 0000 INTCM, 0
0137 0000 STATRY, 0
0140 0000 DATTRY, 0
0141 0000 CHKSAV, 0
0142 0000 FND0SUM, 0
0143 0000 MAXFLD, 0
0144 7607 MAXTIM, 7607
0145 3240 MAXTRK, 3240
0146 3600 BGNBUF, STRBUF
0147 0000 CONSEC, 0
0150 7777 CLKCNT, -1
/
/
0151 0756 DATPOT, DAT1
0152 3522 TIMPOT, 00TM1
0153 3537 STAPOT, D0HRD=3
0154 3512 RUNPUT, D0K0B
/
/
0155 0000 CRCCNT, 0
0156 0000 CRCFLG, 0
0157 0000 DATFLG, 0
0160 0000 SPFLD, 0
0161 0000 SPTRK1, 0
0162 0000 SPTRK2, 0
0163 0000 SPSEC, 0
0164 0000 SPBLK, 0
0165 0000 ERPLG, 0
0166 0000 SAVAC, 0
0167 0000 SVLNK, 0
0170 0000 FINTIM, 0
0171 0000 TRYCNT, 0
0172 3213 XTXT, TEXPC
0173 3142 PRNDAT, TYPDAT
0174 0000 SAVCM, 0
0175 0000 CLNBAK, 0
/
/
0176 3131 BGHLT, BIGHLT
0200 0200
/
/

```

```

/START OF PROGRAM BY OPERATOR:
/AT 0200, TTY INTERMIGATION:
/AT 0201, CHANGE IOT DEVICE CODES:
/AT 0202, RESTART AT SEEK ROUTINE:
/
0200 4777* BGN, JMS APT8 /TO REGULAR TEST
0201 5776* JMP CHNG /CHANGE IOT ROUTINE
0202 5775* JMP RUN
0203 3156 DCA CRCFLG /CLEAR CRC FLAG
0204 6224 RIF
0205 7440 SZA /FIELD 0777?
0206 4576 FLUMLT, JMS I BGMILT /WILL ONLY RUN IN FIELD 0777?
0207 1103 TAD KCDF
0210 3211 DCA ,+1
0211 7402 HLT /MAKE DF=IF
/
/SETUP INTERRUPT SERVICE:
/
0212 1362 TAD ACUCA
0213 3001 DCA 1 /SETUP AC DCA
0214 1250 TAD KRUT
0215 3002 DCA 2 /SETUP ROTATE LINK
0216 1301 TAD LNKDCA
0217 3003 DCA 3 /SETUP SAVE LINK
0220 1360 TAD K5405
0221 3004 DCA 4 /SETUP JMP RETURN
0222 1303 TAD BRKRET
0223 3005 DCA 5 /RETURN POINTER
/
/CLEAR DATA INFORMATION TABLE
/AT END OF PROGRAM:
/
0224 1077 STRTEX, TAD K7700
0225 3111 DCA TRASH1 /CLEAR COUNTER
0226 1774* TAD RANJMS
0227 3773* DCA SWDAT /SET INSTRUCTION SWITCH
0230 7340 CLA CLL CMA
0231 1152 TAD TIMPOT
0232 3010 DCA AUTO10 /LOCATION POINTER
0233 3410 DCA I AUTO10 /CLEAR
0234 2111 ISZ TRASH1
0235 5233 JMP ,+2 /MORE TO CLEAR
0236 3157 DCA DATFLG
0237 5775* SKPNOP, JMP RUN
/
/PRINT PROGRAM NAME AND
/ASK OPERATOR ABOUT AMOUNT
/OF MEMORY:
/
0240 4457 CRLF
0241 4455 PRNTER /PRINT "RK8E/RK8L DATA RELIABILITY"
0242 3307 MES1
0243 4455 PRNTER /PRINT "AMOUNT OF MEMORY"
0244 3346 MES5

```

```

0245 4432 ONEIN
0246 0070 0070 /RECEIVE ONE OCTAL
0247 5243 JMP ,+4 /LIMITS
0250 7004 KRUT, RAL /INPUT ERROR
0251 7006 RTL
0252 7040 CMA
0253 3143 DCA MAXFLD /COMPLEMENT
0254 4772* JMS CLAFLD /MAXIMUM FIELD POINTER
0255 3111 ALLAGN, DCA TRASH1 /CHECK FOR CLASSIC.
0256 1107 TAD M10
0257 3112 DCA TRASH2
0260 3060 DCA AMOUNT /A FEW POINTERS
/
/ASK OPERATOR ABOUT DISK(S) TO TEST:
/
0261 1111 NEXT, TAD TRASH1
0262 1154 TAD RUNPOT
0263 3113 DCA TRASH3 /SAVE RUN POINTER
0264 4455 PRNTER /PRINT "EXERCISE"
0265 3325 MES2
0266 7340 CLA CLL CMA
0267 4455 PRNTER
0270 3332 MES3 /PRINT "DISK"
0271 1067 TAD K0260
0272 1111 TAD TRASH1
0273 4445 TYPE /ADD IN DISK NUMBER
0274 1070 TAD K0277 /TYPE DISK NUMBER
0275 4445 TYPE /TYPE ?
0276 4444 RECEIV /RECEIVE KEY INPUT
0277 4437 YESNO /WAS IT YES OR NO
0300 5255 JMP ALLAGN /NEITHER
0301 5304 JMP ,+3 /WAS A NO
0302 2060 ISZ AMOUNT /AMOUNT OF DISK FOUND
0303 7340 CLA CLL CMA /AC TO 7777 FOR EXISTING DISK
0304 3513 DCA I TRASH3 /SETUP RUN POINTER
0305 2111 ISZ TRASH1
0306 2112 ISZ TRASH2
0307 5261 JMP NEXT /ASK ABOUT NEXT DISK
/
/
/ASK IF ACCEPT MODE:
/
0310 1060 TAD AMOUNT /GET AMOUNT FOUND
0311 7650 SNA CLA /WERE ANY FOUND
0312 5224 JMP STRTEX /OPERATOR ERROR NO DISK INPUT
0313 4455 PRNTER /PRINT "ACCEPT MODE?"
0314 3363 MES6
0315 4444 RECEIV /RECEIVE INPUT
0316 4437 YESNO /YES OR NO777?
0317 5313 JMP ,+4 /NEITHER ALL AGAIN
0320 7610 SKP CLA /MANUAL TEST
0321 5771* JMP ASKSUR /ASK "ARE YOU SURE"
/
/
/IF ACCEPT MODE, INTERAGATE

```

```

/ABOUT CONSTANT FIELD1
/
0322 4455 MANUAL, PRINTER /PRINT "FIELD?"
0323 3404 MES9
0324 4444 RECEIV /RECEIVE Y OR N
0325 4437 YESNO /CHECK FOR Y OR N
0326 5322 JMP MANUAL /NEITHER Y OR N
0327 5345 JMP ASKNX1 /WAS A N, ASK ABOUT NEXT
0330 4431 SPACE /SPACE OUT ONE
0331 4432 ONEIN /GET 1 OCTAL
0332 0070 0070 /LIMITS
0333 5322 JMP MANUAL /INPUT ERROR ASK AGAIN
0334 7104 CLL RAL
0335 7006 RTL
0336 3160 DCA SPFLD /SAVE INPUT
0337 1160 TAD SPFLD
0340 1143 TAD MAXFLD /COMPARE TO MAXIMUM
0341 7700 SMA CLA /U.K.?
0342 5322 JMP MANUAL /INPUT ERROR
0343 7340 CLA CLL CMA
0344 3770 DCA FLOFLG /SETUP FIELD FLAG

```

```

/INTERIGATE ABOUT CONSTANT TRACK1
/
0345 4455 ASKNX1, PRINTER /PRINT "TRACK?"
0346 3410 MES9
0347 4444 RECEIV /RECEIVE Y OR N
0350 4437 YESNO /CHECK FOR Y OR N
0351 5345 JMP ASKNX1 /ERROR, ASK AGAIN
0352 5767 JMP ASKNX2 /N, ASK ABOUT NEXT
0353 4431 SPACE
0354 4432 ONEIN /RECEIVE 1 IN OCTAL
0355 0010 0010 /LIMITS
0356 5345 JMP ASKNX1 /ERROR, ASK AGAIN
0357 5766 JMP SAVE1 /TU SAVE SOME ROOM.

0360 5405 K5405, 5405
0361 3167 LNKDCA, DCA SVLNK
0362 3166 ACUCA, DCA SAVAC
0363 2304 BRKRET, RETURN

```

```

0366 0400
0367 0406
0370 3372
0371 0513
0372 1404
0373 2601
0374 0522
0375 0600
0376 2730
0377 2070
0400

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PAGE

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/INTERIGATE ABOUT CONSTANT
/BLOCK LENGTH1
/
0400 3161 SAVE1, DCA SPTRK1 /SAVE EXTENDED TRACK BIT
0401 4433 FORIN /GET FOUR IN OCTAL.
0402 5777 JMP ASKNX1 /ERROR, ASK AGAIN
0403 3162 DCA SPTRK2 /SAVE CYL., SURFACE, AND SECTOR
0404 7340 CLA CLL CMA
0405 3770 DCA TRKFLG /SETUP TRACK FLAG

0406 4455 ASKNX2, PRINTER /PRINT "BLOCK LENGTH?"
0407 3424 MES11
0410 4444 RECEIV /RECEIVE INPUT
0411 4437 YESNO /CHECK FOR Y OR N
0412 5206 JMP ASKNX2 /ERROR, ASK AGAIN
0413 5225 JMP ASKNX3 /N, ASK ABOUT NEXT
0414 4431 SPACE /Y, SPACE OUT 1
0415 4432 ONEIN /RECEIVE 1 IN OCTAL
0416 0010 0010 /LIMITS
0417 5206 JMP ASKNX2 /ERROR, ASK AGAIN
0420 7640 SZA CLA /SET HALF BLOCK?
0421 7340 CLA CLL CMA /YES
0422 3164 DCA SPBLK /SETUP BLOCK NUMBER
0423 7340 CLA CLL CMA
0424 3775 DCA MLFPLG /SETUP BLOCK FLAG

/
/INTERIGATE ABOUT CONSTANT
/SECTORS1
/
0425 4455 ASKNX3, PRINTER /PRINT "EXTRA SECTORS?"
0426 3414 MES10
0427 4444 RECEIV /RECEIVE INPUT
0430 4437 YESNO /CHECK FOR Y OR N
0431 5225 JMP ASKNX3 /INPUT ERROR
0432 5204 JMP ASKNX5 /N, ASK ABOUT NEXT
0433 4431 SPACE /SPACE OUT 1
0434 4432 ONEIN /RECEIVE 1 IN OCTAL
0435 0010 0010 /LIMITS
0436 5225 JMP ASKNX3 /ERROR, ASK AGAIN
0437 7104 CLL RAL
0440 7006 RTL
0441 3163 DCA SPSEC /SAVE IT
0442 4432 ONEIN /RECEIVE 1 IN OCTAL
0443 0070 0070 /LIMITS
0444 5225 JMP ASKNX3 /INPUT ERROR, ASK AGAIN
0445 1163 TAD SPSEC /ADD IN LAST
0446 3163 DCA SPSEC /SAVE ALL
0447 1164 TAD SPBLK
0450 7640 SZA CLA
0451 5254 JMP .+J /BLOCK LENGTH 0????
0452 1160 TAD SPFLD /NO LIMIT IS 17.
0453 7640 SZA CLA
0454 1065 TAD K0010 /FIELD 0?????
0455 1064 TAD K0007 /LIMIT IS 17.

```

```

0456 7140      CLL CMA
0457 1163      TAD 8PSEC
0460 7630      SZL CLA
0461 5225      JMP ASKNX3
0462 7340      CLA CLL CMA
0463 3774      DCA SECFLG
/
/
/
/INTERIGATE ABOUT "OPERATOR
/SELECT DATA"
/
0464 4455      ASKNX5, PRNTER
0465 3433      MES13
0466 1322      TAD RANJMS
0467 3773      DCA SWDAT
0470 4444      RECEIV
0471 4437      YESNO
0472 5264      JMP ASKNX5
0473 5313      JMP ASKSUR
0474 1346      TAD KSKP
0475 3773      DCA SWDAT
0476 1105      TAD M12
0477 3111      DCA TRASH1
0500 7340      CLA CLL CMA
0501 1151      TAD DATPOT
0502 3010      DCA AUT010
0503 4437      CRLP
0504 4433      FORIN
0505 5264      JMP ASKNX5
0506 3410      DCA I AUT010
0507 2111      ISZ TRASH1
0510 5303      JMP .05
0511 7340      CLA CLL CMA
0512 3157      DCA DATFLG
/
/ASK IF HE'S SURE;
/
0513 4455      ASKSUR, PRNTER
0514 3436      MES14
0515 4444      RECEIV
0516 4437      YESNO
0517 5313      JMP ASKSUR
0520 5772      JMP STRTEX
0521 5771      JMP RUN
0522 4426      RANJMS, GENDAT
/
/THIS ROUTINE TESTS FOR BEING ON APT.
/IF ON APT RETURN IS PLUS ONE, IF NOT RETURN IS PLUS TWO.
/
0523 0000      CHEK22, 0
0524 1022      TAD 22
0525 7700      SNA CLA
0526 2323      ISZ CHEK22
0527 5723      JMP I CHEK22

```

```

/ROUTINE TO NOTIFY APT.
/
0530 0000      KTIME, 0
0531 4424      CHK22
0532 7410      SKP
0533 5730      JMP I KTIME
0534 6002      IOF
0535 6201      CDF 00
/
0536 6272      CIF 70
0537 4741      JMS I K6500
0540 5730      JMP I KTIME
/
0541 6500      K6500, 6500
/
/
/SUBROUTINE TO LOAD COMMAND REGISTER
/
0542 0000      LDCM, 0
0543 3123      DCA CMNEG
0544 1123      TAD CMNEG
0545 6746      IOT6, DLDC
0546 7610      KSKP, SKP CLA
0547 4576      ERHLT6, JMS I BGHLT
0550 1122      TAD EXREG
0551 7110      CLL RAM
0552 7630      SZL CLA
0553 1016      TAD K0200
0554 6740      IOT0, DLSC
0555 7610      SKP CLA
0556 4576      ERHLT0, JMS I BGHLT
0557 5742      JMP I LDCM
/
/
/SUBROUTINE TO ISSUE "DCLR" CLEAR IOT
0560 0000      CLDR, 0
0561 6742      IOT2, DCLR
0562 5760      JMP I CLDR
0563 4576      ERHLT2, JMS I BGHLT
/
0571 0000
0572 0224
0573 2601
0574 3574
0575 3575
0576 3573
0577 0345
0600
PAGE
/
/
/SETUP ADDRESSING, COMMAND,
/AND DATA PARAMETERS;

```

```

/MAKE FIELD1
/
0600 4406 RUN, LAS
0601 0016 AND K0200
0602 7640 SZA CLA
0603 7402 STPHLT, HLT
0604 1777 TAD FLDPLG
0605 7650 SNA CLA
0606 5211 JMP ,+3
0607 1160 TAD SPFLD
0610 5230 JMP HNPLD
0611 7301 CLA CLL IAC
0612 1143 TAD MAXFLD
0613 7650 SNA CLA
0614 5230 JMP HNPLD
0615 4440 RANGEN
0616 0014 AND K0070
0617 7450 SNA
0620 5230 JMP HNPLD
0621 3136 DCA INTCM
0622 1136 TAD INTCM
0623 1143 TAD MAXFLD
0624 7710 SPA CLA
0625 5231 JMP HNPLD+1
0626 1143 TAD MAXFLD
0627 7040 CMA
0630 3136 RNPLD, DCA INTCM
/MAKE BLOCK LENGTH1
/
0631 1776 TAD HLFPLG
0632 7650 SNA CLA
0633 4440 RANGEN
0634 1164 TAD SPBLK
0635 0015 AND K0100
0636 1136 TAD INTCM
0637 3136 DCA INTCM
0640 1136 TAD INTCM
0641 0015 AND K0100
0642 7640 SZA CLA
0643 1016 TAD K0200
0644 1104 TAD K7400
0645 3112 DCA TRASH2
0646 1112 TAD TRASH2
0647 7041 CIA
0650 3114 DCA UPDATE
0651 1136 TAD INTCM
0652 0350 AND A0170
0653 7640 SZA CLA
0654 1065 TAD K0010
0655 1064 TAD K0007
0656 3111 DCA TRASH1
/MAKE AMOUNT OF SECTORS
/TO TRANSFER1

```

```

/GET THE SWITCHES.
/MASK MALT SW.
/TIME TO MALT?
/MALT FROM SWR4=1.
/GET FIELD FLAG
/WAS IT SET?
/NO, USE RANDOM FIELD
/YES, GET OPERATOR FIELD
/GO
/GET MAXIMUM FIELD POINTER
/ANY FIELDS THERE
/NO EXTENDED FIELDS TO USE
/YES, GET A RANDOM FIELD
/MASK
/COULD BE 0
/WAS DON'T HAVE TO CHECK LIMITS
/SAVE FIELD FOUND
/ADD IN MAXIMUM FIELD POINTER
/IN LIMITS????
/YES, USE IT
/NO, USE MAXIMUM IN THE MACHINE

```

```

/
0657 1775 TAD SECPLG
0660 7650 SNA CLA
0661 4440 RANGEN
0662 1163 TAD SPSEC
0663 0111 AND TRASH1
0664 3147 DCA CONSEC
0665 1147 TAD CONSEC
0666 7040 CMA
0667 3111 DCA TRASH1
/MAKE WORD COUNT1
/
0670 1112 TAD TRASH2
0671 2111 ISZ TRASH1
0672 5270 JMP ,+2
0673 3127 DCA WCREG
/MAKE CURRENT ADDRESS1
/
0674 4440 RANGEN
0675 3126 DCA CAREG
0676 1136 TAD INTCM
0677 0014 AND K0070
0700 7640 SZA CLA
0701 5317 JMP FILLUP
0702 1146 TAD BGNBUF
0703 7140 CMA CLL
0704 1126 TAD CAREG
0705 7620 SNA CLA
0706 5315 JMP CONCUR
0707 1127 TAD WCREG
0710 7041 CIA
0711 1126 TAD CAREG
0712 1016 TAD K0200
0713 7630 SZA CLA
0714 5317 JMP FILLUP
0715 1146 CONCUR, TAD BGNBUF
0716 3126 DCA CAREG
/ROUTINE TO FILL AND CHECK SUM BUFFER
/
0717 4425 FILLUP, KTICK
0720 4434 SETGEN
0721 1106 TAD M4
0722 3137 DCA STATRY
0723 4435 REPILL, SETFLO
0724 3325 DCA ,+1
0725 7402 HLT
0726 3141 DCA CHKSAV
0727 4427 NEWRD, RANDAT
0730 3111 DCA TRASH1
0731 1111 TAD TRASH1
0732 3411 DCA I AUTO11
0733 7100 CLL

```

```

/GET SECTOR FLAG
/WAS IT SET????
/USE RANDOM
/GET OPERATOR INPUT
/MASK OUT
/SAVE
/CONSECUTIVE TO DO
/COMPUTE INITIAL WC
/UPDATE BY BUILDER
/INITIAL WORD COUNT ****
/GENERATE RANDOM CA
/SAVE IT
/MASK FIELD BITS
/EXTENDED FIELD????
/INITIAL CA 0,K,****
/LESSER THAN PROGRAM+1
/NO, USE CONSTANT VALUE
/GET WORD COUNT
/ADD IN CA
/WITHIN BOUNDS????
/YES, INITIAL CA 0,K,****
/NO, USE PROGRAM+1
/SAVE IT
/NOTIFY APT IF NEED BE.
/SETUP AND SAVE GENERATOR
/SETUP TRY COUNTER
/FIELD+ BUFTAL+ AUTO 11+ 12
/FIELD TO BUFFER IN AC
/BUF TO BUFFER
/START WITH 0
/GENERATE DATA
/SAVE OUTPUT WOMB
/GET BACK WORD
/STONE IN BUFFER

```

```

0734 1111 TAD TRA0M1 /GET BACK WORD
0735 1141 TAD CHK0AV /ADD IN LAST
0736 7430 SZL /LINK SET??
0737 7001 IAC /ADD IT IN
0740 3141 DCA CHK0AV /SAVE FOR NEXT
0741 2117 ISZ BUPTAL /UPDATE BUFFER TALLY
0742 5327 JMP NEWRD /MORE WORDS TO GO
0743 0201 CDF 0
0744 1165 TAD ERPLG
0745 7650 SNA CLA /ERROR FLAG SET???
0746 5774 JMP POLNEX /POLE DRIVES
0747 5773 JMP REWRT /YES, MUST BE A WRITE ERROR

0750 0170 AB170, 0170
/
/SUBROUTINE TO ISSUE "DSKP" DISK SKIP IOT
/
0751 0000 DSKP, 0
0752 6741 IOT1, DSKP /DISK SKIP IOT
0753 7410 SKP /DID NOT SKIP
0754 2351 ISZ DSKP
0755 5751 JMP I DSKP /EXIT
/
/PLACE FOR DATA IN MANUAL MODE
/
0756 0000 DAT1, 0000
0757 0000 DAT2, 0000
0760 0000 DAT3, 0000
0761 0000 DAT4, 0000
0762 0000 DAT5, 0000
0763 0000 DAT6, 0000
0764 0000 DAT7, 0000
0765 0000 DAT8, 0000
0766 0000 DAT9, 0000
0767 0000 DAT10, 0000
0770 0000 DAT11, 0000
0771 0000 DAT12, 0000
/
0773 1047
0774 1000
0775 3574
0776 3575
0777 3572
1000 PAGE
/
/ROUTINE TO SELECT DRIVE NO.
/SEQUENTIAL SELECTION 0,1,2,3,0,1,ETC.
/
1000 2115 POLNEX, ISZ POLDISK /UPDATE POLE POINTER
1001 1115 TAD POLDISK /GET POINTER
1002 4436 SELCHK /CHECK IF DISK ON SYSTEM,
1003 5200 JMP POLNEX /NO, TRY NEXT DRIVE
1004 1115 TAD POLDISK
1005 7112 CLL RTM
1006 0001 AND K0001

```

```

1007 3122 DCA EXREG /SET EXT, DRIVE BIT
/
/DRIVE COMPLETED, START
/WRITE SEQUENCE,
/SELECT DISK ADDRESS,
/
1010 1115 GOTIT, TAD POLDISK /GET DRIVE NO.
1011 0002 AND K0003 /MASK
1012 7104 CLL RAL /MOVE TO 9-10,
1013 1136 TAD INTCH /ADD IN OTHER,
1014 3136 DCA INTCH /SAVE INITIAL COMMAND.
1015 1777 TAD TRKFLG /GET TRACK FLAG
1016 7650 SNA CLA /WAS IT SET???
1017 4440 RANGEN /GET RANDOM DA.
1020 1101 TAD SPTRK1 /GET OPERATOR CONSTANT INPUT.
1021 0001 AND K0001 /MASK EXT, BIT.
1022 1136 TAD INTCH /ADD IN OTHER,
1023 3136 DCA INTCH /SAVE COMPLETE INITIAL COMMAND.
1024 1777 TAD TRKFLG /GET TRACK FLAG
1025 7650 SNA CLA /WAS IT SET???
1026 4440 RANGEN /USE RANDOM
1027 1102 TAD SPTRK2 /GET INPUT
1030 3124 DCA INTDA /SAVE INITIAL DA.
1031 1777 TAD TRKFLG
1032 7640 SZA CLA /INPUT BY OPERATOR?
1033 5247 JMP REWRT /LET HIM FAIL??
1034 1136 TAD INTCH
1035 7010 RAR
1036 7620 SNA CLA
1037 5247 JMP REWRT /EXT BIT SET?
1040 1145 TAD MAXTRK /NO, DON'T LIMIT DA.
1041 1124 TAD INTDA
1042 7630 SZL CLA /BEOND MAXIMUM LIMIT?
1043 5247 JMP REWRT /NO, DONT LIMIT.
1044 1124 TAD INTDA
1045 7040 CMA
1046 3124 DCA INTDA /YES, SET TO LEGAL LIMIT.
/
/WRITE INFORMATION!
/CLEAR BUFFER ON THE FLY!
/
1047 4441 REWRT, DISKGO /GO WRITE
1050 4400 4400 /WRITE DATA POINTER
1051 5263 JMP REREAD /WRITE 0,K.
1052 7340 CLA CLL CMA
1053 3165 DCA ERPLG /SET WRITE ERROR FLAG
1054 4442 RESRAN /RESET GENERATOR
1055 2137 ISZ STATRY /UPDATE WRITE RE-TRY
1056 5776 JMP REFILL /TRY AGAIN
/
/CHECK FOR LOOP ON WRITE!
/
1057 4406 LAS /GET SWITCH 0
1060 7710 TRYIM, SPA CLA /LOOP ON WRITE???
1061 5775 JMP REFILL=2 /YES, TRY WRITE AGAIN

```

```

1062 5351      JMP      STRREL      /RESNORE ALL UNIVES ON ERROR
1063 1260      REREAD, TAD      TRYTIM
1064 3171      DCA      TRYCNT      /SETUP FOR SOFT ERROR RETRY
1065 3165      DCA      ERFLG      /CLEAR ERROR FLAG
1066 1106      TAD      M4
1067 3137      DCA      STATRY      /SETUP TRY COUNTER
1070 1106      TAD      M4
1071 3140      DCA      DATTRY      /SETUP TRY COUNTER
1072 3155      DCA      CRCCNT      /CLEAR CRC COUNTER!!!!

/
/READ INFORMATION/
/CHECK BUFFER ON THE FLY/
/

1073 4441      RDRY,   DISKGO      /READ DATA
1074 0400      0400      /READ DATA POINTER
1075 7610      SKP      CLA      /DATA READ O.K.
1076 5305      JMP      RDRSTA      /STATUS ERROR
1077 3155      DCA      CRCCNT      /CLEAR CRC COUNTER/

/
/CHECK DATA ON NO STATUS ERRORS/
/

1100 4774*     JMS      DTCHK      /CHECK DATA
1101 5324      JMP      REMUN      /DATA O.K.
1102 2140      ISZ      DATTRY      /UPDATE READ RE-TRY
1103 5273      JMP      RDRY      /TRY AGAIN
1104 5323      JMP      RERUN-1     /TRY TO SEEK IT
1105 1121      RDRSTA, TAD      STREG      /GET STATUS READ
1106 0065      AND      K0010      /MASK CRC
1107 7450      SNA      /CRC ERROR???
1110 5320      JMP      UPTRY      /NO, TRY READ AGAIN
1111 3156      DCA      CRCFLG      /YES, SET FLAG
1112 2155      ISZ      CRCCNT      /UPDATE CRC POINTER

/
/CHECK DATA AFTER CRC ERROR/
/

1113 4774*     JMS      DTCHK      /CHECK DATA
1114 7610      SKP      CLA      /IS A HARD ERROR?
1115 7340      CLA      CLL      CMA      /SET RETRY COUNTER/
1116 3165      DCA      ERFLG      /SETUP FOR 64 RETRYS IF AC=7777
1117 7410      SKP      /CHECK ON RETRY!!!!
1120 3155      UPTRY,   DCA      CRCCNT
1121 2137      ISZ      STATRY      /UPDATE TRY POINTER
1122 5273      JMP      RDRY      /TRY AGAIN
1123 3165      DCA      ERFLG      /IS A HARD ERROR
1124 3155      RERUN,   DCA      CRCCNT      /CLEAR CRC COUNT
1125 3156      DCA      CRCFLG      /CLEAR CRC FLAG
1126 4773*     JMS      CKTIM      /CHECK TIME POINTERS
1127 1165      TAD      ERFLG
1130 7650      SNA      CLA
1131 5334      JMP      .+3
1132 2171      ISZ      TRYCNT      /NO DON'T BOTHER
1133 5266      JMP      REREAD+3    /YES, UPDATE RETRY COUNTER
                                      /TRY AGAIN

/
/CHECK FOR LOOP ON READ/
/

```

```

1134 4406      LAS
1135 7104      CLL      RAL      /GET SWITCH 1
1136 7710      SPA      CLA
1137 5263      JMP      REREAD      /LOOP???
1140 1137      TAD      STATRY      /YES, LOOP
1141 7650      SNA      CLA      /TEST FOR HARD ERROR
1142 5351      JMP      STRREL      /YES
1143 3165      DCA      ERFLG      /CLEAR ERROR FLAG

/
/CHECK FOR TYPE STATUS
/REPORT/
/

1144 4406      LAS
1145 0017      AND      K0400      /MASK
1146 7640      SZA      CLA      /TYPE STATUS REPORT???
1147 4772*     JMS      TPSTA      /YES
1148 5771*     JMP      RUN      /DU NEXT DRIVE

/
/RESTORE DRIVE AFTER ERROR
/

1151 1123      STRREL, TAD      CMREG      /GET DRIVE NO.
1152 4443      RECAL      /RESTORE
1153 5771*     JMP      RUN      /START NEXT DRIVE

/
/ROUTINE TO DETERMINE IF TIMING NEEDS TO BE FOR APT SYSTEM,
/

1154 0000      XKTICK, 0
1155 4424      CHK22
1156 7410      SKP
1157 5754      JMP      I      XKTICK      /TEST FOR APT
1160 6201      CDF      0      /NO, RETURN TO NORMAL RUN
1161 2150      ISZ      CLKCNT      /LONG COUNTER FOR APT
1162 5366      JMP      EXTICK      /NORMAL RETURN
1163 1100      TAD      K7760      /INIT COUNTER
1164 3150      DCA      CLKCNT
1165 4770*     JMS      KTIME      /NOTIFY APT
1166 6201      EXTICK, CDF      0
1167 5754      JMP      I      XKTICK

1170 0530
1171 0600
1172 3000
1173 2450
1174 1000
1175 0721
1176 0723
1177 3573
1200
1201

PAGE
/
/ERROR HANDLER/
/UPDATE "SOFT" ON "HARD" TALLYS/
/PRINT ERROR TEXT AND DATA/
/CHECK INHIBIT ERRNR SW/
/

ERNO, 0
IAC
/UPDATE AC FLAG

```

```

1202 3374      DCA      PCNTR2      /SAVE NON-RECOVERABLE POINTER
/
/COMPUTE WAY TO "HARD"/"SOFT" TALLYS
/
1203 1377      TAD      K7773
1204 3375      DCA      PCNTR3      /LINE COUNTER
1205 1123      TAD      CMREG      /GET LAST COMMAND
1206 0063      AND      K0006      /MASK DRIVE NUMBER
1207 7170      CLL CML CMA RAR
1210 3373      DCA      PCNTR1      /SETUP COUNTER
1211 1062      TAD      K0003
1212 2373      ISZ      PCNTR1
1213 5211      JMP      .+2      /COMPUTE WAY TO BUFFER
1214 1133      TAD      STAPOT
1215 3373      DCA      PCNTR1      /POINTER TO BUFFER
/
/DETERMINE IF ERROR IS "HARD" OR "SOFT"
/
1216 1156      TAD      CRCFLG      /GET CRC FLAG
1217 7650      SNA CLA      /CRC ERROR???
1220 5251      JMP      NTSOFT      /NO, WAS DEFINITELY A HARD ERROR
1221 1600      TAD I      ERRO      /GET ERROR POINTER
1222 7650      SNA CLA      /WAS IT FIRST TIME?
1223 5255      JMP      NTERR      /NO ERROR, ADDITIONAL CRC DATA
1224 1125      TAD      DAHEG      /COMPARE FAILING SECTOR TO
1225 0066      AND      K0017      /SECTOR WHERE DATA ERROR
1226 7041      CIA
1227 1131      TAD      ASHEG      /OCCURRED
1230 7640      SZA CLA      /SAME SECTOR?
1231 5251      JMP      NTSOFT      /NO, "HARD" ERROR
1232 7340      CLA CLL CMA
1233 1155      TAD      CRCCNT      /GET CRC COUNTER
1234 7450      SNA
1235 5245      JMP      SOFT      /WAS THIS FIRST POSSIBLE "SOFT"?
1236 1110      TAD      K7775      /YES, UPDATE "SOFT" TALLY
1237 7650      SNA CLA      /CHECK IF NONRECOVERABLE "SOFT"
1240 2373      ISZ      PCNTR1      /WAS IT?
1241 1773      TAD I      PCNTR1      /NO, DUMP "SOFT" TALLY
1242 7440      SZA
1243 1101      TAD      K7777      /OTHERWISE DUMP "HARD" TALLY
1244 5254      JMP      NTERR-1      /DUNT GO BACK WARDS!!!!!!
1245 1101      SOFT,      TAD      K7777      /DUMP APPROPRIATE TALLY
1246 1773      TAD I      PCNTR1      /REDUCE HARD ERROR COUNT
1247 3773      DCA I      PCNTR1
1250 2373      ISZ      PCNTR1      /YES, UPDATE POINTER
1251 1101      NTSOFT,   TAD      K7777      /UPDATE ERROR COUNT
1252 2773      ISZ I      PCNTR1
1253 7610      SKP CLA
1254 3773      DCA I      PCNTR1      /HOLD AT 7777
/
/CHECK INHIBIT SW
/
1255 4423      NTERR,    JMS I      KAERRO      /REPORT ERROR TO APT IF REQUIRED
1256 4406      LAB
1257 7106      CLL RTL

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```

1260 7710      SPA CLA      /INHIBIT ERRORS???
1261 5356      JMP      ERROEX+1      /YES
/
/CHECK FOR NO HEADER ON SECOND DATA ERROR
/
1262 1600      DOMEAD,   TAD I      ERRO      /GET TEXT POINTER
1263 7650      SNA CLA      /DATA ERROR?
1264 5355      JMP      ERROEX      /EXIT
/
/TYPE ERROR MESSAGE
/
1265 4457      CRLF
1266 4457      CRLF
1267 1374      TAD      PCNTR2      /GET NON-RECOV. FLAG
1270 7640      SZA CLA      /WAS IT SET
1271 5275      JMP      .+4      /NO DON'T TYPE IT
1272 7340      CLA CLL CMA
1273 4455      PRNTER
1274 3335      MES4      /PRINT "NON-RECOVERABLE "
1275 1600      TAD I      ERRO      /GET TEXT POINTER
1276 1376      TAD      MEDITAD      /MAKE ERROR HEADER POINTER
1277 1320      DCA      PCREG      /SAVE POINTER
1278 1520      TAD I      PCREG      /GET CORRECT TEXT
1281 3304      DCA      .+3
1282 7340      CLA CLL CMA
1283 4455      PRNTER
1284 7402      HLT      /PRINT HEADER
1285 7340      CLA CLL CMA
1286 4455      PRNTER
1287 3303      MES0      /PRINT "ERROR"
1290 4457      CRLF
1311 1200      TAD      ERRO
1312 3120      DCA      PCREG      /SAVE PC
1313 2200      ISZ      ERRO
1314 1600      TAD I      ERRO
1315 3371      DCA      ESAYE
1316 2200      ISZ      ERRO
1317 1172      TAD      XTEXT      /UPDATE FOR RETURN
1320 3374      DCA      PCNTR2
1321 1372      TAD      XREG
1322 3010      DCA      AUTO10
1323 1105      TAD      M12
1324 3373      DCA      PCNTR1      /COUNTER FOR # OF HEADS
1325 1371      STNAUT,   TAD      ESAYE      /GET TEXT POINTER
1326 7500      SMA
1327 5363      JMP      NOTEX      /NOT THIS ONE
1330 7104      CLL RAL
1331 3371      DCA      ESAYE
1332 2375      ISZ      PCNTR3      /UPDATE LINE FILL COUNTER
1333 7610      SKP CLA      /NO CRLF
1334 4457      CRLF
1335 1374      TAD      PCNTR2      /GET TEXT MESSAGE POINTER
1336 2374      ISZ      PCNTR2
1337 2374      ISZ      PCNTR2
1340 3343      DCA      .+3      /STONE FOR PRNTER

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1341 7340      CLA CLL CMA
1342 4455      PRINTER
1343 7402      MLT
1344 1410      TAD I      AUTO10
1345 4456      UCTEL
1346 2373      AGAIN,   ISZ   PCNTR1
1347 5325      JMP       STHAUT
1350 1520      TAD I      PCMEG
1351 1106      TAD       M4
1352 7650      SNA CLA
1353 4573      JMS I      PRNDAT
1354 5360      JMP       ,+4
1355 4573      ERROEX,   JMS I      PRNDAT
1356 2200      ISZ       ERNO
1357 2200      ISZ       ERNO
1360 7301      CLA CLL   IAC
1361 4454      CLRALL
1362 5600      JMP I      ERNO
1363 7104      NOTEX,   CLL RAL
1364 3371      DCA       ESAVE
1365 2374      ISZ       PCNTR2
1366 2374      ISZ       PCNTR2
1367 2010      ISZ       AUTO10
1370 5346      JMP       AGAIN

/
1371 0000      ESAVE,   0
1372 0117      XREG,    PCREG=1
1373 0000      PCNTR1,  0
1374 0000      PCNTR2,  0
1375 0000      PCNTR3,  0
1376 1377      HEDTAD,  BUFPNT=1
1377 7773      K7773,   7773

/
1400          PAGE
/
/POINTERS FOR TEXT INFORMATION:
/
1400 3247      BUFPNT,  ERTX1
1401 3255      ERTX2
1402 3264      ERTX3
1403 3276      ERTX4

/
/ROUTINE TO CHECK FOR CLASSIC AND LIMIT
/TRANSFERS TO FIELD 0 IF AVAILABLE.
/
1404 0000      CLAFLO,  0
1405 1022      TAD       22
1406 0017      AND       K0400
1407 7650      SNA CLA
1408 5604      JMP I      CLAFLO
1409 7340      CLA CLL   CMA
1410 3143      DCA       MAXFLO
1411 5604      JMP I      CLAFLO

/ON CLASSIC?
/NO.
/LIMIT TO FIELD 0.

/ROUTINE TO DO CRLF

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```

1414 0000      UPONE,   0
1415 7300      CLA CLL
1416 1224      TAD       K0215
1417 4445      TYPE
1420 1225      TAD       K0212
1421 4445      TYPE
1422 4445      TYPE
1423 5614      JMP I      UPONE

/
1424 0215      K0215,   0215
1425 0212      K0212,   0212

/ROUTINE TO PRINT FOUR OCTAL
/
1426 0000      FROCT,   0
1427 7006      RTL
1430 7006      RTL
1431 3214      DCA       UPONE
1432 1106      TAD       M4
1433 3250      DCA       PRN
1434 1214      TAD       UPONE
1435 0064      AND       K0007
1436 1067      TAD       K0260
1437 4445      TYPE
1440 1214      TAD       UPONE
1441 7006      RTL
1442 7004      RTL
1443 3214      DCA       UPONE
1444 2250      ISZ       PRN
1445 5234      JMP       ,+11
1446 4431      SPACE
1447 5626      JMP I      FROCT

/
/SUBROUTINE TO PRINT TEXT
/
1450 0000      PRN,     0
1451 7650      SNA CLA
1452 4457      CRLF
1453 1650      TAD I      PRN
1454 2250      ISZ       PRN
1455 3226      DCA       FROCT

/TYPE CRLF
/YE!!!!
/GET POINTER

1456 7300      MPRN,    CLA CLL
1457 1626      TAD I      FROCT
1460 0077      AND       K7700
1461 7450      SNA
1462 5304      JMP       EXIT
1463 7500      SNA
1464 7020      CML
1465 7001      IAC
1466 7012      RTR
1467 7012      RTR

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1470 7012      RTR
1471 4445      TYPE
1472 1626      TAD I   FRUCT
1473 0102      AND     K0077
1474 7450      SNA
1475 5304      JMP     EXIT
1476 1313      TAD     K3740
1477 7500      SNA
1500 1074      TAD     K4100
1501 4431      SPACE   /SPACE OUT 1
1502 2226      ISZ     FROCT
1503 5256      JMP     MRPRN      /MORE TO PRINT
1504 7300      EXIT,    CLA CLL
1505 5650      JMP I   PRN
/
/ROUTINE TO SPACE OUT 1
1506 0000      SPAC,    0
1507 1312      TAD     K0240
1510 4445      TYPE
1511 5706      JMP I   SPAC
/
1512 0240      K0240, 240
1513 3740      K3740, 3740
/
/THIS ROUTINE WILL BE A SKIP INSTRUCTION FOR SYSTEMS WITHOUT CLASSIC
/OTHERWISE IT WILL EXECUTE THE NEXT INSTRUCTION IN FIELD 0 AND THEN
/SKIP THE INSTRUCTION AFTER THAT ONE.
/
1514 0000      CLASIK, 0
1515 3345      DCA     SAVEAC      /SAVE CURRENT AC.
1516 1714      TAD I   CLASIK      /GET INSTRUCTION TO EXECUTE.
1517 3344      DCA     ROUTMP      /SAVE IT.
1520 2314      ISZ     CLASIK
1521 1022      TAD     OP2
1522 0017      AND     K0400
1523 7640      SZA CLA      /ARE WE ON CLASSIC?
1524 5327      JMP     ,+3         /YES.
1525 1345      TAD     SAVEAC      /NO, THEN
1526 5714      JMP I   CLASIK      /EXIT.
1527 2314      ISZ     CLASIK
1530 6211      COF     10
1531 1020      TAD     SWR
1532 3777      DCA I   (SWR)      /SAVE SWITCH REGISTER.
1533 1021      TAD     OP1
1534 3776      DCA I   (OP1)      /SAVE CONTROL 1.
1535 1022      TAD     OP2
1536 3775      DCA I   (OP2)
1537 1344      TAD     ROUTMP
1540 3774      DCA I   (ROUTMP)   /SAVE ROUTINE IN FIELD 1.
1541 1345      TAD     SAVEAC      /GET BACK AC.
1542 6212      CIF     10
1543 5774      JMP I   (ROUTMP)   /GO AND EXECUTE INSTRUCTION.
/
1544 0000      ROUTMP, 0

```

```

1545 0000      SAVEAC, 0
/
/ROUTINE TO GET THE SWITCHES.
/
1546 0000      MYLAS, 0
1547 4407      CLASIK      /CHECK FOR CLASSIC.
1550 4425      C0CKSW      /GET SWITCHES.
1551 7604      7604
1552 5746      JMP I   MYLAS      /EXIT.
/
/ROUTINE TO RESET REGISTERS FOR ERROR PRINTIN
/
1553 0000      SETREG, 0
1554 1073      TAD     K4000      /GET STATUS
1555 3121      DCA     STREG      /SAVE FOR ERROR PRINTER
1556 7340      CLA CLL CMA      /DECREASE BY 1
1557 1111      TAD     TRASH1     /GET SECTOR POINTER
1560 0066      AND     K0017
1561 1112      TAD     TRASH2
1562 3125      DCA     DAMEG      /ADD IN ADDRESS
1563 1170      TAD     FIKTIM     /SAVE FOR ERROR PRINTER
1564 7640      SZA CLA      /CHECK IF FIRST SECTOR?
1565 5753      JMP I   SETREG      /IF SO, DON'T UPDATE COMMAND!
1566 1174      TAD     SAVCM      /NO, DON'T!
1567 3123      DCA     CMREG      /GET COMMAND REG.
1570 5753      JMP I   SETREG      /SAVE FOR ERROR PRINTER
/
/RETURN
/
1574 1302
1575 0022
1576 0021
1577 0020
1600 1000      PAGE
/
/ROUTINE TO CHECK DATA READ
/
1600 0000      DTCHK, 0
1601 1156      TAD     CRCFLG      /GET CRC FLAG
1602 7640      SZA CLA      /WAS IT SET?
1603 5212      JMP     WRDCHK      /YES, THEN WORD BY WORD CHECK!!!
1604 1142      TAD     FNDUSUM      /GET CHECK SUM FOUND
1605 7041      CIA
1606 1141      TAD     CHKSAV      /COMPARE TO GOOD VALUE SAVED
1607 7650      SNA CLA      /WERE THEY THE SAME
1610 5600      JMP I   DTCHK      /YES, DATA O.K.
1611 7340      CLA CLL CMA
1612 3446      WRDCHK, DCA I XEMRO      /SETUP CHECKSUM ERROR FLAG
1613 1123      TAD     CMREG
/
1614 0015      AND     K0100
1615 7640      SZA CLA      /HALF BLOCK SET??
1616 1016      TAD     K0200      /YES!
1617 1104      TAD     K7400
1620 3112      DCA     TRASH2
1621 1112      TAD     TRASH2
1622 7040      CMA

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1623 3314      DCA  MSKER
1624 7340      CLA  CLL  CMA
1625 3142      DCA  FNDOSUM
1626 4442      RESRAN
1627 1130      TAD  FNMEG
1630 4435      SETFLD
1631 3246      DCA  GOCDF
1632 1112      TAD  TRASH2
1633 3361      DCA  RSRAN
1634 1124      TAD  INTDA
1635 3353      DCA  STGEN
1636 1361      DTR1, TAD  RSRAN
1637 0314      AND  MSKER
1640 3132      DCA  WAREG
1641 1353      TAD  STGEN
1642 0066      AND  K0017
1643 3131      DCA  ASREG
1644 4427      RANDAT
1645 3134      DCA  DGREG
1646 7402      GOCDF, HLT/CDP
1647 1411      TAD  I  AUTO11
1650 6201      CDF  0
1651 3135      DCA  DBWEG
1652 1011      TAD  AUTO11
1653 3133      DCA  ADREG
1654 1135      TAD  DBREG
1655 7041      CIA
1656 1134      TAD  DGREG
1657 7650      SNA  CLA
1660 5272      JMP  NOERR
1661 2142      ISZ  FNDOSUM
1662 5310      JMP  NTRKRS
1663 1156      TAD  CRCFLG
1664 7650      SNA  CLA
1665 1140      TAD  DATTRY
1666 2200      ISZ  DTCHK
1667 4446      ERROR
1670 0004      0004
1671 7760      7760
1672 2361      NOERR, ISZ  RSRAN
1673 5300      JMP  +5
1674 2353      ISZ  STGEN
1675 7000      NOP
1676 1112      TAD  TRASH2
1677 3361      DCA  RSRAN
1678 2117      ISZ  BUFTAL
1679 5236      JMP  DTM1
1682 1446      TAD  I  XENRO
1683 7650      SNA  CLA
1684 3155      DCA  CRCCNT
1685 2446      ISZ  I  XENRO
1686 5600      JMP  I  DTCHK
1687 4576      BAUHLT, JMS  I  BGHLT
1688 4446      NTRKRS, ERROR
1689 0000      0000

```

```

/SET FIRST TIME FLAG
/NO, SETUP RANDOM GENERATOR
/GET FINAL WC
/GET AUTO11+ BUFTAL+ FIELD
/SAVE FIELD CDF

/GENERATE DATA
/SAVE GOOD DATA POINTER
/CDP TO BUFFER FIELD
/GET BAD DATA WORD
/HOME DP
/SAVE BAD WORD
/GET ADDRESS
/SAVE FOR PRINTER
/GET DATA READ

/COMPARE TO GOOD VALUE
/WERE THEY THE SAME
/YES, NO ERROR
/FIRST TIME PRINT???
/NO, JUST ADDRESS AND DATA
/GET CRC FLAG
/IF SET NO NON-RECOVERABLE.
/NO, GET NON-RECOVERABLE FLAG.
/UPDATE FOR ERROR RETURN
/ERROR DATA
/POINTER
/POINTER

/UPDATE BUFFER TALLY
/MORE WORDS TO CHECK
/GET ERROR INDICATOR;
/WAS THERE AN ERROR?
/NO, CLEAR CRC COUNTER
/CHECK FOR COMPUTER ERROR?
/ALL O.K.
/COMPUTER MUST BE DOWN, CHECKSUM
/OTHER ERRORS IN BUFFER

```

```

1712 0000      0000
1713 5272      JMP  NOERR
1714 0000      MSKER, 0
/ROUTINE TO GENERATE RANDOM NUMBERS
RANDOM, 0
1715 0000      CLA  CLL  IAC
1716 7301      TAD  RAD1
1717 1373      TAD  RAD2
1720 1374      TAD  RAD3
1721 1375      DCA  RAD1
1722 3373      RAL
1723 7004      TAD  RAD1
1724 1373      TAD  RAD2
1725 1374      TAD  RAD3
1726 1375      DCA  RAD2
1727 3374      RAL
1730 7004      TAD  RAD1
1731 1373      TAD  RAD2
1732 1374      TAD  RAD3
1733 1375      DCA  RAD3
1734 3375      TAD  RAD3
1735 1375      JMP  I  RANDOM
1736 5715      /EXIT, RANDOM NUMBER IN AC

/GENERATOR FOR RANDOM DATA
GNDAT, 0
1737 0000      CLA  CLL  IAC
1740 7301      TAD  MAN1
1741 1367      TAD  MAN2
1742 1370      CLL  RTL
1743 7106      DCA  MAN1
1744 3367      TAD  MAN2
1745 1370      RTR
1746 7012      TAD  MAN1
1747 1367      DCA  MAN2
1750 3370      TAD  MAN2
1751 1370      JMP  I  GNDAT
/ROUTINE TO SAVE RANDOM GENERATOR
STGEN, 0
1753 0000      TAD  MAN1
1754 1367      DCA  SAV1
1755 3371      TAD  MAN2
1756 1370      DCA  SAV2
1757 3372      JMP  I  STGEN
/ROUTINE TO RESET RANDOM GENERATOR
RSRAN, 0
1761 0000      TAD  SAV1
1762 1371      DCA  MAN1
1763 3367

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```

1764 1372      TAD      SAV2
1765 3370      DCA      RAN2
1766 5761      JMP I    RSKAN
/
1767 1234      RAN1,    1234
1770 5670      RAN2,    5670
/
1771 0000      SAV1,    0
1772 0000      SAV2,    0
1773 1234      RAD1,    1234
1774 5670      RAD2,    5670
1775 4321      RAD3,    4321
/
/
2000          PAGE
/ROUTINE TO WAIT FOR KEY FROM OPERATOR.
/
2000 0000      WAIT,    0
2001 6032      KCC
2002 6031      KSF
2003 5202      JMP ,=-1
2004 6036      KRB
2005 0234      AND      K177
2006 1016      TAD      K0200
2007 3235      DCA      CHKYN
/SAVE CHARACTER
2010 1022      TAD      22
/CHECK FOR CLASSIC
2011 0017      AND      K0400
/MASK CLASSIC BIT
2012 7650      SNA CLA
/CLASSIC=NON ZERO
2013 5226      JMP      WAIT1
2014 1235      TAD      CHKYN
/RESTORE CHAR. FOR CLASSIC
2015 6211      CDF      10
2016 3777*     DCA      C0CHAR
/SAVE CHARACTER.
2017 2776*     ISZ      INMODE
2020 1777*     TAD      C0CHAR
/GET BACK AC.
2021 6201      CDF      0
2022 4407      CLASIC
/CHECK FOR CLASSIC.
2023 4427      C0CNTR
/ROUTINE TO EXECUTE.
2024 7000      NOP
2025 7300      CLA CLL
/CLEAR CLASSIC AC RETURN
2026 1235      WAIT1,   TAD      CHKYN
/RESTORE CHARACTER
2027 6046      TSB
2030 6041      TBF
2031 5230      JMP ,=-1
2032 6042      TCF
2033 5000      JMP I    WAIT
/EXIT
2034 0177      K177,    0177
/ROUTINE TO CHECK FOR YES OR NO
/
2035 0000      CHKYN,    0
2036 3200      DCA      WAIT
/SAVE POINTER
2037 1235      TAD      CHKYN
/GET PC STORED
2040 3260      DCA      CHKPT0
/SAVE IT

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2041 1200      TAD      WAIT
2042 2235      ISZ      CHKYN
2043 7041      CIA
2044 1257      TAD      K0316
2045 7650      SNA CLA
/WAS IT A NO
2046 5635      JMP I    CHKYN
/YES
2047 1200      TAD      WAIT
2050 2235      ISZ      CHKYN
2051 7041      CIA
2052 1256      TAD      K0331
2053 7650      SNA CLA
/WAS IT A YES
2054 5635      JMP I    CHKYN
/YES
2055 5660      JMP I    CHKPT0
/WAS NEITHER
/
/
2056 0331      K0331,    0331
2057 0316      K0316,    0316
/ROUTINE TO CHECK DISK RUN POINTERS
/
2060 0000      CHKPT0, 0
2061 0064      AND      K0007
2062 1154      TAD      RUNPT0
2063 3200      DCA      WAIT
2064 1600      TAD I    WAIT
/GET RUN POINTER
2065 7640      SZA CLA
/RUN THIS DRIVE
2066 2260      ISZ      CHKPT0
/NO
2067 5660      JMP I    CHKPT0
/EXIT
/ROUTINE TO TEST FOR APT AND SET UP APPROPRIATE
/REGISTERS IN UN THE SYSTEM.
/
2070 0000      APT0,    0
2071 4424      CHK22
/TEST FOR APT
2072 5301      JMP ,+7
/YES
2073 4407      CLASIC
2074 4431      C0SWIT
2075 7000      NOP
2076 1355      TAD      K7000
2077 3775*     OCA      SKPNOP
2100 5351      JMP      EXAPT0
/EXIT
2101 1022      TAD      OP2
2102 0354      AND      K7377
/NOF CONSULE PACKAGE
2103 3022      DCA      UP2
2104 1355      TAD      K7000
2105 3774*     DCA      MYLAS+3
/NOF SWITCH REGISTER
/NO OPERATOR INTERVENTION ALLOWED
2106 1022      TAD      OP2
2107 0064      AND      K0007
/GET # OF DRIVES
2110 3111      OCA      TRASH1
2111 1022      TAD      OP2
2112 0015      AND      K0100
2113 7650      SNA CLA
/DOUBLE DRIVE = NON ZERO AC
2114 5325      JMP      M0USKS
/NO.

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2115 7301          CLL CLA IAC
2116 3060          DCA AMOUNT          /ONLY ONE DRIVE
2117 1111          TAD TRASH1          /GET DRIVE NUMBER
2120 1154          TAD MUNPOT
2121 3111          DCA TRASH1
2122 7340          CLL CLA CMA
2123 3511          DCA I TRASH1          /UU THIS DRIVE
2124 5342          JMP MEMSET
2125 1111          MOD8KS, TAD TRASH1
2126 7040          CMA
2127 3112          DCA TRASH2          /SAVE THE NUMBER OF DRIVES
2130 3111          DCA TRASH1
2131 1111          TAD TRASH1
2132 1154          TAD MUNPOT          /ESTABLISH DRIVE
2133 3113          DCA TRASH3
2134 7340          CLL CLA CMA
2135 3513          DCA I TRASH3          /UU THIS DRIVE
2136 2111          ISZ TRASH1
2137 2040          ISZ AMOUNT
2140 2112          ISZ TRASH2          /DONE?
2141 5331          JMP MOD8KS+4          /MORE TO DO
2142 1021          MEMSET, TAD Z1
2143 7012          RTR
2144 0064          AND K0007
2145 7104          CLL RAL
2146 7006          RTL
2147 7040          CMA
2150 3143          DCA MAXFLD          /NEGATIVE AMOUNT OF FIELDS.
2151 2270          EXAPTS, ISZ APT8
2152 2270          ISZ APT8
2153 5670          JMP I APT8
/
2154 7377          K7377, 7377
2155 7000          K7000, 7000
/
/THIS ROUTINE WILL NOTIFY APT OF AN ERROR.
/ONLY THE DRIVE IN ERROR IS ESTABLISHED.
/
2156 0000          AENROR, 0
2157 4424          CHK22
2160 7410          SKP          /CHECK FOR APT=8.
2161 5756          JMP I AENROR          /EXIT
2162 6002          IOF
2163 7200          CLA
2164 1115          TAD FOLD8K          /DRIVE NUMBER
2165 0064          AND K0007
2166 6201          CDF 00
2167 6272          CIF 70
2170 5772          JMP I K6520          /NOTIFY APT
2171 7402          HLT          /SOMETHING WENT WRONG IF IT GETS HERE
/
2172 6520          K6520, 6520
2174 1551
2175 0237
2176 1076

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2177 1075
2200 PAGE
/
/ROUTINE TO WRITE OR READ SECTORS SELECTED
/
2200 0000 DSKGO, 0
2201 7340 CLA CLL CMA
2202 3170 DCA FINTIM /SETUP FIRST TIME POINTER
2203 3156 DCA CRCFLG /CLEAR CRC FLAG
2204 1126 TAD CAREG /GET INITIAL CURRENT ADDRESS
2205 4452 LOCUR /LOAD CURRENT ADDRESS
2206 1127 TAD WCNEG
2207 3130 DCA FWNEG /SETUP FINAL WC
2210 1124 TAD INTDA /GET INITIAL STARTING SECTOR
2211 3111 DCA TRASH1 /SAVE
2212 1124 TAD INTDA /GET DISK ADDRESS
2213 0100 AND K7760 /MASK
2214 3112 DCA TRASH2 /SAVE
2215 1136 TAD INTCM /GET INITIAL COMMAND
2216 1600 TAD I DSKGO /GET READ OR WRITE
2217 4451 LOCMD /LOAD COMMAND
2220 1123 TAD CMNEG
2221 1075 TAD K1000
2222 3174 DCA SAVCM /MAKE READ ALL OR WRITE ALL
2223 1111 TAD TRASH1 /SAVE FOR SWITCH TO CONSECUTIVE MODE
2224 0066 AND K0017 /SECTOR TO DO
2225 1112 TAD TRASH2 /MASK
2226 4453 LDADD /ADD TO TRACK
2227 6001 IDN /LOAD AND GO
/TURN INTERRUPT ON
/
/ROUTINE TO CLEAN OR CHECK SUM BUFFER ON THE FLY
/
2230 3777* GOBAK, DCA TIMER2 /CLEAR LONG TIMER
2231 3142 DCA FNUSUM /CLEAR SUM CHECK
2232 4435 SETFLD /GET FIELD TO BUFFER
2233 3254 DCA CMNCDF /SAVE CDF
2234 1170 TAD FINTIM
2235 7650 SNA CLA /TIME TO GO
2236 5241 JMP STNWRK /YES!!!!
2237 4776* JMS TIME /WAIT FOR FIRST INTERRUPT
2240 5234 JMP .+4 /NOT HERE YET
2241 1117 STNWRK, TAD BUPTAL
2242 7041 CIA
2243 1130 TAD FWNEG
2244 7450 SNA /COMPARE TO SOFTWARE FINAL
2245 5274 JMP WRKDON /WAIT FOR DISK???
2246 7041 CIA /YES!!!!
2247 3175 DCA CLNBAK /SAVE DIFFERENCE
2250 1175 TAD CLNBAK
2251 7041 CIA
2252 1117 TAD BUPTAL
2253 3117 DCA CMNCDF /UPDATE BUFFER TALLY
2254 7402 HLT /CUF TO BUFFER FIELD
2255 1123 TAD CMNEG
2256 7700 SNA CLA /READ OR WRITE

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2257 5264      JMP      WASRD,      /HAS A READ!!
2260 3411      GOCLR, DCA I AUTO11 /HAS A WRITE, CLEAR BUFFER
2261 2175      ISZ      CLMBAK    /UPDATE TALLY
2262 5260      JMP      GOCLR     /MORE TO CLEAR
2263 5274      JMP      WRKDON    /DONE WITH SOME
2264 1142      WASRD, TAD      FNDSUM
2265 7100      GOCHK, CLL      /
2266 1411      TAD I AUTO11      /GET WORD
2267 7430      SZL
2270 7001      IAC
2271 2175      ISZ      CLMBAK    /UPDATE CLEAR POINTER
2272 5265      JMP      GOCHK    /MORE TO CHECKSUM
2273 3142      DCA      FNDSUM    /SAVE IT
2274 6201      WRKDON, CDF      0
2275 1117      TAD      BUFTAL
2276 7650      SNA CLA          /LAST WORD DONE????
2277 5302      JMP      DSKEK    /EXIT
2300 4776      JMS      TIME    /TIME AND WAIT
2301 5241      JMP      STRWRK  /WAIT FOR INT, OR DONE!!!!
2302 2200      DSKEK, ISZ      DSKG
2303 5600      JMP I DSKGO      /EXIT
/
/INTERUPT SERVICE
/
2304 6741      RETURN, DSKP     /DISK SKIP IOT
2305 5353      JMP      NODSKP   /NOT THE DISK
2306 2111      ISZ      TRASH1   /UPDATE SECTOR NUMBER
2307 7000      NOP              /IT WON'T WORK WITHOUT IT!
2310 1114      TAD      UPDATE   /UPDATE WORD COUNT
2311 1130      TAD      FWRG     /
2312 3130      DCA      FWRG
2313 6745      STATUS, DRST     /READ STATUS
2314 1073      TAD      K4000
2315 7440      SZA
2316 5337      JMP      STATER   /ONLY DONE FLAG?
2317 1130      TAD      FWRG     /STATUS ERROR
2320 7650      SNA CLA          /LAST TRANSFER?
2321 5365      JMP      TRDONE   /TRANSFER IS DONE
2322 3170      DCA      FINTIM   /CLEAR FIRST TIME POINTER
2323 1174      TAD      SAVCM    /GET READ OR WRITE COMMAND
2324 6746      RDLWRL, DLDC     /LOAD COMMAND REGISTER
2325 1111      TAD      TRASH1   /GET SECTOR TO DO
2326 0066      AND      K0017    /MASK OFF
2327 1112      TAD      TRASH2   /ADD IN TRACK
2328 6743      LOGGO, DLG      /LOAD DISK ANFD GO
2331 1107      RETRN, TAD      SVLNK /GET LINK
2332 7110      CLL RAK
2333 1166      TAD      SAVAC    /GET AC
2334 6244      RHF              /RESTORE FIELDS
2335 6001      ION              /TURN INTERRUPT ON
2336 5400      JMP I 0          /EXIT
2337 4775      STATER, JMS      SETREG /SETUP REGISTERS!
2340 1123      TAD      CMREG
2341 7710      SRA CLA          /WRITE OR READ
2342 7001      IAC              /WRITE

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2343 7801      IAC
2344 3347      DCA      .+3
2345 1137      TAD      STATRY      /MODIFY HEADER POINTER
2346 4446      ERRORR      /GET TRY POINTER
2347 0000      0000      /PRINT MESSAGE
2350 7770      7770      /MODIFIED HEADER POINTER
2351 2200      ISZ      DSKGO      /MESSAGE POINTER
2352 5302      JMP      DSKEK      /UPDATE FOR ERROR
2353 3374      NODSKP, DCA      TIMERS      /EXIT
2354 2374      ISZ      TIMERS
2355 5354      JMP      .-1
2356 4407      CLASIC      /WAIT FOR DISK TO STOP.
2357 4440      CCKCPA      /CHECK FOR CLASSIC.
2360 7000      NOP      /ROUTINE TO EXECUTE.
2361 6031      KSF
2362 4576      INTER2, JMS I      BGMLT      /KEYBOARD FLAG??
2363 6032      KCC      /ILLEGAL INTERRUPT
2364 5331      JMP      RETRN
2365 4775      TROONE, JMS      SETREG      /EXIT BACK.
2366 3170      DCA      FINTIM      /SETUP REGISTERSI
2367 1167      TAD      SVLNK      /CLEAR FIRST TIME POINTERI
2370 7110      CLL      RAM
2371 1166      TAD      SAVAC      /REPLACE LINK
2372 6244      RMF      /REPLACE AC
2373 5400      JMP I      0      /RESTORE MEMORY FIELDS+ FLAGS
                                /RETURN TO BACK GROUND
/
2374 0000      /TIMER3, 0
/
2375 1553
2376 3123
2377 3141
2400 2400      PAGE
/
/ROUTINE TO GET ONE IN OCTAL
OCT1, 0
2400 0000
2401 4444      RECEIV      /RECEIVE
2402 3354      DCA      LOAD      /SAVE IT
2403 1600      TAD I      OCT1      /GET LIMITS
2404 0064      AND      K0007      /MASK
2405 1067      TAD      K0260
2406 7141      CLL      CIA
2407 1354      TAD      LOAD
2408 7620      BNL      CLA      /GET INPUT
2409 5226      JMP      INERR      /IN LIMITS????
2410 1600      TAD I      OCT1      /NO, ERROR EXIT
2411 0014      AND      K0070      /GET LIMITS
2412 7110      CLL      RAM      /MASK
2413 7012      RTR
2414 1067      TAD      K0260
2415 7040      CMA
2416 1354      TAD      LOAD
2417 7630      BZL      CLA      /GET INPUT
2418 5226      JMP      INERR      /IN LIMITS????
2419 1354      TAD      LOAD      /NO, ERROR
                                /GET INPUT

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2424 0064      AND      K0007      /MASK
2425 2200      ISZ      OCT1
2426 2200      INERR,  ISZ      OCT1
2427 5600      JMP      OCT1      /6000 EXIT
/
/ROUTINE TO RECEIVE FOUR OCTAL
/
2430 0000      OCT4,  0
2431 1106      TAD      M4
2432 3341      DCA      RDST      /SETUP COUNTER
2433 3350      DCA      LDCA      /START WITH 0
2434 4432      ONEIN      /RECEIVE ONE OCTAL
2435 0070      0070      /LIMITS
2436 5630      JMP      OCT4      /ENRUR EXIT
2437 1350      TAD      LDCA      /GET LAST
2440 2341      ISZ      RDST      /UPDATE COUNTER
2441 7410      SKP
2442 5246      JMP      .+4      /EXIT
2443 7004      HAL
2444 7006      RTL
2445 5233      JMP      OCT4+3
2446 2230      ISZ      OCT4
2447 5630      JMP      OCT4      /EXIT OCTAL IN AC
/
/ROUTINE TO UPDATE AND CHECK FOR PASS COMPLETE
/
2450 0000      CKTIM,  0
2451 1115      TAD      POLDISK
2452 0064      AND      K0007      /SETUP CURRENT DRIVE #
2453 3341      DCA      RDST      /POINTER
2454 1341      TAD      RDST
2455 1152      TAD      TIMPOT      /GET TIME POINTER
2456 3354      DCA      LDAD      /SAVE IT
2457 7301      CLA CLL IAC      /ONE FOR 0
2460 1147      TAD      CONSEC      /GET AMOUNT DONE
2461 1754      TAD I LDAD      /ADD IN AMOUNT COMPLETED SO FAR
2462 3754      DCA I LDAD      /SAVE IT
2463 7620      SNL CLA      /LINK UP???
2464 5650      JMP I CKTIM      /NO, EXIT
2465 4440      RANGEN
2466 3777      DCA      RAN1      /GET RANDOM NUMBER
2467 4440      RANGEN      /RE-PRIME GENERATOR
2470 3776      DCA      RAN2      /GET RANDOM NUMBER
2471 7100      CLL
2472 1354      TAD      LDAD      /RE-PRIME GENERATOR
2473 1013      TAD      K0004
2474 3354      DCA      LDAD
2475 2754      ISZ I LDAD      /SECOND TIME POINTER
2476 1754      TAD I LDAD      /UPDATE IT
2477 1144      TAD      MAXTIM      /GET COUNT
2500 7620      SNL CLA      /AUD IN FUDGE FACTOR
2501 5650      JMP I CKTIM      /PASS COMPLETE???
2502 3754      DCA I LDAD      /NO, EXIT
2503 1341      TAD      RDST      /ZERO SECOND COUNTER
2504 7040      CMA

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2505 3341      DCA      RDST
2506 1362      TAD      CHPPOT      /SETUP COUNTER
2507 1062      TAD      K0003      /ADD IN POINTER
2510 2341      ISZ      RDST
2511 5307      JMP      .-2      /COMPUTE BUFFER
2512 3341      DCA      RDST      /SAVE ADDRESS POINTER
2513 7340      CLA CLL CMA
2514 2741      ISZ I RDST      /UPDATE PASS COMPLETE POINTER
2515 7610      SKP CLA
2516 3741      DCA I RDST      /HOLD AT 7777
2517 4457      CRLF
2520 4455      PRINTER      /PRINT "DISK"
2521 3477      MES17
2522 1115      TAD      POLDISK      /GET DISK POLE NUMBER
2523 0064      AND      K0007      /MASK
2524 1067      TAD      K0260
2525 4445      TYPE      /TYPE DISK NO,
2526 7340      CLA CLL CMA
2527 4455      PRINTER      /PRINT "PASS COMPLETE"
2530 3502      MES18
2531 4406      LAS
2532 0015      AND      K0100      /MASK
2533 7650      SNA CLA      /PASS COMPLETE DISCONNECT???
2534 5337      JMP      .+3      /NO WAY!!!!
2535 4430      DISCON      /DUMP DRIVE
2536 5775      JMP      RUN      /MORE TO TEST!!!!
2537 4774      JMS      TPSTA      /STATUS-COMplete TYPEOUT
2540 5650      JMP I CKTIM      /EXIT
/
/SUBROUTINE TO READ STATUS REGISTER
/
2541 0000      RDST,  0
2542 6745      IOTS,  DRST      /READ STATUS IOT
2543 7410      SKP
2544 4576      ERMLT5, JMS I BGHLT      /SKIP TRAP
2545 3121      DCA      STREG      /SAVE RESULTS
2546 1121      TAD      STREG
2547 5741      JMP I RDST      /EXIT
/
/SUBROUTINE TO LOAD CURRENT ADDRESS REGISTER
/
2550 0000      LDCA,  0
2551 6744      IOT4,  DLCA      /LOAD CURRENT ADDRESS IOT
2552 4425      KTICK      /NOTIFY APT
2553 5750      JMP I LDCA      /EXIT
/
/SUBROUTINE TO LOAD TRACK ADDRESS REGISTER
/
2554 0000      LDAD,  0
2555 3125      DCA      DAMEG
2556 1125      TAD      DAMEG
2557 6743      IOT3,  DLAG      /LOAD DISK ADDRESS REGISTER
2558 5754      JMP I LDAD      /EXIT
2561 4576      ERMLT3, JMS I BGHLT      /ENRUR SKIP TRAP

```

```

2562 3541 /
2574 3800 CMPPOT, DUCMP-3
2575 0600
2576 1770
2577 1767
2600 2600 PAGE
/
/ROUTINE TO GET RANDOM OR OPERATOR DATA
/
2600 0000 RNWRD, 0
2601 7402 SWDAT, HLT /MODIFIED SWITCH
2602 5600 JMP I RNWRD /EXIT
2603 0201 CDF 0 /HOLD CDF
2604 1412 TAD I AUTO12 /GET DATA
2605 7402 RECD, HLT /BUFFER CDF
2606 2116 ISZ OPMTAL /UPDATE TALLY
2607 5600 JMP I RNWRD /EXIT
2610 3220 DCA PRINT /SAVE WORD
2611 1105 TAD M12
2612 3116 DCA OPMTAL /REPLACE TALLY
2613 7340 CLA CLL CMA
2614 1151 TAD DATPOT
2615 3012 DCA AUTO12 /REPLACE AUTO INDEX
2616 1220 TAD PRINT /GET SAVED WORD
2617 5600 JMP I RNWRD /EXIT
/
/ROUTINE TO TYPE
/
2620 0000 PRINT, 0
2621 3237 DCA DUMP /STORE AC VALUE
2622 4424 CHK22 /SEE IF ON APT
2623 5235 JMP PREXIT /NO, EXIT
2624 1237 TAD DUMP /RETURN AC,
2625 4407 CLASIC /CHECK FOR CLASSIC.
2626 4435 C8TYPE /ROUTINE TO EXECUTE,
2627 7410 SKP
2630 5620 JMP I PRINT /EXIT,
2631 6046 TLF
2632 6041 TSF
2633 5232 JMP ,=1
2634 6042 TCF
2635 7200 PREXIT, CLA
2636 5620 JMP I PRINT
/
/ROUTINE TO DUMP AND REPORT DISK STATUS
/
2637 0000 DUMP, 0
2640 4424 CHK22 /CHECK FOR APT
2641 5637 JMP I DUMP
2642 4455 PRINTER /PRINT "DISK "
2643 3477 MES17
2644 1115 TAD POLOSK
2645 0064 AND K0007 /SETUP CURRENT DRIVE #
2646 3200 DCA RNWRD /SAVE

```

```

2647 1200 TAD RNWRD /GET DISK NUMBER
2650 1067 TAD K0260
2651 4445 TYPE /TYPE DISK NUMBER
2652 7340 CLA CLL CMA
2653 4455 PRINTER /PRINT "DISCONNECTED!"
2654 3445 MES15
2655 4777 JMS TPSTA /TYPE STATUS REPORT
2656 1200 TAD RNWRD
2657 1154 TAD RUNPOT
2660 3200 DCA RNWRD /SAVE POINTER ADDRESS
2661 3600 DCA I RNWRD /CLEAR RUN POINTER
2662 3200 DCA RNWRD
2663 1106 TAD M4
2664 3220 DCA PRINT /CHECK FOR MORE POINTER
2665 1200 TAD RNWRD
2666 4436 SELCHK /CHECK SELECT POINTERS
2667 7610 SKP CLA /DISK NOT HERE
2670 5637 JMP I DUMP /MORE AVAILABLE
2671 2200 ISZ RNWRD
2672 2220 ISZ PRINT /UPDATE POINTERS
2673 5265 JMP ,=6
2674 4457 CRLF
2675 4455 PRINTER /PRINT "DISK"
2676 3477 MES17
2677 7340 CLA CLL CMA
2700 4455 PRINTER /PRINT "SYSTEM DOWN"
2701 3455 MES16
2702 4576 NOO3KB, JMS I BGHLT /ERROR, NO DISK AVAILABLE
/
/ROUTINE TO SETUP FIELD TO BUFFER+ AUTO11+ BUFFER TALLY
/
2703 0000 STFLD, 0
2704 7041 CIA
2705 1127 TAD WMEG
2706 3117 DCA BUPTAL
2707 7340 CLA CLL CMA
2710 1126 TAD CANEG /GET INITIAL CA
2711 3011 DCA AUTO11 /SAVE
2712 1157 TAD DATFLG /GET DATA FLAG
2713 7650 SNA CLA /WAS IT SET???
2714 5322 JMP ,+6 /NO, USE REGULAR
2715 1105 TAD M12
2716 3116 DCA OPMTAL /SETUP SPECIAL TALLY
2717 7340 CLA CLL CMA
2720 1151 TAD DATPOT
2721 3012 DCA AUTO12 /SETUP SPECIAL AUTO INDEX
2722 1136 TAD INTCM /GET LAST COMMAND
2723 0014 AND K0070 /MASK FIELD BITS
2724 1103 TAD KCDF /MAKE BUFFER CDF
2725 3205 DCA RECD /SETUP SPECIAL CDF
2726 1205 TAD RECD /GET BACK CDF
2727 5703 JMP I STFLD /EXIT, FIELD IN AC
/
/ROUTINE TO CHANGE DEVICE IOT CODES
/

```

```

2730 4407  CHANG, CLASIC      /CHECK FOR CLASSIC.
2731 4431  C88WIT           /ROUTINE TO EXECUTE.
2732 7000  NOP
2733 4406  LAS
2734 0071  AND  A0770        /GET SWITCHES
2735 3776  DCA  LDCH         /MASK 3-0
2736 1360  TAD  CHNPOT       /SAVE DESIRED CODE
2737 3111  DCA  TRASH1      /POINTER
2740 1357  TAD  CCNTR1      /ADDRESS POINTER
2741 3112  DCA  TRASH2      /AMOUNT TO DO
2742 1511  CHANGR, TAD I    /SETUP COUNTER
2743 3113  DCA  TRASH3      /GET ADDRESS POINTER
2744 1513  TAD I    TRASH3   /SAVE ADDRESS
2745 0072  AND  A7007       /GET OLD CODE
2746 1776  TAD  LDCH         /MASK OFF OLD CODE
2747 3513  DCA I    TRASH3   /ADD IN DESIRED CODE
2750 2111  ISZ  TRASH1      /RESTORE
2751 2112  ISZ  TRASH2      /UPDATE POINTER
2752 5342  JMP  CHANGR      /UPDATE CHANGE COUNTER
2753 4407  CLASIC          /MORE TO CHANGE
2754 4436  C8ERR           /CHECK FOR CLASSIC.
2755 7402  CHNHLT, HLT      /ROUTINE TO EXECUTE.
2756 5775  JMP  BGN        /IOTS CHANGED, HIT CONTINUE OR
                                   /IF ON CONSOLE PACKAGE
                                   /CONTROL E TO START PROGRAM.

```

```

2757 7765  / CCNTR1, 7765

```

```

2760 2761  / CHNPOT, CHNPOT+1
2761 2304  RETURN
2762 2313  STATUS
2763 2324  ROLWRL
2764 2330  LODGO
2765 0554  IOT0
2766 0752  IOT1
2767 0561  IOT2
2770 2557  IOT3
2771 2551  IOT4
2772 2542  IOT5
2773 0545  IOT6

```

```

2775 0200
2776 0542
2777 3000
3000

```

PAGE

/ROUTINE TO TYPE STATUS REPORT

```

3000 0000  TPSTA, 0
3001 4424  CHK22
3002 5600  JMP I  TPSTA
3003 4457  CRLF
3004 4455  PRINTER
3005 3372  MEB7
3006 1107  TAD  M10
3007 3245  DCA  TSAVE1

```

/PRINT "DISK HARD SOFT COMP"

/MAXIMUM TO DO

```

3010 3246  DCA  TSAVE2
3011 3247  DCA  TSAVE3
3012 1246  CHKRES, TAD  TSAVE2  /CLEAR SOME COUNTERS
3013 1062  TAD  K0003
3014 3246  DCA  TSAVE2
3015 1246  TAD  TSAVE2
3016 1153  TAD  STAPOT
3017 3251  DCA  TSAVE5
3020 1247  TAD  TSAVE3
3021 4436  SELCHK
3022 5241  JMP  NOTSTA
3023 4457  CRLF
3024 4431  SPACE
3025 1247  TAD  TSAVE3
3026 1067  TAD  K0260
3027 4445  TYPE
3030 4431  SPACE
3031 4431  SPACE
3032 7346  CLA CLL CMA RTL
3033 3250  DCA  TSAVE4
3034 1651  TAD I  TSAVE5
3035 4456  OCTEL
3036 2251  ISZ  TSAVE5
3037 2250  ISZ  TSAVE4
3040 5234  JMP  ,=4
3041 2247  NOTSTA, ISZ  TSAVE3
3042 2245  ISZ  TSAVE1
3043 5212  JMP  CHKRES
3044 5600  JMP I  TPSTA

```

/LOCATION OF DISK STATUS

/CHECK RUN POINTER
/DISK NOT RUNNING/SPACE OUT ONE
/GET DISK NO./SPACE OUT ONE
/SPACE OUT ONE/COUNTER FOR FOUR WORDS
/GET STATUS
/TYPE IT

/UPDATE DRIVE NUMBER

/MORE TO REPORT
/EXIT

```

3045 0000  TSAVE1, 0
3046 0000  TSAVE2, 0
3047 0000  TSAVE3, 0
3050 0000  TSAVE4, 0
3051 0000  TSAVE5, 0

```

/ROUTINE TO RECALIBRATE SELECTED DRIVE
/DISCONNECT DRIVE ON ERROR

```

3052 0000  RESTOR, 0

```

```

3053 0063  AND  K0006
3054 3200  DCA  TPSTA
3055 1077  TAD  K7700
3056 3341  DCA  TIMER2
3057 2340  ISZ  TIMER1
3060 5257  JMP  ,=1
3061 2341  ISZ  TIMER2
3062 5257  JMP  ,=3
3063 1200  TAD  TPSTA
3064 4451  LDCH
3065 7326  CLA CLL CML RTL
3066 4454  CLRALL
3067 4450  DBKSKP
3070 5267  JMP  ,=1
3071 4447  ROSTAT

```

/SAVE DRIVE NUMBER

/SETUP COUNTER

/WAIT FOR DISK TO COOL OFF

/CURRENT DRIVE
/LOAD COMMAND
/ENABLE RECALIBRATE BIT
/"RECALIBRATE"
/DISK SKIP IOT
/WAIT FOR FIRST DONE FLAG
/MEAU STATUS

```

3072 7500      SMA
3073 5311      JMP      RESERR
3074 0076      AND      K1777
3075 7640      SZA CLA
3076 5311      JMP      RESERR
3077 4454      RESTA, CLRALL
3100 1016      TAD      K0200
3101 1200      TAD      TPSTA
3102 4451      LDCHD
3103 4450      DSKSKP
3104 5303      JMP      .-1
3105 4447      RDSTAT
3106 1073      TAD      K4000
3107 7650      SNA CLA
3110 5652      JMP I     RESTOR
3111 7300      RESERR, CLA CLL
3112 4446      ERROR
3113 0003      0003
3114 7500      7500
3115 4457      CRLF
3116 4457      CRLF
3117 4455      PRINTER
3120 3174      MES19
3121 4430      DISCON
3122 5652      JMP I     RESTOR

/ROUTINE TO TIME AND WAIT
3123 0000      TIME, 0
3124 2340      ISZ      TIMER1
3125 5723      JMP I     TIME
3126 2341      ISZ      TIMER2
3127 5723      JMP I     TIME
3130 4576      INTER1, JMS I 0GHLT

/ROUTINE TO COMBINE ERROR HALTS,
/WHEN THE COMPUTER HALTS THE AC
/WHICH EQUAL THE PC ON THE FAILING
/HAUT INSTRUCTION,
3131 0000      BIGHLT, 0
3132 7300      CLA CLL
3133 1331      TAD      BIGHLT
3134 4407      CLASIC
3135 4436      CBERR
3136 7402      BIGSTP, HLT
3137 5332      JMP      .-5

3140 0000      TIMER1, 0
3141 0000      TIMER2, 0

/ROUTINE TO TYPE OUT DATA INFORMATION
3142 0000      TYPDAT, 0
3143 4455      PRINTER

```

/PRINT"RECALIBRATE ERROR DISCONNECT"

/DISCONNECT DISK
/MORE DISK AVAILABLE

/EXIT

/EXIT

/NO INTERRUPT OCCURRED, I GUESS!

/LOAD AC WITH PC.
/CHECK FOR CLASSIC.
/ROUTINE TO EXECUTE.
/AC=PC.
/NON-RECOVERABLE.

/PRINT "A01"

```

3144 3235      TEXAS
3145 1131      TAD      ASREG
3146 4456      OCTEL
3147 7340      CLA CLL CMA
3150 4455      PRINTER
3151 3237      TEXWA
3152 1132      TAD      WAREG
3153 4456      OCTEL
3154 7340      CLA CLL CMA
3155 4455      PRINTER
3156 3241      TEXAD
3157 1133      TAD      ADREG
3160 4456      OCTEL
3161 7340      CLA CLL CMA
3162 4455      PRINTER
3163 3243      TEXDG
3164 1134      TAD      DGREG
3165 4456      OCTEL
3166 7340      CLA CLL CMA
3167 4455      PRINTER
3170 3245      TEXDB
3171 1135      TAD      DBREG
3172 4456      OCTEL
3173 5742      JMP I     TYPDAT

3174 2205      MES19, TEXT  "RECALIBRATE ERROR DISCONNECT!"
3175 0301
3176 1411
3177 0222
3200 0124
3201 0540
3202 0522
3203 2217
3204 2240
3205 0411
3206 2303
3207 1716
3210 1605
3211 0324
3212 4100

/
3213 2003      TEXPC, TEXT  "PC:"
3214 7200
3215 2324      TEXST, TEXT  "ST:"
3216 7200
3217 0530      TEXEX, TEXT  "EX:"
3220 7200
3221 0315      TEXCM, TEXT  "CM:"
3222 7200
3223 1101      TEXIA, TEXT  "IA:"
3224 7200
3225 0401      TEXDA, TEXT  "DA:"
3226 7200
3227 0301      TEXCA, TEXT  "CA:"
3230 7200

```

3231	2703	TEXWC, TEXT	"WC1"
3232	7200		
3233	0027	TEXFW, TEXT	"FW1"
3234	7200		
3235	0123	TEXAS, TEXT	"AS1"
3236	7200		
3237	2701	TEXWA, TEXT	"WA1"
3240	7200		
3241	0104	TEXAD, TEXT	"AD1"
3242	7200		
3243	0407	TEXDG, TEXT	"DG1"
3244	7200		
3245	0402	TEXDB, TEXT	"DB1"
3246	7200		
/			
3247	2205	ERTX1, TEXT	"READ STATUS"
3250	0104		
3251	4023		
3252	2401		
3253	2425		
3254	2300		
3255	2722	ERTX2, TEXT	"WRITE STATUS"
3256	1124		
3257	0540		
3260	2324		
3261	0124		
3262	2523		
3263	0000		
3264	2205	ERTX3, TEXT	"RECALIBRATE STATUS"
3265	0301		
3266	1411		
3267	0222		
3270	0124		
3271	0540		
3272	2324		
3273	0124		
3274	2523		
3275	0000		
3276	0411	ERTX4, TEXT	"DISK DATA"
3277	2313		
3300	4004		
3301	0124		
3302	0100		
/			
3303	4005	MESS, TEXT	" ERROR"
3304	2222		
3305	1722		
3306	0000		
3307	2213	MESS1, TEXT	"MK0E/RK0L DATA RELIABILITY"
3310	7005		
3311	5722		
3312	1370		
3313	1440		
3314	0401		
3315	2401		

3316	4022		
3317	0514		
3320	1101		
3321	0211		
3322	1411		
3323	2431		
3324	0000		
3325	0530	MESS2, TEXT	"EXERCISE"
3326	0522		
3327	0311		
3330	2305		
3331	0000		
3332	4004	MESS3, TEXT	" DISK"
3333	1123		
3334	1300		
3335	1617	MESS4, TEXT	"NON-RECOVERABLE "
3336	1655		
3337	2205		
3340	0317		
3341	2605		
3342	2201		
3343	0214		
3344	0540		
3345	0000		
3346	0530	MESS5, TEXT	"EXTENDED R/W MEMORY(0-7)?"
3347	2405		
3350	1604		
3351	0504		
3352	4022		
3353	5727		
3354	4015		
3355	0515		
3356	1722		
3357	3150		
3360	6055		
3361	6751		
3362	7700		
/			
3363	0103	MESS6, TEXT	"ACCEPT MODE?"
3364	0305		
3365	2024		
3366	4015		
3367	1704		
3370	0577		
3371	0000		
3372	0423	MESS7, TEXT	"DSK HARD SOFT COMP"
3373	1340		
3374	1001		
3375	2204		
3376	4023		
3377	1706		
3400	2440		
3401	0317		
3402	1520		

3403	0000		
3404	0611	ME88, TEXT	"FIELD?"
3405	0514		
3406	0477		
3407	0000		
3410	2422	ME89, TEXT	"TRACK?"
3411	0103		
3412	1377		
3413	0000		
3414	0530	ME810, TEXT	"EXTRA SECTORS?"
3415	2422		
3416	0140		
3417	2305		
3420	0324		
3421	1722		
3422	2377		
3423	0000		
3424	0214	ME811, TEXT	"BLOCK LENGTH?"
3425	1703		
3426	1340		
3427	1405		
3430	1607		
3431	2410		
3432	7700		
3433	0401	ME813, TEXT	"DATA?"
3434	2401		
3435	7700		
3436	0122	ME814, TEXT	"ARE YOU SURE?"
3437	0540		
3440	3117		
3441	2540		
3442	2325		
3443	2205		
3444	7700		
3445	4004	ME815, TEXT	" DISCONNECTED!"
3446	1123		
3447	0317		
3450	1616		
3451	0503		
3452	2405		
3453	0441		
3454	0000		
3455	2331	ME816, TEXT	"SYSTEM SHUT DOWN, NO DISKS TO RUN!"
3456	2324		
3457	0515		
3460	4023		
3461	1025		
3462	2440		
3463	0417		
3464	2716		
3465	5440		
3466	1617		
3467	4004		
3470	1123		
3471	1323		

3472	4024		
3473	1740		
3474	2225		
3475	1641		
3476	0000		
3477	0411	ME817, TEXT	"DISK "
3500	2313		
3501	4000		
3502	4020	ME818, TEXT	" PASS COMPLETE!"
3503	0123		
3504	2340		
3505	0317		
3506	1520		
3507	1405		
3510	2405		
3511	4100		
/			
3512	0000	D8K00, 0	
3513	0000	D8K10, 0	
3514	0000	D8K20, 0	
3515	0000	D8K30, 0	
3516	0000	D8K40, 0	
3517	0000	D8K50, 0	
3520	0000	D8K60, 0	
3521	0000	D8K70, 0	
/			
3522	0000	D0TM1, 0	
3523	0000	D1TM1, 0	
3524	0000	D2TM1, 0	
3525	0000	D3TM1, 0	
3526	0000	D4TM1, 0	
3527	0000	D5TM1, 0	
3530	0000	D6TM1, 0	
3531	0000	D7TM1, 0	
3532	0000	D0TM2, 0	
3533	0000	D1TM2, 0	
3534	0000	D2TM2, 0	
3535	0000	D3TM2, 0	
3536	0000	D4TM2, 0	
3537	0000	D5TM2, 0	
3540	0000	D6TM2, 0	
3541	0000	D7TM2, 0	
/			
3542	0000	D0MRD, 0	
3543	0000	D0SOP, 0	
3544	0000	D0CMP, 0	
3545	0000	D1MRD, 0	
3546	0000	D1SOP, 0	
3547	0000	D1CMP, 0	
3550	0000	D2MRD, 0	
3551	0000	D2SOP, 0	
3552	0000	D2CMP, 0	

3553 0000 D3MRD, 0
3554 0000 D38OF, 0
3555 0000 D3CMP, 0
3556 0000 D4MRD, 0
3557 0000 D48OF, 0
3560 0000 D4CMP, 0
3561 0000 D5MRD, 0
3562 0000 D58OF, 0
3563 0000 D5CMP, 0
3564 0000 D6MRD, 0
3565 0000 D68OF, 0
3566 0000 D6CMP, 0
3567 0000 D7MRD, 0
3570 0000 D78OF, 0
3571 0000 D7CMP, 0
/
3572 0000 FLOFLG, 0
3573 0000 TRKFLG, 0
3574 0000 SECFLG, 0
3575 0000 HLFFLG, 0
/
/
/
3600 PAGE
/
3600 STNBUF=,
/
SSSS

0000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0100 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111110
0200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0300 11111111 11111111 11111111 11111111 11111111 11111111 11110011 11111111
0400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0500 11111111 11111111 11111111 11111111 11111111 11111111 11110000 01111111
0600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0700 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11011111
1000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1100 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1500 11111111 11111111 11111111 11111111 11111111 11111111 11111111 10001111
1600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1700 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111100
2000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2100 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11101111
2200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2500 11111111 11111111 11111111 11111111 11111111 11111111 11110000 00001111
2600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2700 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11110111
3000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
3100 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
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3500 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111100
3600
3700

4000
41004200
43004400
45004600
47005000
51005200
53005400
55005600
57006000
61006200
63006400
65006600
67007000
71007200
73007400
75007600
7700

AB170	0750	C8GET	0624	CONCUR	0715
AB770	0071	C8HANG	1122	CONSEC	0147
AT007	0072	C8INQU	4437	CONSOL	0000
AC0CA	0362	C8OCTA	4432	CRCCNT	0155
ACL	7701	C8PAS3	4424	CRCLFG	0156
ACSAVE	1345	C8PAUS	4441	CRLF	4457
ADREG	0133	C8PRNT	4430	D8CMP	3544
AEERROR	2156	C8RDP3	0606	D8HRU	3542
AGAIN	1346	C8RETO	0614	D8SOF	3543
ALLAGN	0255	C8RETR	0536	D8TM1	3522
AMOUNT	0060	C8SETD	0613	D8TM2	3532
APT8	2070	C8SET8	0535	D1CMP	3547
ASKNX1	0345	C8SWIT	4431	D1HRU	3545
ASKNX2	0406	C8SWST	0745	D1SOF	3546
ASKNX3	0425	C8TMP1	1021	D1TM1	3523
ASKNX5	0464	C8TTYI	4426	D1TM2	3533
ASKSUR	0513	C8TYPE	4435	D2CMP	3552
ASREG	0131	CAF	6007	D2HRU	3550
AUTO10	0010	CAREG	0126	D2SOF	3551
AUTO11	0011	CENR1	2757	D2TM1	3524
AUTO12	0012	CHANG	2750	D2TM2	3534
BADMLT	1707	CHANGR	2742	D3CMP	3555
BGHLT	0176	CHEK22	0523	D3HRU	3553
BGN	0200	CHK22	4424	D3SOF	3554
BGNBUF	0146	CHKCLA	1200	D3TM1	3525
BIGHLT	3131	CHKPOT	2000	D3TM2	3535
BIGSTP	3136	CHKRE8	3012	D4CMP	3560
BRKRET	0363	CHKSAV	0141	D4HRU	3556
BUFPNT	1400	CHKYN	2035	D4SOF	3557
BUFTAL	0117	CHNCDF	2254	D4TM1	3526
BYRETR	0506	CHNHLT	2755	D4TM2	3536
C8BY1	0230	CHNPOT	2760	D5CMP	3563
C8BY2	1300	CKCOUT	0232	D5HRU	3561
C8BY3	1061	CKT1M	2450	D5SOF	3562
C8BY4	0515	CLAFLD	1404	D5TM1	3527
C8BY5	1116	CLABIC	4407	D5TM2	3537
C8CHAR	1075	CLASTK	1514	D6CMP	3566
C8CKP	1022	CLDR	0560	D6HRU	3564
C8CKPA	4440	CLKCNT	0130	D6SOF	3565
C8CKSW	4425	CLRALL	4454	D6TM1	3550
C8CNTR	4427	CLRBK	0175	D6TM2	3540
C8CONT	1145	CLRTRN	1315	D7CMP	3571
C8CRLF	4433	CHPPOT	2502	D7HRU	3567
C8DO1	0310	CHREG	0123	D7SOF	3570
C8DO10	1262	CNTRLC	0551	D7TM1	3551
C8DO11	0607	CNTRLD	0600	D7TM2	3541
C8DO2	1033	CNTRLE	0545	DAREL	0125
C8DO3	0350	CNTRLL	0537	DAT1	0756
C8DO4	1006	CNTRLQ	0500	DAT10	0767
C8DO7	0527	CNTRLR	0511	DAT11	0770
C8ECMO	4434	CNTRL3	0521	DAT12	0771
C8ERR	4436	CNTVAL	0252	DAT2	0757

DAT3	0760
DAT4	0761
DAT5	0762
DAT6	0763
DAT7	0764
DAT8	0765
DAT9	0766
DATFLG	0157
DATPOT	0151
DATTRY	0140
DBREG	0135
DCLR	6742
DGREG	0134
DISCON	4430
DISKGO	4441
DLAG	6743
DLCA	6744
DLDC	6746
DLSC	6740
DOCNT	0247
DOMEO	1262
DONEA	0426
DOPACK	0212
DOSET	0251
DRST	6745
D8K00	3512
D8K10	3513
D8K20	3514
D8K30	3515
D8K40	3516
D8K50	3517
D8K60	3520
D8K70	3521
D8KEX	2302
D8KGO	2200
D8KP	6741
D8KSKP	4450
DTCHK	1600
UTR1	1636
DUMP	2637
ENDIT	0742
ERFLG	0165
ERHLT0	0556
ERHLT2	0563
ERHLT3	2561
ERHLT5	2544
ERHLT6	0547
ERR1	0736
ERRM15	1520
ERR0	1200
ERRDEX	1555
ERROR	4446

ERTX1	3247	K0001	0061	M12	0105	NTWRKS	1710
ERTX2	3255	K0003	0062	M4	0106	UCT1	2400
ERTX3	3264	K0004	0013	MANUAL	0322	UCT4	2430
ERTX4	3276	K0006	0063	MAXFLD	0143	OCTEL	4456
ESAVE	1371	K0007	0064	MAXTIM	0144	ONEIN	4432
EXAPT8	2151	K0010	0065	MAXTRK	0145	OP1	0021
EXIT	1504	K0017	0066	MEMSET	2142	OP2	0022
EXITA	0440	K0070	0014	MES0	3303	OPRTAL	0116
EXREG	0122	K0077	0102	MES1	3307	PASCNT	0250
EXTICK	1166	K0100	0015	MES10	3414	PCLF	6662
F1OP1	0021	K0200	0016	MES11	3424	PCNTR1	1373
F1OP2	0022	K0212	1425	MES13	3433	PCNTR2	1374
F18NR	0020	K0215	1424	MES14	3436	PCNTR3	1375
FILCNT	1040	K0240	1512	MES15	3445	PCREG	0120
FILLER	1037	K0260	0067	MES16	3455	PCSAVE	1344
FILLUP	0717	K0277	0070	MES17	3477	PNTBUF	1120
FIRTIM	0170	K0316	2057	MES10	3502	POLDISK	0115
FLDPLG	3572	K0331	2056	MES19	3174	POLNEX	1000
FLDHLT	0206	K0400	0017	MES2	3325	PREXIT	2635
FLSAVE	1347	K1000	0075	MES3	3352	PRINT	2620
FNDOSUM	0142	K177	2034	MES4	3355	PRN	1450
FORIN	4433	K1777	0076	MES5	3346	PRNDAT	0173
FROCT	1426	K3740	1513	MES6	3363	PRNTER	4455
FWREG	0130	K4000	0073	MES7	3372	P8IE	6665
GENDAT	4426	K4100	0074	MES8	3404	P8KE	6663
GETCH1	0703	K5405	0300	MES9	3410	P8KF	6661
GETDAT	0456	K6500	0541	MES1A	0747	P8TB	6664
GNDAT	1737	K6520	2172	MESAC	1533	PTSTOR	0536
GOBAK	2230	K7000	2135	MESFL	1541	RAD1	1773
GOCDF	1646	K7377	2154	MESMAN	1146	RAD2	1774
GOCNK	2265	K7400	0104	MESMU	1336	RAD3	1775
GOCRL	2260	K7700	0077	MESPAS	0253	RAN1	1767
GOITA	0443	K7760	0100	MESPC	1330	RAN2	1770
GOTIT	1010	K7773	1377	MODSKS	2125	RANDAT	4427
GOTOA	0454	K7775	0110	MGA	7501	RANDOM	1715
GTF	0004	K7777	0101	MGL	7421	RANGEN	4440
HEDTAD	1376	KAERRO	0023	MOSAVE	1346	RANJMS	0522
HLFFLG	3575	KCDF	0103	MRRPN	1456	RDLMRL	2324
INDEXA	0455	KROT	0230	M8KER	1714	ROST	2541
INERR	2426	K8KP	0546	MYAC	1317	RDSTA	1105
INMODE	1076	KTICK	4425	MYLAB	1546	RDSTAT	4447
INTCM	0136	KTIME	0530	NEWRU	0727	RDTRY	1073
INTDA	0124	LAS	4406	NEXT	0261	REALPC	1316
INTER1	3130	LOAD	2554	NODSKP	2353	RECAL	4443
INTER2	2362	LOADD	4453	NODSKS	2702	RECDP	2605
IOG	0554	LOCA	2550	NOERR	1672	RECEIV	4444
IOI1	0752	LDCM	0542	NOSET	0242	REDOA	0415
IOI2	0561	LDCMD	4451	NOTEX	1363	REFILL	0723
IOI3	2557	LDCUR	4452	NOY8TA	3041	REREAD	1063
IOI4	2551	LNKDCA	0361	NTCLAS	1270	RERUN	1124
IOI5	2542	LDDGO	2330	NTERN	1255	RESERR	3111
IOI6	0545	M10	0107	NTSOFT	1251	RESRAN	4442

RESTA	3077	TABLB	0471	XC8CNL	1023
RESTOR	3052	TEXAD	3241	XC8ECH	1063
RETRN	2331	TEXAS	3235	XC8EMR	1207
RETURN	2304	TEXCA	3227	XC8ING	0635
REWRT	1047	TEXCM	3221	XC8OCT	1000
RNFLD	0630	TEXDA	3225	XC8PAS	0200
RNWRD	2600	TEXDB	3245	XC8PAU	0337
ROUINS	1302	TEXDG	3243	XC8PNT	0303
ROUTMP	1544	TEXEX	3217	XC8P8W	0656
RSRAN	1761	TEXPW	3233	XC8SW	0262
RUN	0600	TEXIA	3223	XC8TTY	0272
RUNPOT	0154	TEXP	3213	XC8TYP	1077
SAV1	1771	TEXST	3215	XC8K22	0024
SAV2	1772	TEXWA	3237	XC8KYN	0457
SAVAC	0166	TEXWC	3231	XC8PUT	0056
SAVCM	0174	TIME	3123	XC8L8	0007
SAVE1	0400	TIMER1	3140	XC8LX	0054
SAVEAC	1545	TIMER2	3141	XC8LP	0057
SDKP	0751	TIMER3	2374	XDOLMT	1112
SECFLG	3574	TIMPOT	0152	XDOSW	0520
SELCHK	4436	TMPCNT	0746	XDSKWO	0041
SETFLD	4435	TPSTA	3000	XOUMF	0030
SETGEN	4434	TRASH1	0111	XERNU	0046
SETREG	1553	TRASH2	0112	XPRODUCT	0056
SETUP1	1233	TRASH3	0113	XGNDAT	0026
SETUP2	0225	TRDONE	2365	XKTICK	0025
SKPNOP	0237	TRKFLG	3573	XKTICK	1154
SOFT	1245	TRYCNT	0171	XLAS	0006
SPAC	1506	TRYTIM	1060	XLOAU	0053
SPACE	4431	TSAVE1	3045	XLOCA	0052
SPBLK	0164	TSAVE2	3046	XLOCM	0051
SPFLD	0160	TSAVE3	3047	XOCT1	0032
SPSEC	0163	TSAVE4	3050	XOCT4	0033
SPTK1	0161	TSAVE5	3051	XPRINT	0045
SPTK2	0162	TSTCHA	0715	XPRN	0055
STAPOT	0153	TYLPT	1121	XROST	0047
STATER	2337	TYPDAT	3142	XREG	1572
STATRY	0137	TYPE	4445	XNESTR	0043
STATUS	2313	UPAROW	0615	XRNDOU	0040
STFLO	2703	UPDATE	0114	XRNWMD	0027
STGEN	1753	UPONE	1414	XRSRAN	0042
STPHLT	0603	UPTRY	1120	XSDKP	0050
STRAUT	1325	WAIT	2000	XSPAC	0031
STRBUF	3600	WAIT1	2026	XSTFLU	0035
STREG	0121	WAREG	0132	XSTGEN	0034
STRREL	1151	WASRD	2264	XTAOLA	0457
STRTEX	0224	WATHES	0651	XTABLB	0460
STRWRK	2241	WCREG	0127	XTEAT	0172
SVLNK	0167	WRDCHK	1612	XWAIT	0044
SWDAT	2601	WRKDDN	2274	YESNU	4437
SWR	0020	XC8CKP	1041		
TABLA	0461	XC8CNT	0400		

ERRORS DETECTED: 0

LINKS GENERATED: 163

RUN-TIME: 6 SECONDS

3K CORE USED